



Chemosensory Innovation

ECRO XXXVI — Annual Meeting 2026

Poster program

- Session 1 (16 September 2026, 12:30-14:00) — page 3
- Session 2 (16 September 2026, 18:30-20:00) — page 6
- Session 3 (17 September 2026, 12:30-14:00) — page 9
- Session 4 (17 September 2026, 17:30-19:00) — page 12
- Session 5 (18 September 2026, 12:30-14:00) — page 15

Book of Poster Abstracts

- Session 1 (16 September 2026, 12:30-14:00) — page 18
- Session 2 (16 September 2026, 18:30-20:00) — page 58
- Session 3 (17 September 2026, 12:30-14:00) — page 97
- Session 4 (17 September 2026, 17:30-19:00) — page 136
- Session 5 (18 September 2026, 12:30-14:00) — page 175

Poster sessions location:

Salle Marie Curie • Salle Club • Cloître des Cordeliers

Poster format: A0, portrait. Posters must be in place at least 30 min before the session and removed within 30 min after the session.

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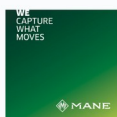
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Session 1 — 16 September 2026, 12:30-14:00

S1-P1: Sustained valence-dependent modulation of sniff volume across sniffs

Andrea Aeimelaesus-Lindström, Frans Nordén, Giulia Ciotti, Irene Zanettin, Artin Arshamian, Johan N. Lundström

→ Abstract page 18

S1-P2: Odors count globally: the number of conscious odor perceptions differs between 17 locations on five continents

Anna Oleszkiewicz, Aleksandra Reichert, Thomas Hummel

→ Abstract page 19

S1-P3: How many odors do we need to smell? A clinical and perceptual perspective

Anne Abriat, Susanne Weise, Evangelia Tsakiropoulou, Iordanis Konstantinidis, Thomas Hummel

→ Abstract page 20

S1-P4: Odor-Evoked Autobiographical Memories Are Not Always Highly Emotional

Berçem Yar, Aslı Aslan, Maria Veldhuizen

→ Abstract page 21

S1-P5: Navigating the molecular space of taste-active peptides

Burcu Özer, Alexandra Steuer, Tuba Bakal, Francesco Ferri, Alexander Otterbach, Silvia Schaefer, Verena Karolin Mittermeier-Kleßinger, Maik Behrens, Corinna Dawid, Antonella Di Pizio

→ Abstract page 22

S1-P6: Unraveling the molecular basis of the Orco-ORx assembly

Camille Meslin, Sridevi Bhamidipati, Zibo Li, Laure Boudier, Nicolas Montagné, Emmanuelle Jacquin-Joly

→ Abstract page 23

S1-P7: Large-scale genetic studies crack the olfactory code

Casey Trimmer, Giulia Papiani, Andreas Keller, Marissa L Kamarck, Leslie B Voss hall, Patrick Pfister, Joel D Mainland

→ Abstract page 24

S1-P8: Episodic Olfactory Autobiographical Memory and expertise across the world

Cédric Manesse, Claire-Sophie Vigneron

→ Abstract page 25

S1-P9: The development of olfactory-mediated disgust in preschool children

Dagmar Schwambergová, Daniela Dlouhá, Marie Štěpánková, Lucie Jelínková, Jan Havlíček

→ Abstract page 26

S1-P10: Olfactory experiences during psychedelic states: a natural language processing study of chemosensory alterations in online communities

Dana Bar-Zvi, Liron Pinchover, Aharon Ravia, Ofer Perl

→ Abstract page 27

S1-P11: Nutrient-Specific Tuning of Sensorimotor Circuits in *Drosophila melanogaster*

Daniel Münch, Rory Beresford, Marcelo Dias, Arina Dygay, Carlos Ribeiro

→ Abstract page 28

S1-P12: Uncovering behavioral signatures of olfactory preference and sex in mice through Keypoint-MoSeq

Donatella Lobraico, Domenico Diacono, Loredana Bellantuono, Claire Martin, Antonio Frigeri, Michele Dibattista

→ Abstract page 29

S1-P13: Aquaporin-4 water channel controls neuronal activity in the olfactory bulb through astrocytes remodeling and affects olfactory behavior

Donatella Lobraico, Pasqua Abbrescia, Ivana Antonia De Marinis, Grazia Paola Nicchia, Antonio Frigeri, Michele Dibattista

→ Abstract page 30

S1-P14: Combinability of odour data for machine learning benchmarks

Florian Anders, Tim Jesgarzewsky, Alexander Crox

→ Abstract page 31

S1-P15: Reproductive state-dependent modulation of odor valence and social behavior via the main olfactory pathway

Jana Marie Sleeboom, Ilona Grunwald Kadow, Annika Cichy

→ Abstract page 32

S1-P16: Characterization of apical-out polarity in murine and human olfactory epithelial organoids

Jinsun Kim, Min-Seok Rha, Hyung-Ju Cho, Chang-Hoon Kim

→ Abstract page 33

S1-P17: AFM uncovers the role of mechanoreception in sweet taste: linking nano-mechanics, cellular signaling, and sensory perception.

Julia Benthin, Melanie Koehler

→ Abstract page 34

S1-P18: Effects of a fragrance intervention on nocturnal arousal: a large-scale randomized controlled trial

Kaori Takahashi, Kanetoshi Ito, Hisashi Mihara, Tsuyoshi Kobayashi, Yukio Goto

→ Abstract page 35

S1-P19: The Subjective of Space. How Atmosphere Shapes Olfactory Imagination.

Lena Trost, Ilja Croijmans

→ Abstract page 36

S1-P20: The impact of familiar and stranger body odors on social anxiety: an eye-tracking and ERP-EEG study protocol

Letizia Zurlo, Elisa Dal Bò, Cinzia Cecchetto, Claudio Gentili

→ Abstract page 37

S1-P21: Role of experience in the construction of flavor and the emergence of aroma-induced taste enhancement

Lucie Ben Sussan, Thibault Chabin, Thierry Thomas-Danguin

→ Abstract page 38

S1-P22: Differential neural processing of sweet and sour food odors: evidence from psychophysics, fMRI activation, and connectivity analysis

Mariano Mastinu, Dmitry Desser, Hitomi Sato, Jonathan Warr, Antje Haehner, Iole Tomassini Barbarossa, Thomas Hummel

→ Abstract page 39

S1-P23: Neurophysiological correlates of attention and creativity in candle-induced olfactory environments

Marine Coeugnet, Zeynep Yenigun, Thomas Fontaine, Yvonne Delevoye-Turrell, Marion Vincent

→ Abstract page 40

S1-P24: Interaction of sweet tasting compounds with saliva proteins is related to their sensory properties

Markus Rechl, Sabrina Schild, Christian Krupitzer, Silke Hillebrand, Jakob Ley, Barbara Lieder

→ Abstract page 41

S1-P25: Using SCED to test the effects of olfactory awakening in young children

Maryse Delaunay, Elise Gendronneau, Odran Guilbert, Ana Reutimann, Marie Anquetil

→ Abstract page 42

S1-P26: Water Matters: Boosting Aroma to Improve Appetite in Senior Dogs

Melanie Trehou

→ Abstract page 43

S1-P27: Pleasant olfactory stimuli can reduce cigarette craving: Implications for smoking cessation

Michael Sayette, Rachel Herz

→ Abstract page 44

S1-P28: Neurophysiological driving mechanisms in olfactory perception

Nick Menger, Sarah Steigleder, Yuri Pavlov, Boris Kotchoubey

→ Abstract page 45

S1-P29: Rat olfactory bulb vasopressin system shows low connectivity within intraglomerular and interglomerular network

Nicolas Reichardt, Veronica Egger

→ Abstract page 46

S1-P30: The Silent Symptom: Olfactory Dysfunction at the Onset of Alzheimer's Disease

Raffaella Pia Gatta, Ivana Antonia De Marinis, Claudia Palazzo, Antonio Frigeri, Michele Dibattista

→ Abstract page 47

S1-P31: How could personalization of olfactory stimuli enhance cognitive performance? Theoretical and empirical evidence for an integrative approach.

Roxane Bartoletti, Ambre Denis-Noël, Dirk Steiner, Xavier Corveleyn

→ Abstract page 48

S1-P32: Functional interaction between olfactory receptors and TRPV1 studied in HEK293T cells

Sakura Moriyama, Shun'ichi Kuroda

→ Abstract page 49

S1-P33: Diverse Expression of Odorant Receptors (ORs) in the Olfactory Organ of Turtles

Shoko Nakamuta, Zicong Zhang, Atsuhiro Sakuma, Masato Nikaido, Hideaki Kato, Masanori Mori, Tomomi Matsumoto, Masanori Kurita, Masayoshi Takeda, Hideki Noda, Takuya Yokoyama, Nobuaki Nakamuta

→ Abstract page 50

S1-P34: Role of Oxytocin Receptor Signaling in the Detection of Pup-Released Molecules by Vomeronasal Sensory Neurons in Male Mice

Thiago Nakahara, Rafael Gotteris-Cerisuelo, Adrien Acquistapace, Elliott Trives, Maxime Meunier, Chantal Porte, Hélène Vacher, Ignacio Caballero, Benoît Piegu, Markus Rothermel, Emmanuel Valjent, Lucie Pelissier, Fernando Martínez-García, Pablo Chamero

→ Abstract page 51

S1-P35: Functional analyses of bitter taste receptors in Horses (*Equus ferus caballus*)

Thomas Delompré, Christine Belloir, Adeline Karolkowski, Cléo Omphalius, Samy Julliard, Loïc Briand

→ Abstract page 52

S1-P36: A low-calorie rebaudioside D-glucose-sodium sweetener matches sucrose in palatability and neural responses in a functional MRI study of healthy adults.

Yuko Nakamura, Rika Takahashi, Tadahiro Ohkuri

→ Abstract page 53

S1-P37: Harder is not better: item difficulty does not enhance discrimination in olfactory identification tests

Zheni Tian

→ Abstract page 54

S1-P38: Pyroxamide as a potential therapeutic option to counteract the α -synuclein preformed fibrils-induced damage in human nasal epithelial cells.

Marina De Miguel, Paz Cartas-Cejudo, Leire Extramiana, Mercedes Lachén-Montes, Silvia Romero-Murillo, Djordje Gveric, Joaquín Fernández-Irigoyen, Enrique Santamaria

→ Abstract page 55

S1-P39: Smelling in Color and Sound: Crossmodal Smell Training

Fabienne Huebener, Christelle Porcherot Lassalette, Maria José Sanchez, Maria Lorente, Sven Henneberg, Jean A McEwan, Ragnhild Eg, Ellie Attack, Monica Borgogno, Noelle Beno

→ Abstract page 56

S1-P40: Context-dependent modulation of danger perception in the mouse Grueneberg ganglion

Flavio Ferreira, Kelly de Sousa, Noah Gilliland, Youssef Rammeh, Julien Brechbühl, Marie-Christine Broillet

→ Abstract page 57

Session 2 — 16 September 2026, 18:30-20:00

S2-P1: Relaxation through olfaction: the soothing power of essential oils to support aging

Amélie Puech, Jacob Fournier, Mailys Le Bris, Johannes Frasnelli, Jane Plailly

→ Abstract page 58

S2-P2: Endoplasmic reticulum specialization predefine sensory neuron maturation in the vomeronasal organ of rodents

Carlsy Ybanez, Paolo Forni

→ Abstract page 59

S2-P3: Electrical stimulation of the human brain elicits odor perception: non-canonical brain structures may contribute to olfactory processing

Coralie Mignot, Susanne Weise, Georg Leonhardt, Stephan B. Sobottka, Dietmar Krex, Dino Podlessek, Moustafa Bensafi, Thomas Hummel

→ Abstract page 60

S2-P4: A step into children's olfactory world: a qualitative analysis of spontaneous odor representations in early childhood

Daniel Marek, Marta Rokosz, Daniel Jędrzejczyk, Anna Oleszkiewicz

→ Abstract page 61

S2-P5: One receptor, two binding sites: ligand-specific activation of TAS2R14 by pharmaceutical drugs

Einav Malach, Shir Eyal, Nitsan Dallal, Alon Rainish, Evgenii Ziaikin, Masha Niv

→ Abstract page 62

S2-P6: Does body mass shape sensory experience and its linguistic expression?

Ekaterina Rzhetskaya, Ilja Croijmans, Laura Speed, Margot van Mulken

→ Abstract page 63

S2-P7: Obesogenic diet alters astrocytes function in the olfactory bulb

Federica Genovese, Julien Castel, Anthony Ansoult, Bahar-Sadat Mousavi, Serge Luquet, Claire Martin

→ Abstract page 64

S2-P8: Olfactory bulb gamma oscillations may represent a valence scaffold for the next sniff

Frans Nordén, Anja Winter, Artin Arshamian, Mikael Lundqvist, Johan Lundström

→ Abstract page 65

S2-P9: Towards Artificial Olfaction

Halima Mouhib

→ Abstract page 66

S2-P10: Age-group differences in effective connectivity between brain regions during salty taste perception in humans.

Hitomi Sato, Hirotaka Wada, Hideki Matsumoto, Kenji Tokumori, Akira Kunimatsu, Osamu Abe, Hiroki C. Tanabe, Tazuko K. Goto

→ Abstract page 67

S2-P11: Real-time chemical sampling at the olfactory epithelium shows sub-optimal receptor access

Irene Zanettin, Frans Nordén, Mikael Lundqvist, Artin Arshamian, Johan Lundström

→ Abstract page 68

S2-P12: Histological and neurochemical characterization of the bovine olfactory and vomeronasal systems during the early fetal stages

Irene Ortiz Leal, Lucía López García, Sara Ruiz Rubio, Renato Barrionuevo, Jacobo Álvarez, Antía Acción, Luis Quintela, Pablo Sanchez Quinteiro

→ Abstract page 69

S2-P13: Detection of neurodegenerative and inflammatory biomarkers in human olfactory mucosa obtained by nasal brushing: a feasibility study

Ji-Hyeon Shin, Hyun Jung Yang, Mark Albers, Priyanka Sinha, Byung Guk Kim, Do Hyun Kim

→ Abstract page 70

S2-P14: Contextual variability in odour quality representations: learning the identity of ambivalent odours

Leonie Seidel, Zhengxiao Feng, Janina Seubert

→ Abstract page 71

S2-P15: Spatial representations in the piriform cortex emerge after odor-context associative learning

Lucca Salomon, Noel Federman, Sebastián Romano, Antonia Marin-Burgin

→ Abstract page 72

S2-P16: Investigating the physiological properties of olfactory bulb granule cells' basal dendrites

Manon Leygnier, Veronica Egger

→ Abstract page 73

S2-P17: Beyond taste: Unveiling the multisensory sensation of creamy mouthfeel by synergizing biophysical, molecular and food biology approaches

Marina Wiesenfarth, Julia Benthin, Sanjai Karanth, Melanie Koehler

→ Abstract page 74

S2-P18: Memory consolidation during wakeful rest: could odors play a role?

Marion Walter, Olivier Martinez, Max Lacaze, Samuel Garcia, Jane Plailly, Nathalie Buonviso

→ Abstract page 75

S2-P19: What children know about smells: linking spontaneous odor reports to olfactory salience

Marta Rokosz, Daniel Marek, Tim Jesgarzewsky, Daniel Jędrzejczyk, Ilona Croy, Anna Oleszkiewicz

→ Abstract page 76

S2-P20: Perception and use of food fatty-acids odours on flies

Martine Berthelot-Grosjean, Cindy Ménagé, Gérard Manière, Ana Depetris-Chauvin, Marie Juge, Marlyse Bourguignon, Solena Canale Parola, Marion Canale, Yaël Grosjean

→ Abstract page 77

S2-P21: Characterization of an olfactory-specific extracellular Glutathione Transferase involved in odorant degradation in the moth *Spodoptera littoralis*

Martine Maïbèche, Eléonore Braun, Nelly Gareil, Nicolas Durand, Muriel Jager, Jean-Marie Heydel, Mathieu Schwartz, Fabrice Neiers, Thomas Chertemps

→ Abstract page 78

S2-P22: Neuroethology of taste in the honey bee *Apis mellifera*: from behaviour to the neural circuits

Melissa Pitzalis, Jean-Christophe Sandoz, Julie Carcaud

→ Abstract page 79

S2-P23: State-dependent modulation of decision making in olfactory-driven conflicting situations

Michelle Eloise Gutierrez Le Barbier, Ilona Grunwald Kadow, Annika Cichy

→ Abstract page 80

S2-P24: Sticky Choices: How the non-clustered protocadherins shape accessory olfactory system circuitry

Nikki Dolphin, Noah LeFever, Paolo Forni

→ Abstract page 81

S2-P25: Neural correlates of olfactory training-induced improvement in odor-hedonic perception and emotional behavior in mice subjected to early-life adversity

Pascal Bichere, Jérôme Brunelin, Nathalie Mandairon

→ Abstract page 82

S2-P26: Flavour-specific mapping of food odours in the insular cortex

Putu Agus Khorisantono, Apostolia Filippopoliti, Hilda Lindén, Maria G Veldhuizen, Janina Seubert

→ Abstract page 83

S2-P27: Infrared thermography as a physiological indicator of the impact of human emotional odors on sheep (*Ovis aries*).

Karim Amziane, Lina Bouly, Fabrice Damon, Alexandra Destrez

→ Abstract page 84

S2-P28: The olfactory processing of social cues in females is highly influenced by prolactin

Rebeca Corona, Verónica Viñuela-Berni, Teresa Morales

→ Abstract page 85

S2-P29: Stimulus temperature changes the psychophysical dimensions of salt taste.

Roberto Vincis, Alan Spector, Mariela Marques

→ Abstract page 86

S2-P30: The nasal cycle as a marker of central nervous system activity

Rotem Arbetman, Ofer Perl, Noam Sobel

→ Abstract page 87

S2-P31: The lingual GLP-1 system modulates sweet taste detection and glucose metabolism.

Savanah Buiret, Pierre-Jean Augagneur, Marie-Annick Maire, Nicolas Poirier, Corinne Leloup, Sandrine Chometton

→ Abstract page 88

S2-P32: Processing mechanisms of temporally complex odours across olfactory circuits

Shambhavi Phadnis, Tobias Ackels

→ Abstract page 89

S2-P33: Sensory dysfunction in the 3xTg-AD mouse model: Investigating the degeneration of visual and olfactory systems in early Alzheimer's Disease

Simona Francia, Su-Chun Huang, Caterina Michetti, Caterina Montani, Serena Riccitelli, Luca Tagliafico, Negin Bigliari, Antonio Uccelli, Matteo Pardini

→ Abstract page 90

S2-P34: Data-driven exploration of functional domains and functions of human olfaction

Typhaine de Simone, Alessio Ragno, Céline Robardet, Marc Plantevit, Eric Angelini, Malick Diop, Moustafa Bensafi

→ Abstract page 91

S2-P35: Cross-modal associations between odors and intranasal sensations

Xi Wang, Halina. S. Stanley, Marc Thevenet, Moustafa Bensafi

→ Abstract page 92

S2-P36: Olfaction: a key player in cognitive development and degeneration

Yu-Nan Chen

→ Abstract page 93

S2-P37: Gastric bitter taste receptors differentially regulate essential and toxic metal ion homeostasis

Hanna Noreen Orth, Veronika Somoza

→ Abstract page 94

S2-P38: Shifted perceptual space across major datasets in olfaction research

Ido Mellul, Kobi Snitz, Tali Weiss, Noam Sobel

→ Abstract page 95

S2-P39: The natural statistics of human olfactory experience: A multi-national project

Barr Dean Herrnstadt, Danielle Honigstein, Rotem Arbetman, Johan Lundström, Jonathan Williams, Danica Kragic

→ Abstract page 96

Session 3 — 17 September 2026, 12:30-14:00

S3-P1: Investigating the activation mechanism of insect odorant receptors

Albert Thorhallsson, Arthur Comte, Maxence Lalis, Sai Zhang, Nicolas Montagné, Jérémie Topin, Emmanuelle Jacquin-Joly, Sébastien Fiorucci

→ Abstract page 97

S3-P2: From sensors to senses: a longitudinal study of personal pollution exposure, olfactory threshold, and well-being

Aleksandra Reichert, Marta Hendler, Anna Oleszkiewicz

→ Abstract page 98

S3-P3: Sensory Drivers of Oral Nutritional Supplement Acceptability in Restrictive Anorexia Nervosa and Restrained Eaters

Ambre Godet, Gabriela Ferreira-Avelino, Anna Claret, Stéphanie Chambaron, Marie-Claude Brindisi, Charlotte Sinding

→ Abstract page 99

S3-P4: Representation and transformation of temporally complex odours in the mouse olfactory bulb

Anantu Sunil, Tobias Ackels

→ Abstract page 100

S3-P5: Investigating the effects of olfactory training in individuals with specific anosmia to R(-)-carvone

Annika Sachs, Anna Kristina Hernandez, Thomas Hummel

→ Abstract page 101

S3-P6: Sequence analysis and modeling of vomeronasal type-2 receptors (V2Rs) in mammals

Aurore Lamy, Rajesh Durairaj, Alessandro Cozzi, Angelo Gazzano, Patrick Pageat

→ Abstract page 102

S3-P7: Olfactory training shows no positive effect on olfactory working memory in older people

Barbara Żyżelewicz, Anna Oleszkiewicz, Michał Pieniak

→ Abstract page 103

S3-P8: Did Proust need to taste his madeleine? Oral cues offer no advantage over orthonasal olfactory cues in evoking autobiographical memories.

Berçem Yar, Maria Geraldine Veldhuizen

→ Abstract page 104

S3-P9: Long COVID, Smell and Taste Dysfunction in Adults: the 2024 U.S. National Health Interview Survey

Chuan-Ming Li, Howard J. Hoffman, Le Chen, Shristi Rawal, Susan E. Coldwell, Diana Fisher, John E. Hayes, Valerie B. Duffy

→ Abstract page 105

S3-P10: Olfactory predictors of generalized anxiety in a non-clinical sample of young adults.

Daniel Jędrzejczyk, Michał Pieniak

→ Abstract page 106

S3-P11: Probing how the temporal structure of odour plumes informs mouse navigation behaviour

Dyutika Banerjee, Amirreza Gholivand, Anantha Padmanabhan, Tobias Ackels

→ Abstract page 107

S3-P12: Role of a Glutathione Transferase in the olfactory system of *Drosophila melanogaster*

Eléonore Braun, Nicolas Petiot, Nelly Gareil, Fabrice Neiers, Martine Maibèche, Thomas Chertemps

→ Abstract page 108

S3-P13: Rethinking olfactory imagery as a multidimensional experience

Eléonore Giraudet, Asifa Majid, Olivier Collignon

→ Abstract page 109

S3-P14: Inflammation modulates neurogenesis in the olfactory epithelium of adult zebrafish following olfactory bulb lesions

Erika Calvo-Ochoa, Lexus Putt, Rebecca Post, Olivia Wiley

→ Abstract page 110

S3-P15: Newly identified amino acid substitutions in the C-terminal domain further

improve functional expression of difficult-to-express ORs

Fiona Buchli, Roger Emter, Andreas Natsch

→ Abstract page 111

S3-P16: Taste receptor–inflammation crosstalk predominates in gustatory tissue in a cafeteria diet model

Isabella Kimmeswenger, Andrej Feješ, Veronika Somoza, Katarína Šebeková, Barbara Lieder

→ Abstract page 112

S3-P17: Functional characterization of juxtaglomerular cells in the mouse accessory olfactory bulb

Johanna Flesch, Andres Hernandez-Clavijo, Marc Spehr

→ Abstract page 113

S3-P18: Identification of key pheromones and neural mechanisms involved in bumblebee reproduction

Jorn Oppersma, Zsolt Karpati, Thomas Schmitt, Julie Carcaud, Jean-Christophe Sandoz

→ Abstract page 114

S3-P19: Chemical Intolerance and Smell: Meta-analysis Reveals Altered Response Bias Rather Than Enhanced Sensitivity

Kazushi Yoshida*, Elisa Dal Bo**, Pamela H. Dalton, Valentina Parma

→ Abstract page 115

S3-P20: Gustatory ERP-Based Classification of Taste Dysfunction and Normogeusia Using Machine Learning

Kwangsu Kim, Alissa Boberg, Hitomi Sato, Thomas Hummel

→ Abstract page 116

S3-P21: Deciphering the molecular basis for activation and inhibition of olfactory receptors.

Marc El Hasbany, Chulwon Choi, Marie-Charlotte Dumargne, Romain Barrès, Hee-Jung Choi, Jérémie Topin

→ Abstract page 117

S3-P22: Multi-dimensional study on sex-specific patterns of taste deficits, α -synuclein, and Supervised Learning for diagnosis and disease stratification in Parkinson's disease

Melania Melis, Lala Chaimae Naciri, Giorgia Sollai, Roberto Crnjar, Iole Tomassini Barbarossa

→ Abstract page 118

S3-P23: Assessing olive oil adulteration: a combined study of human olfaction and nanostructured gas sensor responses

Michele Astolfi, Martina Masin, Mario Barbieri, Irene Donelli, Leda Montesi, Cesare Malagù, Giorgio Rispoli, Angela Pignatelli

→ Abstract page 119

S3-P24: Prenatal development of the rabbit vomeronasal and olfactory systems at embryonic day 20: early differentiation within a 30-day gestation

Pablo Sanchez-Quinteiro, Xoana Rivas López, Renato Barrionuevo Ponce, Uxía Yáñez Ramil, Sara Ruiz Rubio, Jacobo Álvarez, Luis Quintela, Irene Ortiz-Leal

→ Abstract page 120

S3-P25: A wide sensory window between odor and irritation thresholds enables both safety and consumer acceptability of fragrances

Pamela Dalton, Nikaeta Sadekar

→ Abstract page 121

S3-P26: Exploring the mechanisms and causes of VNO inflammation: differences between domestic pigs and sheep

Pietro Asproni, Violaine Mechin, Marine Verbaere, Tolga Karaali, Alessandro Cozzi, Patrick Pageat

→ Abstract page 122

S3-P27: Loss of smell and taste among recipients of hematopoietic stem cell transplantation

Preet Bano Singh, Mahnoor Nazar, Alix Young, Per Ole Iversen

→ Abstract page 123

S3-P28: Depicting Evoked field potentials to Olfactory Stimuli using on-scalp MEG: Preliminary Results for a Smell Oddball Task

Samet Albayrak, Christoph Pfeiffer, Janina Seubert

→ Abstract page 124

S3-P29: Macrophages are dispensable for taste bud regeneration upon nerve injury repairSuyoung Ki, Yongtaek Jeong

→ Abstract page 125

S3-P30: ItypOR41 utilizes universal verbenoid anchors, whereas the stereochemical structure of (S)-cis-verbenol induces unique receptor activation.Tammy Shim, Jothi Kumar Yuvaraj, Maxence Lalis, Jérémie Topin, Martin N Andersson

→ Abstract page 126

S3-P31: Quantifying the cross-linguistic translation appropriateness of odor descriptors using large-scale corpus analysisTim L. Jesgarzewsky, Alexander Croy, Ilona Croy

→ Abstract page 127

S3-P32: The time-dependent effects of benign taste experience on neural plasticity, learning, and taste discrimination.Veronica Flores, Makayla Ellison, Dallas Shuman

→ Abstract page 128

S3-P33: Stimulus-driven modulation of sensory sensitivity in mouse olfactory sensory neuronsVictoria K. Switacz, Lisa M. Moeller, Daniela R. Droese, Marc Spehr

→ Abstract page 129

S3-P34: Orexin neurons contribute to state dependent control of olfaction.Willem Koppenol, Janice Bulk, Sophie Steculorum

→ Abstract page 130

S3-P35: Application of an Electronic Nose in the Noninvasive Identification of Parkinson's DiseaseXiaolu Li, Thomas Hummel, Jesús Lozano Rogado, Cristina Brugera Salas

→ Abstract page 131

S3-P36: Proof-of-concept trial of pre-meal olfactory nudging versus habituation in women with overweight or obesityYao Zhao*, Surabhi Bhutani*, Emma Johnson, Victoria Esparza, Valentina Parma

→ Abstract page 132

S3-P37: How chemistry and temporal perception shape acidity language in wineQian Janice Wang

→ Abstract page 133

S3-P38: Beyond OR51E2: systematic profiling reveals novel dysregulated olfactory receptors in prostate cancerAna Catarina Duarte, Marília Figueira, Andreia Lopes, Ana Raquel Costa, Isabel Gonçalves, Sílvia Socorro, Manuel Lemos, José Eduardo Cavaco, Cecília Santos

→ Abstract page 134

S3-P39: Kokumi peptide RHE under a low-salt condition enhances salt perception and modulates EEG responses in humansChuang Yu, Piao Yi, Tao Jiang

→ Abstract page 135

Session 4 — 17 September 2026, 17:30-19:00

S4-P1: A hindbrain neuronal population mediates water perception and consumption

Akiyuki Taruno

→ Abstract page 136

S4-P2: Transcutaneous vagus nerve stimulation at olfactory bulb-relevant frequencies modulates human olfactory function

Anja L. Winter, Andrea Aejmelaesus-Lindström, Kristen Gregory, Gregory Francis, Cristina Jaen, Evelina Thunell, Artin Arshamian, Johan N. Lundström

→ Abstract page 137

S4-P3: Semantic learning outpaces perceptual change during perfumery training

Anne-Lise Saive, Jane Plailly, Stéphanie Chambaron, Youness Hourri, Carla Monin, Zeineb Nhouchi, Justine Belay, Pauline Chalut, Farnaz Hanaei, Nadine Vallet

→ Abstract page 138

S4-P4: A microbiome-derived olfactory signal regulates inter-male aggression and social dominance in mice

Annika Cichy, Adam Dewan, Zhuoling He, Cassandra Fitzgerald, Madison Ratkowski, Jakub Krasewicz, Vaikhari Ozarkar, Sarah Kaye, Jingji Zhang, Paul Feinstein, Thomas Bozza

→ Abstract page 139

S4-P5: Olfactory sensitivity and division of labor in the honey bee *Apis mellifera*

Camille Faber, Julia Mariette, Jean-Christophe Sandoz, Julie Carcaud

→ Abstract page 140

S4-P6: GLP-1 signaling in the olfactory system modulates glucose metabolism in obese mice: effects of olfactory bulb versus intranasal exendin-4 administration

Camille Gros, Mireia Montaner, Jessica Denom, Christophe Magnan, Hirac Gurden

→ Abstract page 141

S4-P7: Modulating the modulators: chemogenetic dissection of bulbar

dopaminergic contributions to olfactory behaviour.

Chloé Guillaume, Pablo Fernandez-Garcia, Connor Doyle, Edina Horvath-Gulacsi, Elisa Galliano

→ Abstract page 142

S4-P8: Non-monotonic psychometric functions in human olfaction

E. Leslie Cameron, Brijana Labiak, Richard L. Doty

→ Abstract page 143

S4-P9: Axon-bearing, GABAergic/dopaminergic interneurons in the rat olfactory bulb: a cellular substrate for directed lateral inhibition between glomeruli?

Esteban Pino, Nuha Parammal Koller, Sebastian Eder, Magdalena Springer, Martin Nawrot, Veronica Egger

→ Abstract page 144

S4-P10: Auditory and visual object processing in olfactory cortex of individuals with life-long olfactory deprivation

Evelina Thunell, Moa G. Peter, Fahimeh Darki, Johan N. Lundström

→ Abstract page 145

S4-P11: A flexible, semi-naturalistic arena for quantification of complex mouse behavior

Fabian Quicken, Susan A. V. Jennings, Christopher Wiesbrock, Thomas Clavel, Marc Spehr

→ Abstract page 146

S4-P12: Chemosensory function and ingestive behavior in adults using GLP-1 receptor agonists

Giorgia Rutigliani, Richard D. Mattes

→ Abstract page 147

S4-P13: Desensitization, Adaptation and Habituation to Odors: Reconciling Terminology, Mechanisms, and Methods in Human Olfactory Research

Javier Albayay, Kathrin Ohla, Richard Hohenberger

→ Abstract page 148

S4-P14: Emotional and sensory processing, but not resting blood pressure, are associated with variations in olfactory perception

Jessica Gaby, Lauryn Ross, Sylvia Vonderwell, James Loveless

→ Abstract page 149

S4-P15: Dwarfing glutamic acid: Discovery of glutamic acid derivatives as highly potent umami receptor (TAS1R1/TAS1R3) modulators

Joseph Fotsing, Melissa S. Wong, Lan Zhang

→ Abstract page 150

S4-P16: Obesity selectively impairs palatable taste-guided licking behaviour and is reversed by GLP-1 receptor agonists

Junko Inoko, Nawarat Rattanajearakul, Ken-ichiro Nakajima

→ Abstract page 151

S4-P17: Olfactory communication and biocontrol potential in the European water vole

Kévin Poissenot, Andréa Cavaliuc, Céleste Terres-phi, Ambre Fafournoux, Chantal Porte, Jean-philippe Dubois, Hélène Vacher, Virginie Lattard, Matthieu Keller

→ Abstract page 152

S4-P18: Initial Evaluation of Generative AI Responses within the Hub4Smell Smell Test Data Repository Compared to Human Expert Analyses

Danielle Reed, Cailu Lin, Vicente Ramirez, Rollo Sicoco, Valentina Parma, Pamela Dalton, Patricia Lucas-Schnarre, RJ Kedziora

→ Abstract page 153

S4-P19: Flavor-flavor and flavor-nutrient conditioning with odors: competition or amplification?

Lucile Rey, Janina Seubert

→ Abstract page 154

S4-P20: Evolution of pheromone detection in grain weevils

Ludvine Brajon, Milla Menouillard, Pauline Depierrefixe, Nicolas Montagné

→ Abstract page 155

S4-P21: Environmental pollution in a dish: Modelling pollution-induced effects on taste and nutrient sensing using taste bud organoids

Lynn-Christin Saborowski, Jochen Dobner, Andrea Rossi

→ Abstract page 156

S4-P22: Decoding sensory memories of smells and sounds in a Virtual Reality environment.

Malina Szychowska, Teodor Jernsäther, Marie K. X. Low, Cecilia Salzmann, Jonas K. Olofsson

→ Abstract page 157

S4-P23: GraphNOSE: a graph transformer in olfaction for single molecule and mixture prediction

Mrityuniay Sharma, Sarabeshwar Balaji, Valentina Parma, Ritesh Kumar

→ Abstract page 158

S4-P24: Genomic and behavioral evidence for social olfaction in white-faced capuchins

Muhammad Shoaib Akhtar, Nelle K. Kulick, Alice C. Poirier, Hadjira Hamou, Carlota F. Galán Plana, Suheidy Romero Morales, Gabriel W. Benson, Catalina Chaves-Cordero, Raquel Hernandez Rojas, Valeria Villero, Meg Crofoot, Thore Bergman, Joseph D. Orkin, Brendan J. Barrett, Jenny Tung, Katherine M. Jack, Amanda D. Melin

→ Abstract page 159

S4-P25: Trajectory of retronasal olfactory function in patients with chronic rhinosinusitis with nasal polyps treated with biologics: a prospective longitudinal study

Paolo Boscolo-Rizzo, Luca Raimondo, Giacomo Spinato, Giancarlo Ottaviano

→ Abstract page 161

S4-P26: Olfactory discrimination in four year old children using a simplified two alternative forced choice task - preliminary results

Paulina Gołab, Barbara Żyżelewicz, Anna Oleszkiewicz

→ Abstract page 162

S4-P27: Associative learning between intranasal electrical stimulation patterns and olfactory concepts

Philipp Schäfer, Susanne Weise, Halina Stanley, Moustafa Bensafi, Romain Dubreuil, Thomas Hummel, Coralie Mignot
→ Abstract page 163

S4-P28: Machine Learning-Driven Discovery of Behaviorally Active Odorants Across Insect Species and Humans

Anandasankar Ray
→ Abstract page 164

S4-P29: The impact of Sex Hormones on Nasal Respiratory Patterns in Women

Reut Weissgross, Timna Soroka, Lior Gorodisky, Danielle Honigstein, Aharon Weissbrod, Noam Sobel
→ Abstract page 165

S4-P30: A diet experience with two metabolically distinct sugars reprograms the oral glucosensing.

Sandrine Chometton, Lindsey A. Schier
→ Abstract page 166

S4-P31: Exploring barriers to seeking support for Smell and Taste Disorders (SATDs) amongst ethnic minority communities in the UK.

Sanjoli Mathur, Ateeqah Yussuf-Shabi, Tabitha James, Gabija Klyvyte, Mohammed Jawad, Erika Sims, Zhu Yeap, Helen Rhodes, Duncan Boak, Carl Philpott
→ Abstract page 167

S4-P32: Object or cake? Selective processes in the human brain for the recently discovered food area in the ventral visual stream

Sümevra Nur Atalay, Gizem Özen, Putu Agus Khorisantono, Burcu Ayşen Ürgen, Maria Geraldine Veldhuizen
→ Abstract page 168

S4-P33: High-frequency transcutaneous auricular vagus nerve stimulation improves olfactory function in patients with idiopathic and post-infectious olfactory dysfunction

Susanne Weise, Sarah Elsner, Thomas Hummel
→ Abstract page 169

S4-P34: How glia shape adaptive plasticity in the mouse olfactory bulb

Tess Stanley, Elisa Galliano
→ Abstract page 170

S4-P35: Decoding semantic complexity in natural vs. non-natural odor descriptions

Ulrich Ambassa Menyie, Christophe Deroo, Marie-Eugénie Bouge, Christophe Marin, Jérémie Topin, Sébastien Fiorucci
→ Abstract page 171

S4-P36: Does your compound stimulate GLP-1 release? A new screening tool.

Valentina Casà, Sara Montelatici, Laura Stucchi, Camilla Trovesi, Paola Bonetti, Stefania De Cesare, Alice Segnali, Loredana Redaelli, Viviana Agus, Alberto Di Silvio, Marcel Winnig
→ Abstract page 172

S4-P37: Selecting wine stimuli with consistent sensory descriptions across expertise levels

Zuzanna Dedyk, Jérémie Lafraire, Anne-Lise Saive
→ Abstract page 173

S4-P38: Deconstructing the olfactory crosstalk between acetylation and phosphorylation in Alzheimer's disease

Silvia Romero-Murillo, Aroa Roncal, Paz Cartas-Cejudo, Marina De Miguel, Leire Extramiana, Joaquín Fernández-Irigoyen, Enrique Santamaría
→ Abstract page 174

Session 5 — 18 September 2026, 12:30-14:00

S5-P1: Effects of physical exercise and olfactory training with pleasant and unpleasant odors on verbal fluency and depression

Agnieszka Sabiniewicz, Aleksandra Reichert, Anna Oleszkiewicz, Antje Haehner, Thomas Hummel

→ Abstract page 175

S5-P2: Processing history influences bitterness and astringency in rapeseed protein

Aina B. Gil-González, Ching Yue Chow, Marie Sofie Møller, Wender L.P. Bredie

→ Abstract page 176

S5-P3: Bitter taste signalling as a chemosensory gatekeeper at the blood-brain barrier

Ana R. Costa, Diogo D. Gaspar, Isabel Gonçalves, Ana C. Duarte, Cecília R. A. Santos

→ Abstract page 177

S5-P4: Experience-dependent plasticity shapes odorant mixture representations in the antennal lobe via local interneurons

Lautaro A. Duarte, Martín Klappenbach, Nicolás Pérez, Fernando F. Locatelli

→ Abstract page 178

S5-P5: Understanding the attractiveness of human body odors: the link between perception and chemical composition

Camille Ferdenzi, Valentine Brémond Bostoen, Damien Steyer, Nicolas Barthès, Bruno Buatois, Laurent Dormont

→ Abstract page 179

S5-P6: Grounding olfactory perception in language: Benchmarks and models for generating natural language odor descriptions from molecular structure

Cyrille Mascart, Khue Tran, Khristina Samoilova, Liam Thomas Storan, Tung kai Liu, Alexei Koulakov

→ Abstract page 180

S5-P7: Perception of Caloric-Richness Influences the Metabolic Dose-Response Relationship of Sugar-Sweetened Beverages

Dilan Deniz Yilmaz, Maria Geraldine Veldhuizen, Zeynep Altinkaya, Ugur Dal

→ Abstract page 181

S5-P8: Sodium chloride and sour taste in humans – a pilot study

Emilia Leszkowicz, Julia Figat, Manuela Fronda, Krystian Marszałek, Jakub Kosiński, Thomas Hummel

→ Abstract page 182

S5-P9: A low-dimensional code for perceptual similarity in olfaction.

Walter Bast, Cina Aghamohammadi, Priyanka Gupta, Tatiana Engel, Dinu F. Albeanu

→ Abstract page 183

S5-P10: Neural Integration of Peripheral Taste and Internal State in Drosophila

Youngseok Lee

→ Abstract page 184

S5-P11: Olfactory epithelial remodeling in eosinophilic chronic rhinosinusitis: a murine model study

Agnès Dekeyser, Marylène Lecocq, Alessandra Camboni, Sophie Gohy, Charles Pilette, Brian Lin, Caroline Huart, Valérie Hox

→ Abstract page 185

S5-P12: Synthesized human fear odor and linguistic processing: Cognitive and physiological evidence from an eye-tracking study

Ilija Croijmans, Javier Gómez Pérez, Ilona Croy, Jasper de Groot

→ Abstract page 186

S5-P13: What ligand binding reveals about mammalian odorant-binding protein function.

Jérémye Topin, Maxence Lalis, Christine Belloir, Nykola Jones, Søren Hoffman, Cornelia Meinert, Loïc Briand

→ Abstract page 187

S5-P14: Koku compounds modulate fat and viscosity perception in milk: a temporal sensory study

Jonas Junge, Caroline Agerschou, Konstantina Sfyra, Valentina Parma, Ulla Kidmose

→ Abstract page 188

S5-P15: Representations of odor mixtures in the antennal lobe of *Aedes aegypti*

Madhurima Roy, Pranjul Singh, Amritmay Biswas, Abin Thomas, Nitin Gupta

→ Abstract page 189

S5-P16: Attentional modulation of human single-neuron odour representations

Marcel Kehl, Naz Dikecligil, Valeri Borger, Rainer Surges, Jay Gottfried, Florian Mormann

→ Abstract page 190

S5-P17: OSRNs and Resource Editing through Subcellular Cluster Mapping in Fish

Subrata Kumar De, Gour Maity, Swaraj Kumar Sarkar

→ Abstract page 191

S5-P18: Olfactory experience and behavioural processing as a human factors frontier in spaceflight.

Caroline McMillan, Jarin Chu, Colleen McLeod Garner, Mindy Yang

→ Abstract page 192

S5-P19: Impact of olfactory and threat priming on human odor detection thresholds

Michał Pieniak, Daniel Jędrzejczyk, Michał Stefanczyk

→ Abstract page 193

S5-P20: Crossmodal Correspondences between Sound and Smell Mediated by Affect

Michele El Khoury, Olivier Houix, Celine Manetta, Nicolas Misdariis, Patrick Susini

→ Abstract page 194

S5-P21: Beyond smell loss: cognitive and affective alterations in post-COVID olfactory dysfunction and implications for digital rehabilitation

Moritz Engelhardt, Eva Morawa, Yesim Erim, Jessica Freiherr

→ Abstract page 195

S5-P22: Integration of main and accessory olfactory pathways in the rodent BAOT and avMeA

Moritz M Nessler, Leonie Büsching, Maja van den Broek, Marc Spehr

→ Abstract page 196

S5-P23: Survival of the senses: Trp73 regulation in olfactory neuron development

Noah LeFever, Nikki Dolphin, Paolo Forni

→ Abstract page 197

S5-P24: How didn't you notice it: are human social chemosignals unconscious or unattended?

Roxanne Mangampo, Benjamin Young

→ Abstract page 198

S5-P25: An in-house directory of vomeronasal type-1 receptor structures in domestic animals: a computational approach

Samia Laib, Rajesh Durairaj, Aurore Lamy, Alessandro Cozzi, Patrick Pageat

→ Abstract page 199

S5-P26: Olfactory Function in Tracheostomy Patients

Sarah Steigleder

→ Abstract page 200

S5-P27: Early olfactory dysfunction and alteration in neural lipids in a Niemann-Pick type C1 mouse model

Simona Lobasso, Maria Grazia Fioriello, Donatella Lobraico, Raffaella Pia Gatta, Valeria Leone Macina, Johannes Reisert, Antonio Frigeri, Michele Dibattista

→ Abstract page 201

S5-P28: Bitter taste receptor expression and evolution in the eucalypt-eating koalas

Takashi Hayakawa, Kotaro Kondo, Elspeth McLennan, Carolyn Hogg, Katherine Belov

→ Abstract page 202

S5-P29: Development of an Olfactory Stimulation Device Focused on Contamination and Its Performance Evaluation Using PTR-TOF/MS

Tatsu Kobayakawa, Naomi Gotow, Hirotaka Kaneda

→ Abstract page 203

S5-P30: Unveiling the flavor network distribution and specialization for a population living with obesity using functional MRI Multivoxel Pattern Analysis.

Thibault Chabin, Guilbert Anaïs, Alain Lalande, Alexandre Comte, Marie-Claude Brindisi,

Frédéric Ricolfi, Thierry Thomas-Danguin, Jean-Marie Bonny, Charlotte Sinding

→ Abstract page 204

S5-P31: Olfactory receptor diversification across ontogeny in the hydrothermal vent shrimp *Rimicaris exoculata*

Thomas Chertemps, Yuna Boscher, Nicolas Montagné, Juliette Ravaux, Magali Zbinden

→ Abstract page 205

S5-P32: Odor vocabularies diverge on codability and abstraction but converge on valence and edibility

Thomas Hörberg, Hilario de Sousa, Murathan Kurfali, Malina Szychowska, William Fredborg, Nira Cedres, Rohan Raj, Mats E Nilsson, Jonas K Olofsson

→ Abstract page 206

S5-P33: Tasting order, not concentration, governs detection of the kokumi peptide γ -EVG in a simple umami matrix

Valentina Parma, Ulla Kidmose, Jonas Junge

→ Abstract page 207

S5-P34: The Lateral Habenula Integrates Sexual Olfactory Cues to Regulate Male Mouse Behavior

Vincent Hellier, Louise Sionneau, Chantal Porte, Pablo Chamero, Helene Vacher

→ Abstract page 208

S5-P35: Do the living environment and the genetic background influence the mouse VNO histological structure? Preliminary results.

Violaine Mechin, Sara Fuochi, Valentina Gazzano, Marine Verbaere, Simona Capsoni, Carlo Cantile, Patrick Pageat, Pietro Asproni

→ Abstract page 209

Session 1 — 16 September 2026, 12:30-14:00

S1-P1

Sustained valence-dependent modulation of sniff volume across sniffs

Andrea Aeimelaesus-Lindström¹, Frans Nordén¹, Giulia Ciotti¹, Irene Zanettin¹, Artin Arshamian¹, Johan N. Lundström^{1,2,3}

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Multiple studies have demonstrated that sniff parameters are modulated by an odor's perceived valence in that negative odors trigger a smaller sniff. The robustness of this valence-related modulation is underscored by its use as the basis for a clinical assessment of olfactory dysfunction, known as the Sniff magnitude test. To date, however, valence-dependent modulation has only been demonstrated for the initial sniff of an odor; whether similar effects extend to subsequent sniffs during continued odor exposure remains unknown. Participants (n=41) smelled 4 iso-intense odors (2 clearly pleasant, 2 clearly unpleasant) presented in jars, 3 times of the same ongoing odor, while their nasal airflow was assessed using a spirometer with nasal canula. Valence ratings were obtained for all sniffs, and individual ratings were used for subsequent analyses. Each odor was presented a total of 15 times. Skin conductance served as a distractor to mask the focus on respiration and ensure natural sniffing behavior. As expected, negative odors produced lower sniff volumes, when compared to positive odors, for the first sniff ($p < .001$). The clear difference in sniff volumes between valences was sustained for the two following sniffs (both $p < .001$). Interestingly, contrary to previous reports, this was not a universal effect. 46% of participants did not have a significant valence-dependent modulation of their sniff response. In conclusion, we can replicate and extend effects of valence-dependent sniff modulation also for subsequent sniffs, but the individual differences bring the usability of the previously proposed sniff magnitude test into question.

S1-P2

Odors count globally: the number of conscious odor perceptions differs between 17 locations on five continents

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¹ University of Wrocław

² TU Dresden

People around the world perceive odors in similar ways; however, it remains unclear whether they are equally aware of those present in the ambient air. Previous research on odor awareness has predominantly relied on self-report measures. To address this, we conducted a large-scale, global study to develop a comprehensive, multilevel model that would explain the variability in the frequency of conscious odor perception across geographic regions. Our model incorporates both individual and location-related factors. The study involved specialized chemosensory laboratories from 17 countries worldwide and final sample of 550 participants. For the first time at a global scale, we employed a behavioral measure of odor awareness based on odor counting performed 24 hours after the testing session. Our results revealed that location accounted for 15.2% of the variability in odor counts, while individual participant characteristics explained an additional 6.1%. Self-assessed odor awareness significantly predicted the number of consciously perceived odors; however, age, gender, and chemosensory sensitivity were not associated with odor counts. This study provides the first global scale exploration of variability in behavioral odor awareness.

S1-P3

How many odors do we need to smell? A clinical and perceptual perspective

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¹ *The Smell & Taste Lab*

² *Technische Universität Dresden*

³ *Aristotle University Thessaloniki*

⁴ *Aristotle University Thessaloniki*

The olfactory system encodes a vast chemical space: Models predict up to 40 billion odorous molecules from the GDB-17 database of 166 billion candidates. Despite this complexity the number of primary odor categories suggested between 20-30. Understanding which odors are perceptually and clinically relevant, particularly for patients with olfactory dysfunction, is critical for the development of future therapeutic tools including olfactory implants. A survey was conducted in 74 patients with olfactory dysfunction. In complement, 71 normosmic controls were surveyed. Using paper and pencil participants rated the importance of odors across 10 predefined cat. and identified the most and least important specific odorants within each category. Patients reported a mean of 15.8 odor cat, vs to 11.2 for controls. Patients with sudden olfactory dysfunction reported more odors compared to patients with gradual onset (16.4 vs. 15.5). Regarding odor cat prioritization per dysfunction onset type, patients with sudden-onset prioritized food (72%), environment (58%) and danger odors (48%). It was a similar case for gradual-onset patients. They granted 0% of importance to danger odors. Across patients and controls, certain odors per cat stand out. Patients and control groups generally agreed on the most and least important odor per odor group, with exceptions for the most important social odor (body odor). These findings suggest that a limited set of 20–200 odors may be sufficient to cover a satisfactory olfactory experience in daily life, consistent with the observation that each food engages only 3 to 40 key odorants from a palette of 400 functional OR. For clinical applications such as olfactory implants, prioritizing food, environmental, and safety-related odors appears aligned with patient needs. The gap between the theoretical richness of olfactory space and patients' requirements informs of targeted olfactory rehabilitation strategies. Funded with Thessaloniki and Dresden's univ in house grants.

S1-P4

Odor-Evoked Autobiographical Memories Are Not Always Highly Emotional

Berçem Yar¹, Aslı Aslan², Maria Veldhuizen¹

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² Anadolu University

Researchers have described the “Proust phenomenon” as autobiographical memories evoked by odors being older, more vivid, and more emotional than those triggered by other sensory modalities. Odors are potentially effective in eliciting emotional memories due to their direct connection with brain regions involved in memory and emotion. However, findings on the emotional characteristics of this phenomenon are mixed. This study compares the emotional characteristics of autobiographical memories evoked by different sensory modalities. The study was conducted with 108 participants (52 women; $M = 30.95$, $SD = 5.96$, range = 25–46) and used word, photograph, and odor cues in a between-subjects design to examine memory retrieval. In each of the three conditions, participants were presented with cues corresponding to 10 objects in a random order (odors: actual scented objects; photographs: color images of the objects; words: the names of the objects displayed on a computer screen). After the presentation of each cue, participants were asked whether any specific memory was evoked. If a memory was successfully retrieved, participants were asked to write about it in as much detail as possible and to rate the evoked memory based on its experiential characteristics. In the present study, the emotionality dimension was assessed using two separate ratings: emotionality at the time of the event and emotionality during recall. The results indicated that autobiographical memories evoked by odor, photograph, and word cues did not differ significantly in terms of emotionality. In conclusion, autobiographical memories evoked by odors are not necessarily experienced as more emotional. In light of the existing literature, it may be important to incorporate personally meaningful cues in order to elicit emotionally salient memories through odor cues. Funding Source: This project is funded by the TÜBİTAK 1002-A program.

S1-P5

Navigating the molecular space of taste-active peptides

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Understanding the molecular drivers of taste is essential for food science and nutrition. To map the molecular characteristics of taste-active peptides, we are developing the Taste Peptide Space (TPS)—a curated dataset designed to categorize peptides by their sensory profiles. This project evolves from the Bitter Peptide Space (BPS-1000), <https://bps1000.leibniz-lsb.tum.de/> (1). BPS-1000 consists of experimentally validated bitter and non-bitter peptides. It integrates sensory data, bitter taste thresholds (BTTs), and bitter taste receptor (TAS2R) activity when available. Sequence-based and structure-based analyses reveal the influence of hydrophobicity, molecular size, and specific amino acid composition in distinguishing bitter from non-bitter peptides. BPS is now being expanded into TPS to include additional peptide classes, such as umami and sweet peptides, thereby broadening the scope and deepening the understanding of peptide properties across different taste modalities. Funding: This work was funded by the Deutsche Forschungsgemeinschaft (DFG).

References:

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S1-P6

Unraveling the molecular basis of the Orco-ORx assembly

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Olfaction is one of the oldest sensory systems insects use to interpret their environment. To detect airborne chemicals, insects rely on several types of receptors, including olfactory receptors (ORs). In winged insects, the OR signaling complex is a heterotetramer composed of two subunits: the highly conserved co-receptor Orco and a more rapidly evolving tuning OR. While Orco shows strong conservation across insect species, tuning ORs can diverge by up to 80% within a single species' repertoire. Despite this variation, Orco can still assemble functional complexes with many different ORs. Recent structural studies have revealed the overall architecture of Orco and ORs, identifying both the interacting domains between subunits and the ligand-binding site. However, it remains unclear how Orco accommodates such diverse OR partners to form active complexes. Our research seeks to address this question by integrating phylogenetic analyses of Orco across insect lineages, computational modeling of Orco-OR complexes, and electrophysiological recordings. By comparing the interactions between Orco and a range of divergent, functionally distinct ORs, we aim to uncover mechanisms that enable complex formation despite high sequence variability. This work will advance our understanding of the molecular principles that govern assembly of the Orco-ORx signaling complex.

S1-P7

Large-scale genetic studies crack the olfactory code

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Translating olfactory receptor (OR) activation into human odor perception remains poorly understood, due in part to limitations in the functional in vitro expression of ORs and a lack of evidence directly linking OR response to sensory outcome. Natural variation in the OR gene family offers a powerful approach to this problem, as loss-of-function variants can reveal which perceptual features are affected and to what degree. Over 500 human subjects rated the intensity and pleasantness of 50 odorants across concentrations. Genetic variation at specific OR loci was significantly associated with the perception of 25 odors (50% of those tested) ($p < 0.05$, with FDR correction), suggesting that the combinatorial code is not highly redundant for many odorants. To identify causal ORs, we tested all major variants for their response to the associated odor in vitro. In 50% of cases (12 odors), reduced receptor function correlated with reduced perceived intensity, which was accompanied by shifts in pleasantness for 8 odors. Our results demonstrate that, despite the combinatorial nature of the olfactory code, variations in a single receptor substantially alter odor perception. These results move toward a set of generalizable rules relating OR genotype to receptor function and sensory outcome and enable our efforts to modulate OR activation for targeted fragrance solutions, such as malodor control.

S1-P8

Episodic Olfactory Autobiographical Memory and expertise across the world

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¹ TAKASAGO TEPL Emotion Lab

The impact of expertise in olfaction on odor memorization and description has been extensively studied over the years. Previous works provided evidence of cortical reorganization with expertise in brain areas involved in odor processing, and consistently showed that expertise enhances identification skills, working odor memory, and long-term odor memorization. However, the link between olfactory expertise and odor associated autobiographical memory remains relatively underexplored. Although it is generally accepted that personal experience influences odor-evoked autobiographical memory, there is no consensus about why some odor trigger such memories while others don't. Furthermore, there is no clear evidence of how olfactory expertise might influence this process. That said, it is well established that expertise affects both verbalization skills and mental imagery. In addition, cultural factors are also known to play a role in odor verbalization. Taken together, and considering that verbalization skills affect odor processing as well as autobiographical memory and odor naming though verbal overshadowing, it is reasonable to hypothesize that both expertise and culture may modulate the way odors trigger autobiographical memories. To address this hypothesis, the present study relies on expertise from professionals in the perfume industry to investigate olfactory childhood experiences. Results presented here were obtained from across nine countries (United States, Germany, France, Singapore, India, China, Indonesia, Brazil, and Japan), involving 750 experts (at different levels). To our knowledge, this represents one of the largest expert-based investigations of childhood-associated odors across cultures.

S1-P9

The development of olfactory-mediated disgust in preschool children

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Disgust is often understood as a mechanism that helps humans avoid pathogens; however, its development over time remains unclear. Although smells are strong triggers of affective responses, most research has focused on visual stimuli in adults, leaving early childhood—especially the preschool years—relatively understudied. This study examined changes in olfactory-mediated disgust sensitivity over two years by testing the same group of children at ages four and five. A total of 345 children (171 girls and 174 boys) participated in the study, evaluating 11 odour samples in 2024 and again in 2025 within public kindergartens. The set included five pleasant and five unpleasant/disgusting odours, categorized based on prior adult ratings, along with one odourless control. Children rated each sample's pleasantness using a five-point scale with emoticons. The data were first analyzed at the level of individual odours and then grouped into two categories using cluster analysis, which aligned with the adult-based classifications. In pleasant odours, we found that five-year-old children reported statistically significantly higher pleasantness than at age four ($F = 39.44$, $p < 0.001$, $\beta = 0.30$). Conversely, unpleasant odours were rated as less pleasant by five-year-olds compared to four-year-olds ($F = 85.38$, $p < 0.001$, $\beta = -0.43$). Finally, boys consistently rated unpleasant odours as less pleasant than girls across both years ($F = 6.69$, $p = 0.01$, $\beta = 0.19$). Overall, these findings highlight the development of olfactory-mediated disgust during the preschool years and show that both age and sex are important factors shaping early disgust responses. This study was supported by a grant from the Czech Science Foundation (GAČR 24-10955S).

S1-P10

Olfactory experiences during psychedelic states: a natural language processing study of chemosensory alterations in online communities

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Psychedelics are known to alter states of consciousness and perception. However, scientific literature and first-person reports have focused on visual, auditory, and, to a lesser extent, tactile phenomena, while olfactory experiences remain poorly characterized. Studying psychedelic-related olfactory alterations may therefore provide insight both into the biology of psychedelic states and into mechanisms underlying human olfaction. Here, we systematically examined olfactory experiences during psychedelic states using a large population-scale dataset of naturalistic, anonymous self-reports. We focused on five classical serotonergic psychedelics: psilocybin, LSD, DMT, 5-MeO-DMT, and mescaline. Reports containing olfactory content were identified and classified through a three-step pipeline combining natural language processing, Large Language Model classification, and final human verdict. Olfactory-related reports constituted less than 1% of the overall psychedelic discourse. These reports were classified into descriptions anchored in external reality and phenomenological categories: olfactory hallucinations, synesthetic smell experiences, and distortions of ordinary perception. Most olfactory references described physical odors rather than drug-induced perceptual changes, suggesting that genuinely phenomenological olfactory experiences are uncommon. Within the phenomenological subset, hallucinations were the dominant category across substances, while synesthesia and distortion were less frequent. Substance-level comparisons revealed a consistent gradient, with DMT-associated reports showing the highest relative phenomenological rate, followed by psilocybin, 5-MeO-DMT and substantially lower rates for LSD and mescaline. Analyses of thematic and emotional qualities of these experiences are currently underway. In sum, olfactory phenomenology is a rare but identifiable component of psychedelic states, with prevalence and qualitative features that vary across substances.

S1-P11

Nutrient-Specific Tuning of Sensorimotor Circuits in *Drosophila melanogaster***Daniel Münch¹, Rory Beresford², Marcelo Dias², Arina Dygay², Carlos Ribeiro²**¹ *Institute of Animal Physiology, Justus Liebig University, Giessen, Germany*² *Behavior and Metabolism Laboratory, Champalimaud Research, Lisbon, Portugal*

Animals adapt their behavior to meet changing internal needs, such as the requirement for specific nutrients to maintain homeostasis. While it is known that internal states like nutrient deprivation or reproductive status drive appetite for foods that are rich in particular nutrients, the neural mechanisms by which these needs shape behavior remain incompletely understood. Building on our previous work that established a functional atlas of the subesophageal zone (SEZ) and identified SEZ regions involved in protein appetite, we now focus on how carbohydrate deprivation alters sensory processing. Using volumetric 2-photon imaging, behavior, and genetic manipulation, we show that carbohydrate deprivation engages distinct SEZ regions, separate from those modulated by protein hunger. Furthermore, we identify candidate driver lines and neurons that innervate these carbohydrate sensitive regions and contribute to carbohydrate specific appetite. These findings indicate that nutrient specific deficits activate separable neural pathways to flexibly adjust sensorimotor processing based on internal need, highlighting how the brain dynamically adapts sensory processing to guide nutrient specific food choice and maintain physiological balance.

S1-P12

Uncovering behavioral signatures of olfactory preference and sex in mice through Keypoint-MoSeq

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The odor perception represents a sophisticated interplay between innate biological preferences, environmental context, and olfactory learning. Despite the importance of olfaction, the precise behavioral dynamics that transform an odorant stimulus into a structured motor response remain poorly characterized. In this study, we bridged this gap by decoding the signatures of olfactory preference, with also a focus on identifying how sex serves as a biological variable in shaping the olfactory behavior. By breaking down complex animal behavior into discrete, neuronal-driven sequences of posture transitions, we uncovered the underlying rules governing how mice approach or avoid different olfactory stimuli. Specifically, we analyzed movement patterns of mice exposed to various odors, utilizing Keypoint-MoSeq to track fine-scale motor behavior and decompose actions into distinct, reoccurring motifs known as "syllables." Our analysis demonstrates that olfactory preference is not a stochastic process; rather, it follows highly structured patterns rooted in sensorimotor behavior. We observed that mice exhibit odor-specific behavioral signatures, allowing us to categorize odors into distinct groups of attractive, neutral, or aversive. While the odor category dictates the general direction of behavior, such as the tendency to approach or avoid, a more nuanced layer of complexity emerges in the motor execution. Crucially, our investigation reveals a unique divergence: while the general behavior is largely determined by the identity and hedonic value of the odorant, the transitional organization of the syllable movements is significantly modulated by the sex of the animal. Consequently, this modular organization provides a robust framework for prediction. By analyzing the high-resolution movement data, we could successfully predict the sex of the subject based solely on their behavioral patterns, offering deeper insight into the sex-specific nature of olfactory behavior.

S1-P13

Aquaporin-4 water channel controls neuronal activity in the olfactory bulb through astrocytes remodeling and affects olfactory behavior

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Astrocytes are essential for supporting neuronal function, primarily by modulating ion and water movement through Aquaporin-4 (AQP4) water channel. We investigated the roles of various AQP4 isoforms; the extended forms (AQP4M23ex, AQP4M1ex) and the M23 isoform that forms orthogonal arrays of particles (OAPs), within the olfactory bulb (OB). Although these isoforms are known to shape astrocyte morphology and modulate neuronal activity, their precise contribution to OB function has remained elusive. To clarify these roles, we analyzed the structural properties of astrocytes, and we found that in OAP-null mice, they display reduced branch length and diminished dimensions, highlighting the importance of M23 for structural integrity. Further investigation into neuronal activity within the glomerular layer (GL) was conducted by measuring the c-Fos expression after exposure to amyl acetate. Findings revealed that while OAP-null mice exhibit lower basal neuronal activity, they show an increase in activated cells under stimulation, suggesting that M23 helps calibrate baseline neuronal excitation in the GL, despite the presence of compensatory mechanisms that attempt to recover odorant-evoked responses. Behavioral testing was also performed to link these cellular changes to olfactory function. We found that OAP-null mice are unable to differentiate between two distinct odors. In parallel, AQP4ex-KO mice fail to display a clear olfactory preference observed in wild-type mice. Collectively, these findings demonstrate that specific AQP4 isoforms are critical for maintaining neuronal homeostasis in the OB. By establishing that astrocytes actively control neuronal activity through the AQP4 isoforms, these results provide a useful framework for understanding the glial-neuronal interaction within the OB. Furthermore, our behavioral findings emphasize the importance of deciphering the specific impact of each AQP4 isoform on both olfactory preference and associative learning.

S1-P14

Combinability of odour data for machine learning benchmarksFlorian Anders¹, Tim Jesgarzewsky², Alexander Croy¹¹ Institute of Physical Chemistry, Friedrich Schiller University Jena, Germany² Institute of Psychology, Friedrich Schiller University Jena, Germany

Machine learning methods investigating the structure-odour-relationship often use a curated combination of expert-rated data sets without describing the unification process in detail, despite subjectivity related disagreements even in expert odour label assignments to molecules. In this study, we provide a benchmark, data noise reduction and alignment techniques with dummy classifiers, random forest, and different neural network models in order to investigate the combinability of expert-rated data sets. To this end, a simple label filtering has been employed, while keeping most of the initial data points, in addition to semantic alignment of the perceptual space using a large-language model and subsequent, hierarchical clustering for the final unification of the perceptual space. In view of the subjective nature of olfaction data, we discuss the concept of the structure-odour-relationship, viewing data sets as spanning problem domains and further discuss the applicability of label filtering and clustering to industry related interests. This work is funded by the Volkswagen foundation within project 'Olfactorial Perceptrics' (No. 9B396).

S1-P15

Reproductive state-dependent modulation of odor valence and social behavior via the main olfactory pathway

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The internal state of female organisms is significantly modulated by fluctuating hormones associated with the reproductive cycle. These endocrine changes have a wide impact, not only on internal bodily functions but also on the perception of environmental stimuli as well as on behavioral choices. The olfactory system is an ideal model to study state-dependent modulation because it drives essential behaviors such as reproduction or food acquisition and receives extensive neuromodulatory input. However, how hormonal status shapes processing in the main olfactory system remains poorly understood. Here, we show that the estrous cycle significantly modulates the behavioral valence responses of female mice towards the male-derived odor trimethylamine (TMA). Females in the non-receptive diestrus phase avoid this cue, while estrus females behave neutral - revealing a naturally occurring, neuroendocrine state-dependent valence switch. Moreover, this behavioral switch depends on a specific main olfactory receptor (trace amine-associated receptor 5, TAAR5), which is specifically activated by TMA, as valence responses in females lacking TAAR5 are abolished. In addition, male-female social interaction experiments provide preliminary evidence that estrous females lacking TAAR5 exhibit increased defensive responses towards males compared to estrous wild-type females, potentially indicating an altered interpretation of the male counterpart. To dissect the neuronal mechanisms of this estrous-dependent modulation of odor valence and social behavior, we perform in vivo two-photon calcium imaging of defined olfactory bulb neurons. Odor stimulation with TMA reveals preliminary evidence for functionally distinct neuronal subtypes, characterized by heterogeneous activity profiles, ranging from stimulus-evoked excitation to suppression. Together, these findings suggest that the estrous state modulates odor valence by shaping olfactory processing to drive social behavior in females.

S1-P16

Characterization of apical-out polarity in murine and human olfactory epithelial organoids

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Objectives: Olfactory organoids are 3D spheroids derived from olfactory epithelium (OE) stem cells and provide advanced models for sensory neurobiology and host-pathogen studies. In this study, we characterized olfactory organoids with controlled apical-out, mimicking the native polarity of the OE to evaluate their functional roles and translational potential. **Methods:** Organoids were generated from murine olfactory epithelial cells, with apical-out and basal-out polarity. Cells were expanded in matrigel and differentiated in suspension culture, followed by immunostaining, qPCR analysis and SARS-CoV-2 infection. Human olfactory organoids were also established from minimally invasive brushing-derived cells and characterized by immunofluorescence and qPCR analysis. **Results:** Immunostaining revealed that the K⁵⁺ basal cell marker localized to the inner region, whereas OMP⁺ mOSNs were found in the outer region of apical-out organoids. Following SARS-CoV-2 infection, co-localization of ACE2 and viral NP was observed. Human olfactory organoids were established from brushing-derived samples and maintained in culture. Immunofluorescence staining confirmed the presence of multiple olfactory epithelial cell types in the human organoids. **Conclusions:** In summary, we identified diverse cell types and differentiated mature olfactory sensory neurons within apical-out organoids, demonstrating structural similarity to mouse olfactory tissue. Our findings further indicate that brushing-derived human olfactory organoids can be established and maintained as a complementary translational model. We propose that apical-out olfactory organoids offer significant potential as a platform for olfactory research, highlighting important translational perspectives. **Funding:** This work was supported by the Bio and Medical Technology Development Program of the National Research Foundation (NRF) and funded by the Korean government (MSIT) RS-2024-00443043, RS-2024-00400118, and RS-2024-00344509.

S1-P17

AFM uncovers the role of mechanoreception in sweet taste: linking nano-mechanics, cellular signaling, and sensory perception.

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Flavor perception is widely recognized as a multisensory experience shaped by our five senses. However, the importance of mouthfeel and chemesthesis as additional key pillars is often overlooked. Oral somatosensory cues transmitted via mechanoreceptors and trigeminal pathways not only shape flavor perception but also influence satiety and consumer preferences. In the context of sweet taste, sugar provides pleasurable indulgence in foods and drinks. Yet, although sugar substitutes offer comparable sweetening power, they are often rejected due to an unpleasant mouthfeel, leading consumers to revert to conventional sugar despite its adverse physiological effects. To enable the rational design of healthier, yet sensory-appealing products, elucidating the molecular mechanisms underlying sweet mouthfeel is essential. High-throughput bio-atomic force microscopy (AFM) combined with molecular/ functional cell assays and human sensory trials provides an innovative interdisciplinary approach to uncover oral mechanotransduction underlying sweet taste at the single molecule/cell level. First results demonstrate that in vivo sensory outcomes can be successfully linked to the mechanoreponse (e.g., stiffness) of native oral cells in vitro. Nanomechanical probing, alone or combined with functional Ca^{2+} fluorescence imaging, enables clear discrimination between cell responses driven by physical parameters (viscosity, osmolarity) and those arising from the chemical structure of the sugar(-like) molecule. Notably, sucrose induces a distinct mechanical cell response that is absent for artificial sweeteners. Similarly, sensory panel data reveal significant differences in mouthfeel between sucrose and artificial sweeteners, which can be partially reduced by adjusting the physical properties of the sweetener solution to resemble those of sucrose. This supports the hypothesis that mechanoreception is an important part of the overall sweet perception.

S1-P18

Effects of a fragrance intervention on nocturnal arousal: a large-scale randomized controlled trial

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Objectives: Large-scale sleep studies in general populations are difficult to conduct due to challenges in recruitment and standardization of monitoring conditions. We aimed to evaluate the effect of a fragrance intervention on sleep quality in such a large-scale setting by leveraging the standardized infrastructure of sleeping pod in a capsule hotel (nine hours, Inc), focusing on nocturnal arousal. **Methods:** A randomized controlled trial (RCT) was conducted, with written informed consent obtained from all participants. A total of 2,321 participants were enrolled and randomized to either a single-night exposure of a citrus-floral fragrance accord (Li et al., 2025) developed through EEG-based evaluation to promote sleep or an odorless solvent control, diffused within the sleeping pod throughout the night. Their sleep conditions were monitored via in-pod multi-modal monitoring (mat-type sheet sensors, cameras, and microphones) with a proprietary algorithm yielding objective sleep metrics. Linear mixed models (LMM) incorporating sample $\times$ nationality and sample $\times$ gender interactions were fitted for arousal ratio as the primary outcome and heart rate as secondary, given its sensitivity to olfactory stimulation. **Results:** Descriptive statistics showed lower arousal ratio in the fragrance group among male participants. After adjusting for covariates, LMM revealed a significant overall treatment effect on arousal ratio ($\Delta = -1.00\%$, $p = 0.040$). As a sensitivity analysis, nocturnal wake time yielded a consistent result ($\Delta = -5.0$ min, $p = 0.044$). Heart rate was also reduced overall ($\Delta = -0.4$ bpm, $p = 0.049$), with consistent effects across subgroups. **Conclusions:** This large-scale field RCT demonstrated that a fragrance intervention reduces nocturnal arousal and associated cardiac activity, suggesting that olfactory stimulation throughout the night offers a non-pharmacological approach to improving sleep quality in the general population. **Funding:** Takasago International Corporation.

S1-P19

The Subjective of Space. How Atmosphere Shapes Olfactory Imagination.

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Olfactory perception is increasingly understood as an embodied and situated process shaped by sensory input, bodily engagement, context, memory, and expectation. From this perspective, olfactory imagery—the experience of smells without physical olfactory input—appears closely intertwined with perception. Yet research has largely relied on questionnaires and self-reports, offering limited insight into how such experiences can be evoked through embodied processes. This contribution examines how the relationship between olfactory perception and imagery varies across contexts. In an ongoing within-subject study, we employ Glass Tastings, an art-based performative method in which odorless glass objects act as anchors for imagined smells, activated through language, multisensory cues, and spatial context. Each participant took part in two environments—an art space and a laboratory—to compare contextual effects. Preliminary analyses (N=26) show no significant differences in imagined smell vividness between contexts, despite differences in situational framing. No significant association was found between olfactory perception and imagined smell vividness (art space: $r=.336$, $p=.087$; lab: $r=.245$, $p=.218$). However, a context-dependent pattern emerges for familiarity: olfactory perception (Sniffin' Sticks 16-item identification test) was positively associated with familiarity of imagined smells in the art space ($r=.506$, $p=.005$), but not in the laboratory ($r=.213$, $p=.276$). These findings suggest that contextual framing does not primarily affect imagery vividness, but modulates how specific dimensions of olfactory imagery are grounded in perceptual ability. This points toward the need to further investigate olfactory imagery as a context-sensitive embodied process, for which art-based methods may provide a promising approach.

S1-P20

The impact of familiar and stranger body odors on social anxiety: an eye-tracking and ERP-EEG study protocol

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Individuals with social anxiety disorder often interpret neutral social stimuli as threatening. Beyond well-studied visual cues, body odors (BOs) also convey socially relevant information, such as the sender's identity (familiar vs. stranger). Notably, familiar BOs (FBOs) are typically perceived as calming, whereas stranger BOs (SBOs) may elicit threat-related responses. However, it remains unclear how individuals with elevated social anxiety process these olfactory social cues. This pre-registered experiment, currently underway, addresses this gap using a multisensory paradigm combining olfactory stimulation with a face processing task, to investigate how FBOs and SBOs influence subjective and psychophysiological responses as a function of social anxiety severity. Thirty-five normosmic, right-handed women are recruited. Sample size was determined via a priori power analysis ($f = 0.30$, 90% power, $\alpha = 0.05$). Participants complete a within-subject design with three odor conditions: clean air (CA), a friend's body odor (FBO, donated by a close female friend), and a stranger's body odor (SBO). During the task, participants view 27 neutral faces paired with each odor condition across three blocks. Behavioral (valence, arousal, approachability ratings), eye-tracking (gaze behavior, pupillometry), and EEG (P1/N1, N170, LPP) measures are collected. Social anxiety severity is operationalized as continuous scores on the Liebowitz Social Anxiety Scale (LSAS) and used as the main predictor in linear mixed-effects models, with odor condition as a within-subject variable and an Odor $\times$ LSAS interaction as the primary effect of interest. Preliminary behavioral and psychophysiological results will be presented. The present study is expected to advance our understanding of multisensory processing across varying levels of social anxiety and to highlight the role of olfactory cues in shaping social perception. Funded by the University of Padova.

S1-P21

Role of experience in the construction of flavor and the emergence of aroma-induced taste enhancement

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Flavor is a multisensory construct that emerges when the brain combines taste, retronasal olfaction, and oral-somatosensory inputs into a unified representation of what we are eating or drinking. Understanding how the brain transforms these multisensory food signals into coherent flavor experiences is essential for developing strategies to improve eating behavior and public health. The present study aims to investigate how odors acquire taste qualities through learning and come to enhance taste perception. We argue that odor-induced taste enhancement, whereby an odor paired with sucrose comes to increase perceived sweetness, provides a particularly informative behavioral marker of flavor integration. Most existing studies rely on familiar odor-taste pairings that primarily index pre-existing associations and offer limited insight into the underlying mechanisms. To address this gap, we will use controlled conditioning paradigms in humans, in which unfamiliar, taste-neutral odors are repeatedly paired with sucrose under well-defined conditions, allowing us to track the emergence and temporal dynamics of newly learned odor-taste associations. We will combine these learning manipulations with electroencephalography (EEG) to monitor trial-by-trial changes in neural activity. We expect odors to rapidly acquire sweetness when presented in isolation and to further enhance perceived sweetness when participants adopt a synthetic attentional strategy toward flavor. At the neural level, we anticipate learning-related changes in EEG markers of integration, including modulation of the P300 component, as well as changes in oscillatory activity and phase synchrony between chemosensory regions. These signatures should reveal when and where odors shift from independent cues to components of integrated flavor representations within distributed cortical networks. This project is supported by ANR AROMA funding (n°20-CE21-004).

S1-P22

Differential neural processing of sweet and sour food odors: evidence from psychophysics, fMRI activation, and connectivity analysis

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The association of food-related odors with sweet and sour taste qualities may engage distinct sensory, hedonic, and reward-related brain mechanisms. To investigate whether taste-category associations influence odor perception and its neural representation, we examined psychophysical responses and whole-brain activation in a cohort of 40 healthy participants (50% females) during perception of food odors described as sweet (caramel, marshmallow) and sour (grapefruit, quinine). Participants completed evaluations of odors in a laboratory session and again during fMRI scanning. Linear mixed-effects models revealed a significant main effect of session for intensity, with lower ratings in the scanner, particularly for sour odors. Sweet odors were rated as more pleasant, edible, and drinkable than sour odors ($p \leq 0.017$), while intensity and familiarity remained comparable across odor categories. The contrast sweet > sour odors showed significant activation in the right orbitofrontal cortex, right insula, and right frontal pole ($T \geq 3.30$), regions implicated in sensory integration and reward evaluation. In a separate cohort ($n = 47$, all female), task-based ROI-to-ROI functional connectivity analysis was performed during the same odor paradigm, including non-food reference odors (linen and lavender). Caramel elicited stronger right caudate–right hippocampus connectivity compared with linen, suggesting enhanced mnemonic-reward integration, whereas grapefruit showed weaker right central operculum–left amygdala connectivity ($p \leq 0.045$). Together, these findings indicate that sweet food odors which showed higher ratings in the scanner than sour food odors, are primarily differentiated at the level of regional activation and perceptual evaluation, while functional connectivity is driven more strongly by individual odor identity than by broad taste-category associations.

S1-P23

Neurophysiological correlates of attention and creativity in candle-induced olfactory environments

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Olfactory environments are increasingly used to support well-being and cognitive performance in everyday settings, yet their neurophysiological influence on sustained attention and creativity remains poorly documented. This exploratory study examined the underlying neurophysiological activity of how candle-based olfactory environments may affect creative and attentional performance. Thirteen healthy adults performed a Sustained Attention to Response Task (SART) and an Alternative Uses Task (AUT) under six conditions (three scents: floral, green, neutral; two lighting conditions: lit, unlit). Candles were presented within an olfactory dome to create a fully immersive sensory environment. Odor perception was assessed via Likert scales and the Geneva Emotion and Odor Scale (GEOS), and brain activity was recorded using combined EEG and fNIRS. The floral candle was rated as more pleasant than the neutral one ($\chi^2(2) = 9.114$, $p = .023$) and specifically enhanced creative elaboration compared to green and neutral scents ($\chi^2(2) = 8.048$, $p = .018$, $W = .268$), with no significant effects on sustained attention. At the neurophysiological level, fNIRS revealed greater frontal hemodynamic activity in the floral condition during the final minutes of the cognitive task ($\chi^2(2) = 11.743$, $p = .038$, $W = .391$), and EEG showed significant odor-related modulation of frontal alpha oscillations ($\chi^2(2) = 7.6$, $p = .022$, $W = .64$), with both effects more pronounced when the candle was unlit. These findings show that olfactory environments can shape creativity. The convergence of frontal hemodynamic and alpha oscillation modulations across fNIRS and EEG suggests that different scents engage distinct patterns of frontal resource allocation. The fact that the floral scent was both perceived as more pleasant and associated with enhanced creative elaboration raises intriguing questions about the relationship between odor pleasantness and creativity, which future large-scale studies should

S1-P24

Interaction of sweet tasting compounds with saliva proteins is related to their sensory properties

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While the molecular mechanisms of sweet taste perception are increasingly well understood, the origin of frequently reported unpleasant side tastes of sweeteners remains unclear. One possible explanation is an interaction of sweet-tasting compounds with the oral pellicle, but experimental data to support this hypothesis are lacking. Thus, we investigated the interaction of 15 sweet-tasting compounds with a novel oral pellicle model based on Caco-2 cells, which endogenously express mucin 1. Confluent Caco-2 monolayers were incubated with unstimulated human whole saliva to allow the formation of an in vitro pellicle composed of salivary mucin 5B and the extracellular domain of cell-bound mucin 1. Subsequently, cells were exposed to equisweet concentrations of the test compounds, as determined by prior sensory evaluation. Changes in the abundance of mucin 1 and mucin 5B were quantified by ELISA, and mucin 5B immunostaining was used to assess potential aggregation phenomena. The abundance of mucin 1 and mucin 5B, and the distribution of mucin 5B, were altered to different extents after contact with the tested sweeteners. Some sweeteners also led to an increased aggregation of mucin 5B. A heat map analysis combined with hierarchical clustering revealed distinct similarity patterns among the tested sweeteners. These patterns showed notable correspondence with known sensory descriptors, as test compounds that were rated as sensorially similar frequently clustered together in our analysis, with the closest agreement in sensory ratings observed for astringent taste or aftertaste and metallic notes. Our findings support the concept that sweeteners can differentially modulate pellicle organization, providing a mechanistic basis for their divergent side-taste profiles and contributing to a more comprehensive understanding of oral surface interactions in taste perception. This work was financially supported by the Christian Doppler Research Association and Symrise AG, Germany.

S1-P25

Using SCED to test the effects of olfactory awakening in young children

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In Western societies olfaction is neglected, despite its role in overall development, and particularly in eating, breathing and avoiding poisoning. To date, we lack research examining the impact of olfactory awakening on young children. An intervention carried out while sharing focus attention and the intention to smell together with young children, should help them develop olfactory skills, such as smelling an odorant object rather than mouthing it directly. This is the aim of the present study, that take advantages of Single Case Experimental Designs (SCED). Nine children (4 boys), randomly selected (from 3 different nursery groups in Normandy, France), and aged between 8 and 25 months were followed-up after obtaining ethical and parental agreements. They were videotaped weekly over the course of 3 phases. Phase A, the baseline period, during which unfamiliar objects scented with natural products (floral waters, or spices), were presented to the child without any olfactory awakening, during either 7, 9, or 11 weeks. Phase B, the olfactory awakening intervention, during which an olfactory awakening was realized with natural scented objects (food), during 5 weeks. Then, a test session, similar from phase A, tested the spontaneous use of the skill. Two independent observers used The Observer® XT17 (Noldus) to assess the duration of smelling or mouthing. The observations were reliable ($k > 0.8$). Visual analyses represent the curves of the behavioral responses recorded in phases A and B and during the test. Statistical analyses focus on non-overlap indices (Tau U) and size effects and proved to be significant for smelling in six children ($p \leq 0.05$), but not for mouthing. An infant missed too many sessions in Phase B, and another acquired the skill, but used it mainly in the final test. This study shows that olfactory awakening learns young children to distinguish smelling with from mouthing. It could enable them to avoid poisoning, or even to eat healthily.

S1-P26

Water Matters: Boosting Aroma to Improve Appetite in Senior Dogs

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The global dog population is increasingly aging, driven by advances in veterinary care and nutrition. Aging is associated with mobility limitations, dental issues, and sensory decline, particularly in olfaction and taste, which can negatively impact feeding behavior and reduce appetite. Supporting adequate food intake in senior dogs therefore requires practical and effective solutions. This study investigates a simple, owner-applicable strategy: adding water to dry dog food to enhance aroma release and improve olfactory perception. The hypothesis is that hydration promotes the release of volatile compounds from dry matrices, enhancing the aroma and potentially increasing food attractiveness, stimulating intake. Optimization of water addition parameters (volume, temperature, contact time) was firstly performed using solid-phase microextraction coupled with gas chromatography-mass spectrometry (SPME/GC-MS) to characterize volatiles composition. In parallel, palatability tests were conducted to assess feeding performance and correlate analytical data with animal response. For a deeper understanding of the occurred phenomena, the optimized condition was characterized using proton transfer reaction time-of-flight mass spectrometry (PTR-TOF-MS) for real-time monitoring of aroma release. Results show that water addition significantly enhances the release of volatile compounds, increasing their availability in the headspace and improving olfactory perception. This effect is positively correlated with increased palatability and food intake. Optimal hydration conditions were identified, highlighting the importance of preparation parameters. This study demonstrates that adding water to kibble is simple for pet owners, low-cost, and effective to maintain the food intake in senior dogs, contributing to improved nutrition and overall well-being. The authors are employees of Royal Canin SAS, Mars, Inc., which funded and all the work presented.

S1-P27

Pleasant olfactory stimuli can reduce cigarette craving: Implications for smoking cessation

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Cigarette craving plays a central role in smoking, which remains a leading preventable cause of death worldwide. During moments of temptation the appeal of smoking rises, coping and “quit-smoking” messages may be ignored, and the habit persists. Unfortunately, research has struggled to develop nonpharmacological treatments for craving relief. One promising approach is the strategic use of odors to reduce craving. This presentation describes a research program examining the possibility that exposure to odors can reduce the urge to smoke and potentially aid in smoking cessation as part of a comprehensive intervention. We describe 2 experiments in which nicotine-deprived smokers evaluated the pleasantness of a series of odors. To obtain baseline hedonic responses, they first sampled a menu of scents and evaluated them for pleasantness and intensity. For the manipulation phase, after an initial urge rating, participants were exposed to a robust smoking cue, which reliably increased urge to smoke. Next, they smelled their (a) most pleasant odor OP from the baseline assessment, (b) an unscented control, OC, (c) their most unpleasant odor OU (study 1), or a tobacco-related odor OT (study 2) and again rated their urge to smoke. Results revealed that exposure to a participant’s OP significantly reduced urge to smoke, relative to the OC and OT (F ’s > 4.4, p ’s < .05) which interestingly was similarly effective as their OU (F < 1, ns). We next discuss ongoing studies aimed to (a) understand the neurobiological mechanisms underlying the putative urge-reducing effects of odors and (b) integrate odor delivery into an established quit-smoking mobile app. Our results support the viability of olfactory stimuli as an approach to urge reduction and highlight the promise of conducting (a) interdisciplinary neurobehavioral research and (b) clinical interventions designed to evaluate the impact of odors on smoking cessation. Research supported by NIH: R01CA184779; R34CA286803; R01AT010896

S1-P28

Neurophysiological driving mechanisms in olfactory perception

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The olfactory modality is unique in that it involves respiration to perceive odorants. Respiratory changes affect heart rate and may also be linked to brain-related changes. Our aim was to model these measure-dependent changes in a temporal manner, and to relate them to the valence and arousal of odors. To this end, we analyzed responses from 78 participants to pleasant, unpleasant, and neutral odors across a range of physiological and behavioral measures, including valence and arousal ratings, ECG, PPG, diaphragmatic and nasal respiration, EEG, and facial EMG. We applied temporal decoders to test how physiological signals distinguish between odor conditions over time and identified temporal clusters with peak classification performance. For these clusters, we tested potential driving mechanisms (e.g., respiration–heart rate coupling) and quantified inter-measure relationships using temporally resolved correlations. Results showed that temporal decoders were able to discriminate odor conditions at distinct time points for different measures, with lag analyses indicating temporal dependencies among physiological measures. Using these multi-measure approaches, olfactory perception can be shown in a more comprehensive manner. This could offer more robust differences on a single-subject level, avoiding the inter-individual variability in olfactory perception. The project was funded by the Else Kröner-Fresenius-Stiftung.

S1-P29

Rat olfactory bulb vasopressin system shows low connectivity within intraglomerular and interglomerular networkNicolas Reichardt¹, Veronica Egger¹¹ University of Regensburg

Vasopressin (VP) is a neuropeptide involved in various social behaviors, and its action in the olfactory bulb (OB), the first brain region responsible for olfactory processing, is essential for social discrimination in rats. The OB features an intrinsic vasopressin system, consisting of local vasopressin cells (VPCs). We previously found that these VPCs respond to electrical stimulation of the olfactory nerve in acute OB slices with inhibitory postsynaptic potentials. Moreover, GABAA blockade unmasks barrages of putative excitatory postsynaptic potentials. Application of the neuromodulator acetylcholine (ACh) switches evoked inhibition to action potential firing in the majority of VPCs. To unravel the presynaptic partners of the VPCs, we performed paired whole cell recordings with or without ACh present. Since VPCs innervate their home glomerulus with a dense apical tuft they might be contacted by the bursting glutamatergic external tufted cells (eTCs) and GABAergic periglomerular cells (PGCs) belonging to the same glomerulus. However, neither with nor without ACh we could so far find direct connections eTCs → VPCs (n=12) or PGCs → VPCs (n=10) within the home glomerulus. Since inputs might also terminate on VPC somata or lateral dendrites, we also tested for connections from adjacent glomeruli eTCs → VPCs, with the same result (n =13). This preliminary result is in line with another study describing a cluster of putative VPCs that seems to lack local connectivity (Tavakoli et al 2018). We hypothesize that the observed inhibitory inputs might originate from other glomerular interneuron types more distant to the VPCs' home glomerulus. Funding: DFG FOR5424, DFG GRK2174

S1-P30

The Silent Symptom: Olfactory Dysfunction at the Onset of Alzheimer's Disease

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Neurodegenerative diseases, such as Alzheimer's disease, are characterized by the progressive buildup of toxic proteins, including beta-amyloid and tau. The brain has a natural clearance system, the glymphatic system, which plays a key role in promoting solute removal from the extracellular space, supporting cerebral homeostasis. This pathway facilitates elimination of amyloid- β (A β) and hyperphosphorylated tau (pTau), whose accumulation may drive neurodegeneration. A key component ensuring proper glymphatic function is Aquaporin-4 (AQP4), a water channel expressed in astrocytes. Cerebrospinal fluid flows through the olfactory bulb and reaches the olfactory epithelium, suggesting that glymphatic activity extends to the olfactory pathway and contributes to waste clearance. Early smell impairment is one of the first signs of Alzheimer's disease, often appearing years before memory loss, suggesting that olfactory pathways may reflect early glymphatic dysfunction. The aim of this study is to explore the link between the olfactory system and Alzheimer's disease using the 5xFAD mouse model. Behavioral tests, including the cookie test, performed on 10 month old wild type and heterozygous mice, showed altered olfactory function in the transgenic group. These deficits appeared earlier than other behavioral changes, as the Y-maze test showed no significant differences at the same age, indicating that olfactory impairment may precede broader cognitive decline. To investigate glymphatic involvement, immunofluorescence analysis assessed Aquaporin-4 localization in olfactory bulb sections from 10 month old mice. Results showed altered AQP4 distribution in heterozygous animals compared to controls. Overall, these findings suggest that glymphatic function in olfactory areas is altered in Alzheimer's disease, offering new opportunities to better understand disease progression.

S1-P31

How could personalization of olfactory stimuli enhance cognitive performance? Theoretical and empirical evidence for an integrative approach.

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Objective. Sensory stimuli influence our cognition in many ways, and characteristics of the sensory stimuli, of the task, and of the individual interact with one another. However, these factors have always been studied in isolation. The effects of personalized or imposed music and smells on executive function, reaction times, and perceived performance in young and older adults were investigated. **Method.** Sixty young and older adults were distributed in personalized or imposed sensory environments to perform a cognitive task. Personalized environments used participant-preferred music and smells; imposed used predefined, non-preferred stimuli. The cognitive performances were tested in various sensory conditions: single-sensory (favorable or unfavorable to concentrate), multisensory (congruent/incongruent), and neutral (no diffusion). **Results.** Favorable perceptions of sensory environments had positive effects on perceived performance. Compared with unfavorable and incongruent environments, fatigue and sensory distraction are felt less acutely in the presence of stimuli perceived as favorable, whilst motivation, control, and concentration increased. However, only young adults show stronger effects from favorable environments and greater negative impact from unfavorable ones. Regarding objective performance, whilst perceived favorability and stimulus' nature appeared to influence reaction time, multisensory congruence and the personalization of the environment influence accuracy. **Conclusion.** Favorability, stimulus type, and personalization influence real and perceived performance, but complexity in factors like task difficulty makes precise predictions challenging. To help understand the interactions between these factors, a cognitive model was developed following this study. Future research should refine these factors to better personalize environments for cognitive enhancement. **Funding.** This study has been supported by the CoCoLab and Université Côte d'Azur (UCA, France).

S1-P32

Functional interaction between olfactory receptors and TRPV1 studied in HEK293T cells

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My research investigates the molecular interplay between olfactory receptors (ORs) and the ion channel TRPV1 in olfactory signaling using a reconstituted HEK293T cell system. ORs are G protein-coupled receptors (GPCRs) that detect odors, while TRPV1, known for temperature and pain sensing, is also expressed in olfactory sensory neurons (OSNs). Interestingly, some ORs and TRPV1 are co-expressed in non-olfactory tissues, such as the prostate and gastrointestinal tract, although their functional roles there remain unclear. Some odorants, such as vanillin, activate both ORs and TRPV1, suggesting functional crosstalk. Using intracellular Ca^{2+} and cAMP assays, I demonstrate that OR activation modulates TRPV1 responses (e.g., to capsaicin and eugenol) via cAMP- and PKA-dependent phosphorylation, whereas TRPV1 activation (e.g., capsaicin) suppresses OR51E1-mediated cAMP production through a Ca^{2+} -dependent, GRK-mediated mechanism. These findings are consistent with a bidirectional feedback mechanism that could modulate odor detection at the receptor level. This work clarifies key molecular mechanisms of olfactory signal integration and suggests potential applications in odor sensing and therapeutic delivery. It also provides a foundation for future studies linking molecular mechanisms to behavioral responses to smell. Funding: Japan Science and Technology Agency (JST) JPMJFS2125 (S.M.) JSPS DC2 Research Fellowship 25KJ1737 (S.M.) Kyoto University Foundation (planned, S.M.) KAKENHI Challenging Research (Pioneering) 18H05359, 20K20370, 22K18343 (S.K.) JST A-STEP JPMJTR194C (S.K.) NEDO project JPNP23200460 (S.K.)

S1-P33

Diverse Expression of Odorant Receptors (ORs) in the Olfactory Organ of Turtles

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The turtle olfactory organ consists of the upper chamber epithelium (UCE) and the lower chamber epithelium (LCE). Based on the presence or absence of associated glands and their anatomical locations, the UCE is thought to detect airborne odorants, whereas the LCE is thought to detect waterborne odorants. Turtles inhabit a wide range of environments, and the relative sizes of the UCE and LCE vary among species occupying different habitats. Furthermore, the relative abundance of Class I and Class II odorant receptor (OR) genes, that are classified according to sequence similarity, also differs among turtles from different habitats. This study examined the possible relationship between OR expression sites in the olfactory organs and environmental habitats in seven species of turtles: the red-bellied short-necked turtle *Emydura subglobosa*, the red-eared slider *Trachemys scripta*, the Reeve's pond turtle *Mauremys reevesii*, the snapping turtle *Chelydra serpentina*, the green sea turtle *Chelonia mydas*, the loggerhead sea turtle *Caretta caretta*, and the African spurred tortoise *Centrochelys sulcata*. In the semi-aquatic species, namely *E. subglobosa*, *T. scripta*, *M. reevesii*, and *C. serpentina*, Class I ORs were expressed in the LCE, whereas Class II ORs were expressed in the UCE. In the marine species, including *C. mydas* and *C. caretta*, Class I ORs were expressed in the LCE, while Class II ORs were expressed in both the UCE and LCE. Furthermore, in the terrestrial species, *C. sulcata*, Class I ORs were expressed in both the UCE and LCE, whereas Class II ORs were expressed in the UCE. These results suggest that habitat-related diversity of the turtle olfactory organ is reflected not only in the relative sizes of the UCE and LCE and the ratio of Class I to Class II OR gene numbers, but also in the sites of OR gene expression.

S1-P34

Role of Oxytocin Receptor Signaling in the Detection of Pup-Released Molecules by Vomeronasal Sensory Neurons in Male Mice

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Newborn mice release molecules that are detected by adults and trigger behaviors depending on the social experience of the recipients. In virgin males, such molecules are detected by the vomeronasal organ and trigger pup-directed aggression, while in sexually-experienced males such behavior is not frequently observed. Although recent advances have been made in the study of pup-directed behaviors, the identity of such molecules and the mechanisms underlying pup-directed aggression remains elusive. In this study we combined an ex-vivo calcium imaging strategy with in vivo exposure to previously identified pup-specific molecules to investigate the involvement of the vomeronasal organ (VNO) in the detection of those molecules. Moreover, using conditional knockout models, we also examined the role of the oxytocin receptor (OTR) signaling in the peripheral detection and central processing of pup signals. We show that pup-derived molecules are capable of activating VNO cells both ex vivo and in vivo, and that OTR signaling plays a key role in triggering pup-directed aggression or the silencing of such behavior.

S1-P35

Functional analyses of bitter taste receptors in Horses (*Equus ferus caballus*)

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Horses (*Equus caballus*) can distinguish sweet, salty, sour, and bitter tastants and are especially sensitive to many plant toxins. Horse bitterness sensitivity complicates the oral administration of bitter dietary supplements and medications. Current bitterness-masking strategies (flavorings, sweeteners, or encapsulation) are either ineffective or costly. The equine genome has revealed the presence of ~20 bitter taste receptors (ecTas2rs), enabling detection of potentially harmful dietary compounds. Identifying ecTas2r antagonists could provide an innovative solution to reduce the bitterness of equine nutritional supplements and pharmaceuticals. To functionally characterize equine bitter taste receptors, 6 ecTas2rs were expressed in human HEK293 cells. The cells were stimulated with 53 chemically various bitter compounds, including mineral salts, vitamins, and anthelmintics. All 6 receptors were functional and exhibited different activation profiles, which were broader for ecTas2r1, ecTas2r4, and ecTas2r14 receptors. Dose-response curves revealed that ecTas2r4 was more sensitive to CuSO₄ ($EC_{50} = 1.85 \pm 0.65 \mu M$) than ecTas2r1 ($EC_{50} = 300 \pm 17 \mu M$). Interestingly, ecTas2r1 responded to 2 known bitter anthelmintics, praziquantel and pyrantel, with EC_{50} values of 261 ± 286 and $26 \pm 9 \mu M$, respectively. Behavioral assays using a two-choice preference/aversion test were conducted in 10 adult horses with quinine and CuSO₄ to determine rejection (EC_{20}) and detection (EC_{40}) thresholds, as well as EC_{50} values. EC_{20} and EC_{40} values were similar for quinine (287 and 65 μM) and CuSO₄ (526 and 100 μM). The EC_{50} value for CuSO₄ was 58 mg/L (363 μM), in agreement with in vitro receptor activation. This work provides a prerequisite for identifying new ecTas2r antagonists that could reduce bitterness in oral equine dietary supplements and pharmaceutical.

S1-P36

A low-calorie rebaudioside D-glucose-sodium sweetener matches sucrose in palatability and neural responses in a functional MRI study of healthy adults.

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Non-nutritive sweeteners often taste less palatable than glucose-containing sugars, potentially because they differentially engage sweet taste pathways. We examined a low-calorie rebaudioside D-glucose-sodium (RebD mix) sweetener, formulated based on evidence that its components engage type 1 taste receptors 2 and 3 (T1r2/T1r3) and sodium-glucose cotransporter 1 (SGLT1)-related pathways, with the aim of determining whether the RebD mix matches sucrose in subjective palatability and neural responses. A within-subject fMRI study was conducted in 28 healthy adults (mean age 22.36 ± 5.20 years; 22 males; BMI 20.34 ± 2.22). During scanning, participants received three randomized gustatory conditions with tasteless rinses between trials: sucrose (234 mM), RebD mix (29.2 mM sucrose + 194.3 mM glucose + 0.156 mM RebD + 5 mM sodium gluconate), and RebD mix without Na⁺ (29.2 mM sucrose + 194.3 mM glucose + 0.184 mM RebD). Liking and wanting ratings for the gustatory stimuli were observed at scanning. Imaging analyses focused on orbitofrontal cortex (OFC), a key region associated with palatability, and OFC-centered functional connectivity across the gustatory conditions. The results showed that liking and wanting did not differ across the three conditions. OFC responses to the RebD mix were comparable to those elicited by sucrose, whereas the RebD mix without Na⁺ exhibited greater OFC-postcentral gyrus connectivity than the other two conditions (psBonferroni-corrected < 0.001). Across participants, OFC-postcentral connectivity in the RebD mix without Na⁺ condition was negatively associated with fullness measured during scanning. In conclusion, the RebD mix matched sucrose in subjective palatability and OFC responses, while removal of Na⁺ altered OFC-postcentral gyrus connectivity linked to satiety. These findings suggest that this RebD-based low-calorie formulation can achieve sucrose-comparable palatability.

S1-P37

Harder is not better: item difficulty does not enhance discrimination in olfactory identification tests

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Background: Short odor identification tests are widely used, but reducing item number may affect their sensitivity. It remains unclear whether item difficulty influences the ability of such tests to discriminate between groups. Methods: A 16-item odor identification test was divided into an easy 8-item set and a difficult 8-item set based on item accuracy. Performance differences between the two sets were examined, and their ability to discriminate between groups (gender, age, and self-reported olfactory ability) was evaluated using independent-samples t-tests and repeated-measures ANOVA. Results: Performance on the difficult item set was significantly lower than on the easy item set, confirming the manipulation of difficulty. However, no significant differences between groups were observed for either item set, and no interaction between difficulty and group was found. Conclusion: Increasing item difficulty reduced overall performance but did not improve the ability to discriminate between groups. These findings suggest that item difficulty and discriminatory power should be considered as distinct properties when developing short olfactory tests. This pattern may also be influenced by the characteristics of the current sample, which consisted mainly of participants with relatively homogeneous olfactory abilities.

S1-P38

Pyroxamide as a potential therapeutic option to counteract the α -synuclein preformed fibrils-induced damage in human nasal epithelial cells.

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Objectives: Parkinson's disease (PD), the second most common neurodegenerative disorder, is characterized by dopaminergic neuron loss and early olfactory dysfunction which affects up to 90% of patients and represents a premotor sign. PD processes remain poorly understood, revealing the need for novel biomarkers for early diagnosis, as well as for the evaluation of new therapies. The aim of this study was to identify novel drug candidates capable of reversing the proteomic signature of the olfactory tract of PD subjects by applying a drug repositioning strategy. **Experimental methods:** This work integrates neuropathological diagnosis, quantitative proteomics, physical/functional interaction data, large-scale pharmacogenomics databases and in vitro functional experiments. **Results:** Olfactory tract (OT) identified 87 and 162 differentially expressed proteins (DEPs) across LBD stages regarding the control group. Drug repurposing revealed the HDAC1 inhibitor pyroxamide as one of the top drug candidates. Pyroxamide in vitro treatment induces a protective effect against α -syn preformed fibrils in human nasal epithelial cells (HNECs) demonstrated by a survival increment 3 and 6 hours at 5 nM (***) and 20 nM (**). Mechanistically, pyroxamide induces a cross-talk between HDACs and SIRT families, and a reduction in IL1b (**) and an upregulation of NFE2L2 (***) expression levels. **Conclusions:** These findings confirm the suitability of omics profiles in drug repurposing workflows against PD, offering valuable insights into the potential of HDAC1 inhibitors in the therapeutic pipeline of PD. **Funding:** Grants PID2023-152593OB-I00 funded by MCIU/AEI/ 10.13039/501100011033 / FEDER, UE to ES and JFI and 0011- 1411-2023-000028 (from Government of Navarra) to ES.

S1-P39

Smelling in Color and Sound: Crossmodal Smell Training

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Objective We developed an app-assisted multisensory smell training based on experimentally derived crossmodal associations between odours, sounds, and visual features. This approach aims to enhance traditional smell training, which is known to support olfactory function, cognition, and wellbeing. **Methods** A total of 146 participants associated four odours (lemon, eucalyptus, clove, rose) with visual (shape, brightness, colour) and auditory features (instrument type, tempo, pitch). Significant crossmodal associations emerged; for example, lemon was linked to round shapes, bright colours, fast tempo, and high pitch, whereas clove was associated with angular shapes, brown tones, slow tempo, and woodwind instruments. Based on these findings, six artists created abstract music videos for each odour. A further 200 participants rated the congruence between odours and 16 videos, from which the best-matching videos were selected. Thirty-six participants aged 50–75 then completed a six-week, twice-daily app-assisted multisensory smell training programme. Olfactory function (threshold, discrimination, identification), cognition (verbal fluency), odour awareness (OAS), and wellbeing (WHOQOL) were assessed. Thirty-three participants completed the intervention (92% adherence). **Results** Odour discrimination and identification improved significantly after six weeks. One cognitive measure (verbal fluency) also showed significant improvement. No significant changes were observed in odour threshold, odour awareness, or wellbeing. **Conclusions** App-assisted multisensory smell training based on crossmodal associations is feasible, engaging, and shows high adherence. A three-month follow-up study comparing multisensory training, traditional smell training, and a no-training control condition is currently underway. **Funding** Funded by the European Sensory Network (ESN)

S1-P40

Context-dependent modulation of danger perception in the mouse Grueneberg ganglion

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The mouse Grueneberg ganglion (GG) is a specialized olfactory subsystem involved in the detection of danger-associated volatile cues, triggering stereotyped behavioural and physiological responses such as freezing, risk assessment, and stress hormone release under laboratory conditions. While these responses are well characterized, they are typically studied in controlled environments that poorly reflect the complexity of natural settings. Here, we combine complementary approaches to investigate mouse GG function across multiple levels of organization. Molecular analyses such as RT-qPCR and immunohistochemistries are used to identify candidate signalling components, while ex-vivo calcium imaging enables functional testing of putative ligands. Downstream integration is assessed through central activity mapping (c-Fos), endocrine measurements, and behavioural assays quantifying stress-related responses. However, in natural environments, animals continuously integrate multiple internal states and external cues, including metabolic status, social context, and background olfactory landscapes. These factors are likely to modulate the perception and behavioural relevance of danger signals. Building on established laboratory paradigms, we investigate how such contextual variables shape GG-dependent responses and influence decision-making processes. By bridging controlled experimental approaches with ecologically relevant frameworks, this work highlights that danger perception is not solely stimulus-driven but dynamically shaped by context, and provides a framework to interpret chemosensory communication beyond purely signal-centric models. This work was supported by the Department of Biomedical Sciences at the University of Lausanne and by the Swiss National Science Foundation Grant 310030_185161 (to M.-C.B.).

Session 2 — 16 September 2026, 18:30-20:00

S2-P1

Relaxation through olfaction: the soothing power of essential oils to support aging

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The inhalation of essential oils is increasingly being investigated for its effects on physiological regulation and well-being. Previous studies have shown that pleasant odors can modulate autonomic activity by decreasing heart rate and promoting slow, deep breathing, often associated with increased low-frequency EEG activity. In addition to olfactory perception, odorants may stimulate the trigeminal system, producing sensations such as freshness or tingling through pathways closely linked to cardiac and respiratory regulation. However, the specific contribution of trigeminal stimulation to relaxation remains poorly understood. The Inh'âge project aims to investigate the impact of aging on the interactions between chemosensory perception and autonomic regulation by dissociating olfactory and trigeminal contributions. Human participants from different age groups inhale mint-based odorants delivered via an olfactometer. Pilot studies identified odorants with similar olfactory qualities but distinct trigeminal profiles, allowing controlled comparisons of trigeminal stimulation. During the experiment, physiological responses are recorded from the first sniff and throughout the 30 seconds following odor presentation. Cardiac, respiratory, and subjective measures are analyzed to characterize immediate relaxation-related responses. The main study is currently ongoing. We hypothesize that stronger trigeminal stimulation enhances subjective and autonomic markers of relaxation and that these effects are modulated by age-related declines in olfactory and trigeminal sensitivity, particularly after 60 years of age. This work will improve understanding of the sensory mechanisms involved in respiratory modulation and may help optimize the use of essential oils as non-pharmacological interventions adapted for both young and older adults. This PhD project is funded by Roudnitska. Research conducted in Quebec is supported by MITACS and Neurodis.

S2-P2

Endoplasmic reticulum specialization predefine sensory neuron maturation in the vomeronasal organ of rodents

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The vomeronasal organ (VNO) in mice is a chemosensory structure that detects non-volatile pheromonal and semiochemical signals that regulate sociosexual and interspecies behaviors. The vomeronasal sensory neurons (VSNs) can express from V1R or V2R G protein-coupled receptors which have distinct molecular complexities. V2Rs feature larger extracellular domains and are notoriously challenging to express in heterologous systems. Single-cell sequencing analysis reveals that V2R VSNs are enriched for the expression of endoplasmic reticulum (ER)-related proteins, including the ER stress regulator BiP/HSPA5 and Calreticulin-4 (Calr4) chaperones. The ER plays a central role in protein folding, quality control, and degradation processes. We expect a specialized ER is critical for the maturation and homeostasis of V2R VSNs. To investigate the in vivo role of Calr4, we analyzed a constitutive Calr4 knockout mouse line. We used immunohistochemical and transcriptomic approaches on postnatal days (P)10, 16, and 21. By P16, Calr4KOs showed elevated ER stress, marked by increased Caspase-12 expression, and aberrant adhesion molecule profiles in V2R VSN projections. By P21, V2R VSNs showed a ~35% reduction and a twofold increase in apoptosis. Notably, the V2R receptor V2R2 was undetectable at the lumen of the VNO at all developmental stages examined, suggesting that the VSNs of Calr4KOs are nonfunctional. Our data indicate that the chaperone Calr4 is critical for V2R VSN homeostasis and V2R receptor processing, while V1R VSNs remain unaffected. This study provides the first in vivo characterization of Calr4KOs and demonstrate that ER chaperone diversity supports developmental resilience in distinct neuronal subtypes that demand high proteomic activity and receptor multiplexity, such as the V2Rs. All data presented have N>3. Student's t-test and P<0.05 were set as the threshold for significance. This work was supported by NIH NICHD awards 2R01HD097331 and R01HD114827 (PEF).

S2-P3

Electrical stimulation of the human brain elicits odor perception: non-canonical brain structures may contribute to olfactory processing

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Background. Unlike some other sensory systems, the impaired sense of smell does not yet benefit from a ready-to-use olfactory implant or prosthesis. The principle of an olfactory implant is based on volatile compounds that are captured by sensors mimicking the olfactory epithelium, translating them into electrical stimulation patterns, being transferred to and processed by central brain structures. We present a case series of three patients reporting olfactory percepts after direct electrical brain stimulation. **Methods and results.** One patient with left-sided frontotemporal astrocytoma undergoing awake brain surgery reported a lemon smell after stimulation in the posterior part of the olfactory bulb (fork probe, 45 mm, straight, ball tip with one biphasic stimulation, pulse width 800 μ s, 50 Hz, 6 mA, duration 4 s). Two patients with drug-resistant epilepsy undergoing stereo-encephalography reported spinach odor after stimulation in the amygdala, close to the piriform cortex (alternating bipolar rectangular pulses, pulse width 300 ms, 1.6 mA, 1 Hz, duration 5 s), and an unknown disgusting smell after being stimulated in the temporal pole (alternating bipolar rectangular pulses, pulse width 300 ms, 7 mA, 1 Hz, duration 5 s), respectively. **Discussion.** We discuss these findings having in mind the small sample size and methodological limitations. The type of neural oscillations provoked by the electrical stimulation may be a key factor in the success of olfactory percept elicitation. We underline that the engagement of association areas connected to the primary olfactory cortex could be a potentiation, if not a substitution, for odor sensation induction. **Conclusion.** A comprehensive understanding of stimulation targets and parameters and their is crucial to pave the way for an olfactory implant. It will be necessary to map and collect data across larger human neural networks. **Funding:** European Union's Horizon 2020 FET Open programme, grant agreement No 964529.

S2-P4

A step into children's olfactory world: a qualitative analysis of spontaneous odor representations in early childhood

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From the scent of a forest to a grandmother's kitchen, children encounter a rich olfactory world long before they can name it systematically. While adult odor representations are well documented, reflecting learned semantic structures tied to food, nature, and social life, far less is known about how young children organize and verbalize their olfactory experience. Studying these early representations in naturalistic contexts may reveal aspects of olfactory cognition that controlled laboratory tasks cannot capture (Nováková & Vojtušová Mrzílková, 2016). We invited 151 children aged 3–9 years ($Mage = 6.52 \pm 1.19$; 74 girls) to name as many odors as possible from their daily environment in an open-ended free-listing task. Responses were lemmatized by two independent experts and coded across three dimensions: odor sources, contextual references, and affective descriptors. The analysis yielded 406 odor sources, 153 contextual references, and 26 affective descriptors. Children's representations clustered around three domains: environment (natural, anthropogenic), food, and social contexts, with nature-related items (flower, tree, leaf) and food being the most salient. Contextual references pointed predominantly to outdoor and domestic spaces, while affective language was sparse but skewed positive (fresh, nice), with occasional hedonic avoidance (smelly, sweaty). Taken together, these findings paint a picture of early olfactory cognition anchored in concrete, lived experience. The relative scarcity of social odor references compared to adult patterns may reflect a developmental trajectory in which chemosensory social awareness emerges gradually (Sorokowska et al., 2018). Beyond their theoretical value, these results offer a naturalistic lexical foundation for designing olfactory assessments.

S2-P5

One receptor, two binding sites: ligand-specific activation of TAS2R14 by pharmaceutical drugs

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Bitter taste receptors (TAS2Rs) are G protein-coupled receptors activated by structurally diverse compounds, including many clinically used drugs. TAS2R14, the most broadly tuned member of this family, contains both a canonical extracellular binding site and an intracellular binding pocket recently revealed by cryo-EM structures. While several agonists were shown to bind the intracellular site, the functional contribution of each binding pocket remains unclear. We combined structure-based docking and cell-based functional assays to investigate how pharmaceutical agonists interact with the two binding sites of TAS2R14. Known ligands were computationally classified according to predicted binding-site preference, and representative drugs and mutations were selected for experimental validation. Receptor activation was assessed using IP-One and BRET2 assays in HEK293T cells expressing wild-type or point-mutated TAS2R14. The results reveal ligand-specific effects. Carbimazole responses were more strongly reduced by mutation of the intracellular site, while lidocaine was primarily affected by extracellular mutation. Tamoxifen showed assay-dependent effects, with reduced G-protein recruitment upon intracellular mutation but minimal changes in downstream signaling. These findings demonstrate ligand-specific manner of interaction with TAS2R14 pockets, highlight the importance of different readouts and the complexity of bitter taste receptor activation. Funding: This work was supported by the Israel Science Foundation (grant numbers 1129/19 and 1096/25).

S2-P6

Does body mass shape sensory experience and its linguistic expression?

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Lowered olfactory performance has been previously reported in individuals with high BMI (Body Mass Index) in terms of reduced odor sensitivity and overall olfactory functioning as measured by TDI scores (Threshold, Discrimination, Identification). On the other hand, high BMI is associated with heightened olfactory imagery. However, little is known about the association between BMI, olfactory awareness, and food-related language. Often, high BMI is a reflection of bodily processes including eating behavior and chronic disorders that affect sensory perception. Such factors could potentially impact sensory language. This study investigates whether individuals with high BMI differ from those with healthy BMI in self-reported emotional relationships with food, olfactory awareness, sensory imagery, and sensory language used in food descriptions. Using a cross-sectional, mixed-methods design we explore links between BMI, food-related behavior and thought, and sensory language use. Participants from a healthy weight group (BMI 18.5 – 25.18) and participants from a higher BMI group (BMI 25.56 – 31.89) completed an online survey assessing their emotional relationship with food (Positive Eating Scale), olfactory awareness (Odor Awareness Scale – 6), vividness of sensory imagery (the Plymouth Sensory Imagery Questionnaire), and eating behavior patterns (Three Factor Eating Questionnaire – 18). We also asked them to provide experiential descriptions of a set of food images. We hypothesize that individuals with higher BMI may experience higher awareness of environmental odors, stronger sensory imagery and stronger positive emotional connection to food. This could result in longer food descriptions, and more sensory and valenced language. The present study will deepen our understanding of the role of sensory and emotional factors in sensory language. Funding: Starting grant from Radboud University granted to Dr. Ilja Croijmans

S2-P7

Obesogenic diet alters astrocytes function in the olfactory bulb

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The global obesity epidemic, driven by increased availability of high-fat and high-sucrose foods, is associated with chronic, low-grade systemic inflammation. This inflammatory state extends to the central nervous system, causing dysregulation in several brain regions, including the olfactory bulb (OB)¹. A key pathophysiological hallmark of obesity-induced neuroinflammation is astrocytic dysfunction². In this project, we assess in mouse model if diet-induced obesity alters astrocytic activity in response to odorants in the OB, and how this correlates with the neuronal activity as well as odor driven behaviors. Adeno-associated viral (AAV) vectors were unilaterally injected in the mitral cells layer to express the Ca²⁺ indicator GCaMP6s and jRCaMP1a respectively in GFAP positive astrocytes and CaMKIIa positive excitatory neurons of the OB. Ca²⁺ responses from both cell types were then simultaneously recorded while mice were performing a simple odor preference behavioral task with different odorants. Near the odor source, astrocytes showed increased Ca²⁺ responses, whereas neurons showed reduced activity compared to distant regions. Astrocytic responses to odors were significantly reduced after just 1 week on a high-fat, high-sucrose diet (HFHS: 60% fat, 20% carbohydrates, 20% protein) compared to chow-fed controls. This impairment persisted through 2 months of diet, while neuronal odor responses remained unchanged. Correspondingly, mice fed HFHS showed significantly lower odor preference compared to chow-fed controls. Together, these findings reveal that in the OB, obesity induces early astrocytic dysfunction that precedes neuronal alterations.

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S2-P8

Olfactory bulb gamma oscillations may represent a valence scaffold for the next sniff

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Predictive coding frameworks propose that sensory systems continuously generate top-down predictions that are compared with incoming sensory input. For the visual and auditory domains, gamma-band activity is considered to mainly convey bottom-up sensory evidence, while beta-band activity is thought to reflect top-down predictive signals. Whether the human olfactory system operates under a similar regime remains largely unknown. Building on previous findings demonstrating bidirectional, valence-related communication between the olfactory bulb (OB) and piriform cortex (PC), we recorded activity from both regions simultaneously using the electrobulbogram (EBG) method in 32 healthy participants. Odors varying in valence were presented across two consecutive sniffs, with valence switched in 30% of trials to create a sniff-by-sniff oddball paradigm. When odor valence was repeated, OB gamma activity increased during the second sniff, consistent with a refinement of the sensory representation. However, when odor valence changed between sniffs, this gamma enhancement disappeared, and OB gamma reset to first-sniff levels. This pattern indicates that OB gamma activity reflects a refinement process that depends on the stability of sensory input across sniffs. These findings suggest that the human olfactory system engages gamma-mediated updating mechanisms and may participate in a broader predictive-coding architecture, although its implementation may differ from what is commonly demonstrated in vision and audition

S2-P9

Towards Artificial Olfaction

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In contrast to hearing and vision, the exact function of the olfactory sense remains largely unknown. This impacts many research areas ranging from fragrance chemistry to odor perception, as well as sensor design in artificial noses. Recent advances in AI technologies started to pave the way towards digitizing odors for structure-based odor predictions and broadening our understanding of structure odor relationships. However, there are still many open ends and questions before the sense of smell is successfully digitized and new knowledge can be transferred to other applications such as bio-mimetic sensor design. One problem that is currently holding back the advancement in the field of artificial noses is our fragmentary understanding of the molecular detection mechanisms underlying the olfactory sense. During the talk, I will give an overview of the advances and challenges in the field and show different projects that our team is working on at the VU Bioinformatics group of the Vrije Universiteit Amsterdam. Here, one of our main objectives is to unravel and to quantify the uptake mechanisms of odorants and other volatile organic molecules through binding proteins for applications in bio-mimetic sensor units. In future, due to the increasing number of available data on molecular structures and olfactory properties, machine learning and more data-driven approaches will be necessary to address new computational challenges and thus, in the long term, allow us to move towards the digitalization of the olfactory sense.

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S2-P10

Age-group differences in effective connectivity between brain regions during salty taste perception in humans.

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Objective: Taste function is critical for safety and quality of life in older adults. Although age-related declines in perceived taste intensity have been reported, the underlying neural mechanisms remain unclear. This study aimed to examine age-related differences in brain functional connectivity. **Materials and Methods:** Thirty-six older adults (60–81 years) and 45 young adults (21–39 years) participated. Stimuli consisted of distilled water and NaCl solutions at three concentrations (0.001, 0.3, 0.5 M). Functional MRI data were acquired using a 3T scanner. Effective connectivity was analyzed using dynamic causal modeling (DCM) across the thalamus, insula, and postcentral gyrus. Baseline (fixed) and modulatory connectivity parameters were estimated and compared between groups using parametric empirical Bayes (PEB). Effects with posterior probability > 0.95 were reported. **Results and Discussion:** Activation analysis revealed reduced responses in the thalamus and insula in older adults at 0.3 M and 0.5 M concentrations. DCM analysis revealed age-related differences in effective connectivity. In the presence of gustatory input (pooled across NaCl concentrations), several connections reached significance within each age group, including common positive baseline connectivity from the thalamus to the insula and postcentral gyrus, and common negative connectivity from the postcentral gyrus to the thalamus. Importantly, PEB analysis showed that only the thalamus-to-insula and thalamus-to-postcentral gyrus connections were significantly greater in young adults than in older adults. No age group differences were observed in stimulus-dependent effects. These findings may indicate that age-related declines in thalamic function weaken baseline connectivity originating from the thalamus, highlighting its central role in taste processing. This work was supported by JSPS KAKENHI Grant Numbers 24K16583, 23K22226, 22H00955, 21K21100, 17H04423.

S2-P11

Real-time chemical sampling at the olfactory epithelium shows sub-optimal receptor access**Irene Zanettin¹**, Frans Nordén¹, Mikael Lundqvist¹, Artin Arshamian¹, Johan Lundström^{1,2,3}¹ Department of Clinical Neuroscience, Karolinska Institutet, Stockholm, Sweden² Department of Otorhinolaryngology, Karolinska University Hospital, Stockholm, Sweden³ Monell Chemical Senses Center, Philadelphia, Pennsylvania, USA

Olfactory perception is typically studied under the implicit assumption that the chemical composition and concentration of an odorant at the nasal entrance accurately reflect the stimulus reaching the olfactory epithelium. However, airflow dynamics, mucosal interactions, and physicochemical properties of odors may substantially transform the stimulus before odorant-receptor binding occurs. Here, we introduce a novel approach combining Proton-Transfer-Reaction Time-of-Flight Mass Spectrometry (PTR-TOF-MS) with simultaneous in vivo sampling at the naris and at the human olfactory epithelium to directly characterize the chemical stimulus that reaches olfactory receptors. Using a range of pure odorants, differing in chemical characteristics, we measured real-time odorant concentration at both sampling locations. Preliminary results reveal variance both between recording locations and odorants, indicating that not all odorant molecules reach the receptor surface equivalently. This variance appears to depend on the chemical identity of the odorant, suggesting that physicochemical properties may shape the effective olfactory stimulus. These findings suggest that perceptual or neural differences may arise from peripheral filtering at the level of the nasal cavity and epithelium, prior to any central neural processing. By directly measuring the chemical stimulus at its point of neural transduction, this work opens new paths for investigating how peripheral processes contribute to odor perception, discrimination, and individual variability in olfaction.

S2-P12

Histological and neurochemical characterization of the bovine olfactory and vomeronasal systems during the early fetal stages

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The bovine species relies on a sophisticated chemosensory apparatus for neonatal survival and social behavior. While adult anatomy is well-documented, the prenatal timeline for the functional maturation of the olfactory and vomeronasal organ (VNO) remains under-explored. This study aims to characterize the histological and neurochemical development of these systems during the first stage of bovine gestation to determine their functional status. Bovine fetuses (*Bos taurus*) at gestational days 14, 15, 54, 58, 62, 66, 76, and 90 were processed for histological and immunohistochemical analysis. We targeted olfactory marker protein (OMP) to identify mature sensory neurons and G-protein subunits Gai2 and Gao to map signal transduction pathways. Additionally, lectin histochemistry was performed using *Lycopersicon esculentum* (LEA) and *Ulex europaeus* (UEA-I) to assess the glycoconjugate profile of the primary projections. Our results show the VNO is morphologically and functionally well-developed during the first gestation stage, maintaining communication with the environment through the incisive papilla. Neurochemically, the VNO sensory epithelium showed expression for both Gao and Gai2. Mature sensory axons were confirmed by positive OMP and lectin (LEA, UEA-I) labeling. A significant developmental heterochrony was observed at the central level: while the main olfactory bulb (OB) exhibited clearly differentiated glomeruli by the end of this stage, the accessory olfactory bulb (AOB) remained immature, lacking a defined glomerular layer despite the advanced state of the peripheral VNO. These findings indicate that the bovine main olfactory pathway matures structurally earlier than the accessory system. The presence of Gao in the bovine VNO suggests a complex receptor signaling environment early in development. Despite the peripheral morphological complexity, the AOB completes its maturation later in gestation compared to the main olfactory system.

S2-P13

Detection of neurodegenerative and inflammatory biomarkers in human olfactory mucosa obtained by nasal brushing: a feasibility study

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Objectives: Olfactory mucosa (OM) may offer an accessible peripheral window into brain-related changes in neurodegenerative disease. Nasal brushing is minimally invasive, but limited tissue volume and preservation remain challenges for molecular analysis. This study evaluated whether nasal brushing-derived human OM contains olfactory epithelial cells by qPCR for OMP and TUBB3, compared preservation methods for RNA recovery, and assessed feasibility of measuring neurodegenerative and inflammatory biomarkers at transcript and protein levels. **Methods:** OM samples were obtained by nasal brushing from five adults with normosmia (two in their 50s, three in their 70s; one male with mild cognitive impairment). Samples were preserved in RNAlater or by direct freezing of the cell pellet. Total RNA was assessed by NanoDrop. After cDNA synthesis, PCR, gel electrophoresis, and qPCR were performed for GAPDH, OMP, TUBB3, TARDBP, APP, and MAPT. Protein levels of A β 40, A β 42, tTau, pTau181, CXCL10, and CCL2 were measured using Luminex. No statistical analyses were performed. **Results:** RNAlater produced higher RNA concentration and purity than direct cell pellet freezing. However, OMP and TUBB3 expressions were higher in directly frozen samples. TARDBP expression was detected in directly cell-pellet samples by qPCR and gel electrophoresis. APP was not detected, and MAPT showed a poor melt curve profile. In the Luminex assay, A β 40 and A β 42 were below the detection limit. Mean tTau and pTau181 levels were higher in the 70s group than in the 50s group, whereas CXCL10 was higher in the 50s group and CCL2 was higher in the 70s group. **Conclusions:** These findings suggest that nasal brushing-derived human OM may be used for exploratory molecular profiling, including selected olfactory epithelial, neurodegenerative, and inflammatory markers. Preservation method influenced RNA quality and target detectability. Further optimization and validation in larger cohorts are needed.

S2-P14

Contextual variability in odour quality representations: learning the identity of ambivalent odours

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Odours we experience daily are composed of multiple components, and the perception of an unfamiliar note within a familiar smell depends on situational context. Detecting unusual notes in food odours is essential to avoid ingesting contaminants, yet this caution must be balanced against the costly risk of rejecting safe food. Our previous work showed sharp declines in the perceptual dominance of an attended food component in an odour mixture as the proportion of a contaminant increases. In contrast, the perceived dominance of an attended non-food odour remains comparatively robust. This multi-part study investigates whether this asymmetry depends on knowledge of an odour stemming from an edible or inedible source. In part 1, an odour-learning paradigm was developed in which participants (N = 20) formed associations between odours (food vs. non-food vs. ambivalent) and their sources. Participants rated familiarity, intensity, pleasantness, and edibility before and after label-learning to track induced changes in odour evaluation. Free text labelling and post-learning edibility ratings demonstrate that we successfully manipulated participants across conditions to identify the same ambivalent odours (AO1, AO2) as originating from either an edible or an inedible source (AO1: $b_{\text{condition}} = 48.82$, $p = 0.042$; AO2: $b_{\text{condition}} = -38.67$, $p = 0.048$). Part 2 examines whether perceptual dominance (point of subjective equality, just-noticeable difference) across odour dilution series depends on learned identity, and whether ambivalent odours adopt food- or non-food-like processing following learning. This approach enables testing whether identity-based learning alone can reproduce previously observed differences in food versus non-food odour processing and provides a framework for investigating underlying neural mechanisms. This work was funded by the European Research Council (No. 947886 to JS).

S2-P15

Spatial representations in the piriform cortex emerge after odor-context associative learning

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Olfactory processing depends on prior experience, present context, and animal's internal state. To investigate how learning dynamically reshapes olfactory cortical circuits, we trained mice in a virtual reality-based, context-dependent GO/NO-GO task. Mice explored virtual corridors presenting two odors across two visual contexts, learning to lick for a water reward only for one specific odor-context combination. In previous experiments, we showed that odor-responsive primary piriform cortex (PCx) neurons become mixed-selective after learning, encoding positional, contextual, and associative information. It is still unclear how this shift emerges in the PCx encoding scheme. Here, we performed PCx neuronal recordings throughout the learning process, using behavioral analysis alongside neuronal encoding and decoding models to address this open question. We show that mice learn the task sequentially: they first discriminate odors (intermediate stage) before successfully incorporating contextual information (expert stage). PCx neuronal encoding parallels this behavioral progression. In intermediate sessions, PCx neurons encode odors but lack contextual information. Conversely, expert animals show robust encoding of both parameters. Decoding analyses confirm this shift, revealing that neurons in expert animals exhibit significantly enhanced multiplexing, modulation by a greater number of task parameters, compared to early or intermediate stages. This evidence indicates that the observed contextual encoding in the PCx depends on the associative learning process in which odorant cues are combined with positional information to produce correct behavioral responses in the task. Ongoing experiments focus on the neural mechanisms responsible for making this association in the PCx. Funding: SNSF SPIRIT 216044, PICT 2020-00360, FOCEM-Mercosur (COF 03/11).

S2-P16

Investigating the physiological properties of olfactory bulb granule cells' basal dendrites

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In the olfactory bulb (OB), granule cells (GCs) mediate recurrent reciprocal and lateral inhibitory synaptic interactions between mitral and tufted cells (MTC) via their apical dendrite in the external plexiform layer. GCs also possess a brush of basal dendrites in the GC layer, where they are known to receive inputs from MTC axon collaterals and olfactory cortical areas. Recent studies showed that the inputs at those dendrites contribute to action potential (AP) generation. Yet, the physiological properties of GC basal dendrites were not studied so far. First, we examined the Ca^{2+} dynamics in the GC basal dendrites in response to somatic AP using 2-photon Ca^{2+} imaging (indicator OGB-1 100 μM) and GC whole-cell recording in juvenile rat OB slices. ($\Delta\text{F}/\text{F}$) AP transients in the basal dendrites had a mean amplitude of $17 \pm 8\%$ (mean $\pm$ SD), significantly smaller than in the proximal apical dendrite of the same cell ($24 \pm 31\%$). In contrast to the apical dendrite, ($\Delta\text{F}/\text{F}$) AP did not depend on distance from the soma. We next studied GC responses to excitatory inputs at the basal dendrites either with electrical stimulation in the GC layer or 2-photon glutamate uncaging (TPU) near basal dendritic spines. We compared spontaneous and TPU-evoked Ca^{2+} (($\Delta\text{F}/\text{F}$) spont vs ($\Delta\text{F}/\text{F}$) TPU) and electrophysiological (sEPSP vs uEPSP) events, and their amplitudes were statistically similar. Thus TPU is likely to reproduce physiological events. The probability of obtaining an evoked synaptic or TPU-evoked Ca^{2+} transient was low (0.16 and 0.26 respectively) compared to the apical dendrite (0.63 and 1 respectively) and their $\Delta\text{F}/\text{F}$ amplitude was small on average ($23 \pm 15\%$ and $19 \pm 6\%$ respectively). Thus Ca^{2+} influx does not seem to be a major player at the basal dendritic synapses. AMPA receptor blockade by CNQX abolished uEPSPs completely ($n = 7$ cells). Thus, those results suggest that the functional differences between basal dendrites and apical dendrites are also reflected in their physiology. Funding DFG FOR5424

S2-P17

Beyond taste: Unveiling the multisensory sensation of creamy mouthfeel by synergizing biophysical, molecular and food biology approaches

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Although creamy mouthfeel strongly influences consumer acceptance of (plant-based) dairy, the mechanisms underlying its perception remain poorly understood. Proposed mechanisms discussed in the literature include chemical stimulation of oral chemoreceptors (e.g., CD36, GPR120) by fat molecules and/or mechanical stimulation of mechanosensitive receptors (e.g., Piezo1/2, TMEM63) driven by structural characteristics. However, specific receptors involved and the integration of chemical and mechanical cues are largely unknown. To address this gap, we apply a novel approach combining bio-atomic force microscopy (bioAFM) with molecular biology techniques to probe an oral model cell line. With this, we aim to elucidate the physiological, biochemical and biophysical basis of creamy mouthfeel, with particular focus on oral mechanotransduction from a (nano)mechanical perspective. Initial functional Ca^{2+} -mobilization assays indicated cellular activation by treatments associated with creamy mouthfeel (i.e., palmitic, oleic, and orotic acid), motivating further investigations at the receptor level. Complementary analysis revealed changes in gene expression patterns of fat and mechanoreceptors, suggesting involvement of both chemical and mechanical sensing pathways. To study this mechanistic link, nanomechanical bioAFM measurements showed treatment-induced changes in cell membrane mechanical properties, potentially triggering or modulating mechanoreceptor responses. Ongoing inhibition and knockdown studies aim to resolve individual and synergistic contributions of receptors. Together, these findings suggest that creamy mouthfeel arises from combined activation of fat and mechanosensitive pathways rather than a single receptor type, highlighting the importance of multimodal approaches. This mechanistic insight may provide a deeper understanding of creamy mouthfeel perception and support the development of new low-fat or plant-based alternatives without compromising sensory appeal.

S2-P18

Memory consolidation during wakeful rest: could odors play a role?

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The experience of odors is primarily hedonic, hence their use to enhance well-being. Recent research highlights the olfactory system's ability to promote relaxation through two key mechanisms: pleasant odors can reduce anxiety and stress, impacting physiological markers such as heart and respiratory rates. Additionally, because olfactory receptor cells are sensitive to respiratory airflow, slow breathing has been shown to induce respiration-locked slow oscillations in large cerebral networks, specifically during quiet wakefulness in both rodents and humans. Slow brain oscillations are known to be implicated in memory consolidation during sleep. Interestingly, recent studies suggest that periods of rest following learning can also facilitate memory consolidation compared to engaging in distracting tasks, and that the power of slow EEG oscillations correlates with memory performance during rest. This project investigates whether exposure to personally pleasant essential oils during a post-learning rest period could enhance memory consolidation by inducing a state of calm wakefulness and slowing respiratory rate, as well as EEG oscillations. In a within-participant design, we compared memory scores in two randomized conditions, separated by two weeks: an +odor condition and a -odor condition, while recording EEG, physiological activity. In both conditions, encoding was followed by a 15-minute rest period, during which participants were either exposed to an odor (+odor) or not (-odor). Preliminary results revealed a significant decrease in respiration frequency in the +odor condition compared to the -odor condition, along with an increase in inspiration volume and amplitude. However, no significant difference in memory performance was observed between the two conditions. EEG data analysis is ongoing. Current findings suggest that while pleasant odors may induce deeper and slower breathing, this may not directly impact memory consolidation. Funding: ANR BreathSmellRelax

S2-P19

What children know about smells: linking spontaneous odor reports to olfactory salience

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Olfactory perception is closely linked to memory, language, and experience, yet its development in children remains relatively understudied, particularly in naturalistic, open-ended contexts and in relation to meta-cognitive aspects such as odor awareness. Children's subjective experiences are also rarely compared to objective, linguistically derived measures. In the present study, 122 children (Mage=6.71, SDage=1.16, 62 girls) with a normal sense of smell (measured by the Sniffin' Sticks identification test) were asked to freely report sources of odors. These responses were compared with established olfactory indices: the Olfactory Association Index (OAI) and Olfactory Specificity Index (OSI) proposed by Iatropoulos et al., which quantify the extent to which words are linked to olfactory contexts in large language corpora. OAI reflects how strongly a word is associated with olfaction, whereas OSI captures how specific the odor reference is. A subsample of 60 children (Mage=7.25, SDage=1.36, 30 girls) was retested after 4 weeks to assess temporal stability. Results showed that more frequently reported odor sources were moderately associated with higher OAI and lower OSI values. This suggests that commonly mentioned odors are strongly linked to olfaction but tend to be more general and used in broader olfactory contexts. Temporal stability analysis revealed a moderate, positive correlation. Importantly, stability did not differ substantially depending on whether it was calculated using raw frequency counts or values adjusted by OAI and OSI indices. This suggests that children's representations of odor sources are relatively robust and not strongly dependent on the specific metric applied. Overall, the findings suggest that children's spontaneous odor answers are meaningfully related to olfaction, rather than random. Their general nature may indicate that children's answers are more based in their real-life experiences and less in developed linguistic representations.

S2-P20

Perception and use of food fatty-acids odours on flies

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Odours are a vital source of information that allows animals to avoid a potential danger, to communicate, or to find food. Some volatile components can also trigger reversible anaesthesia. However, their mode of action remains largely unknown.

Here we describe the role of Olfactory Receptor 47b (OR47b) in modulating the Olfactory Sensory Neurons (OSNs) that express the male-specific transcription factor Fruitless, FRUM. We have identified two odorant volatile fatty-acids, propanoic and butyric acids, that are by-products of food, to which OR47b is sensitive. Using calcium imaging, we have shown that the OSNs projecting to the VA1v glomerulus in the *Drosophila* antennal lobe are inhibited by these odours, and that males are more sensitive than females. Depending of the concentration used, these odours are strongly repulsive and cause a reversible anaesthesia. Furthermore, this sex-specific behaviour is restricted to adults as larvae are highly attracted to the odours, and never get anesthetized. This effect is shared by *Drosophila melanogaster* and *Drosophila suzukii* the spotted-wing fly that parasitizes red-fruit crops, although no gender difference is observed in the latter. These findings suggest an evolutionary and stage-dependent sensitivity to some odours through a specific sensory pathway linked to OR47b. And, second, they suggest that the odours could be used to protect crops in an environmentally friendly way, as shown in preliminary field tests. Research in Yaël Grosjean's laboratory is supported by the "Centre National de la Recherche Scientifique", the "Université Bourgogne Europe", the Conseil Régional Bourgogne Franche-Comté (PARI grant), the European Council (ERC starting Grant, GliSFCo-311403), the SATT-Sayens (Drosomous), the "Agence Nationale pour la Recherche" (ANR-2010-JCJC-1401-01, ANR-25-CE20-7404-01). A patent (PTC/EP2020/075386) and an operating license of the technology are based on these findings.

S2-P21

Characterization of an olfactory-specific extracellular Glutathione Transferase involved in odorant degradation in the moth *Spodoptera littoralis*

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Glutathione transferases (GSTs) are ubiquitous multifunctional enzymes involved in detoxification, facilitating the elimination of xenobiotics such as insecticides and plant compounds in insects. Although primarily expressed in detoxification organs, GSTs are also found in antennae, suggesting additional roles in chemosensory processes. Olfaction underlies key behaviors essential for insect survival. Odor detection occurs in sensilla, where odorants are transported by odorant-binding proteins to receptors and subsequently degraded by odorant-degrading enzymes (ODEs), ensuring signal termination. Some antennal GSTs interact with odorants, indicating potential roles in odorant clearance and neuronal protection, although these functions remain poorly characterized. In the agricultural pest *S. littoralis*, transcriptomic analyses revealed a broad repertoire of ODEs, including 33 antennal GSTs⁵. Among them, SlitGSTd2 is highly enriched in adult antennae. Here, we combined molecular, biochemical, and structural approaches to investigate its function. SlitGSTd2 undergoes alternative splicing, generating two isoforms, one containing a signal peptide suggestive of secretion. Experimental data confirmed that both isoforms are expressed in antennae and that one is secreted, an unexpected property for a GST. In situ hybridization localized its expression to olfactory sensilla in both larvae and adults. Functional assays showed binding and enzymatic activity toward various odorants, while structural analysis revealed distinctive features. Altogether, our results support a model in which certain antennal GSTs are secreted into the sensillar lymph, where they contribute to odorant inactivation or sequestration, thereby modulating olfactory signaling. This work advances our understanding of insect olfaction and identifies GSTs as potential targets for disrupting chemoreception in pest control strategies. Funding: INRAE SPE-AAP project 2024

S2-P22

Neuroethology of taste in the honey bee *Apis mellifera*: from behaviour to the neural circuits

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The honey bee *Apis mellifera* is a traditional model for the study of sensory perception, information processing, and learning abilities. Surprisingly, this species' sense of taste remains largely understudied compared to the other sensory modalities. Nonetheless, honey bees' gustatory abilities are crucial for the survival of the species, influence food choices, and enable the avoidance of toxic substances. This work seeks to fill the gaps in the study of honey bee gustation by exploring the mechanisms underlying taste information processing through two complementary approaches: behaviour and neuroanatomy. First, through the adaptation of the Multi-Capillary Feeder assay (MultiCaFe), we inferred the bees' gustatory preferences to different tastants belonging to three gustatory qualities (salts, bitters, and amino acids) at multiple concentrations. Second, we developed a new set-up to perform appetitive associative conditioning experiments, in which bees associate gustatory stimuli with a sucrose reward. This allowed us to demonstrate that bees use taste quality to discriminate between stimuli, even if they have the same hedonic value. Finally, we performed a neuroanatomical study to describe the gustatory pathways in the honey bee brain by means of fluorescent tracer injections, both in the taste appendages (proboscis, antennae, and tarsi) and in the central brain. We showed that the spatial organization of gustatory appendages is centrally reflected in the distribution of gustatory afferents. The overlap of projections is notable among inputs from the same appendage origin (between mouthparts) and remains limited between those from different origins (antennae and mouthparts). In addition, the subesophageal-calycal tract was confirmed as the principal output pathway connecting the SEZ to higher brain centres. Together, these findings advance our understanding of the behavioural and neural mechanisms underlying food evaluation and gustation in *Apis mellifera*.

S2-P23

State-dependent modulation of decision making in olfactory-driven conflicting situations

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Internal states such as pregnancy and hunger have a strong impact on sensory preferences and decision-making in most animals. This becomes particularly apparent in situations of conflicting choice, i.e., when the need to approach food has to be weighed against avoiding a predatory threat, which requires the organism to prioritize one innate behavior over another. This prioritization is dynamically adjusted via neuromodulatory pathways based on the current state of the organism. Although critical for survival, it remains poorly understood how these state-dependent modulations are encoded at the cellular and neuronal circuit level. Here, we are using an olfactory-driven conflicting choice paradigm consisting of food odor, predator odor or a mixture thereof to examine how the states of hunger and pregnancy, either by themselves or in combination, affect decision making. Our preliminary data indicate that metabolic and reproductive state shift behavioral preferences, altering the balance between approach and avoidance. Together, this behavioral conflicting choice paradigm provides the framework to study how internal states dynamically regulate innate behavioral hierarchies at the cellular and neuronal circuit level.

S2-P24

Sticky Choices: How the non-clustered protocadherins shape accessory olfactory system circuitry**Nikki Dolphin^{1,2}**, Noah LeFever^{1,2}, Paolo Forni^{1,2}¹ University at Albany, SUNY² The RNA Institute at University at Albany, SUNY

The rodent accessory olfactory system (AOS), comprising the vomeronasal organ (VNO) and the accessory olfactory bulb (AOB), detects pheromones that drive social behaviors. Vomeronasal sensory neurons (VSNs) are divided into V1R neurons in the apical layer, which project to the anterior AOB, and V2R neurons in the basal layer, which project to the posterior AOB. Although several guidance cues and Kirrel adhesion molecules are known to organize these projections, the function of non-clustered protocadherins has not been studied. Results: Single-cell RNA sequencing of the VNO revealed that *Pcdh7* is predominantly expressed in V1R neurons, whereas *Pcdh9* is mainly found in V2R neurons. This suggests they may play roles in their connectivity. Loss of *Pcdh7*, *Pcdh9*, or both did not statistically affect overall VSN or mitral cell number, but it did significantly alter glomerular number and size, showing defects in AOB organization. Notably, *Pcdh7* knockout mice showed ectopic apical VSN projections into the posterior AOB, similar to the phenotype observed in *neuropilin-2* mutants. While buried cookie test demonstrated absence of olfactory defects, the loss of *Pcdh7* or *Pcdh9* significantly altered VNO-dependent stereotypical behaviors eg. male v male aggression and opposite sex odor preference. 123Cre mediated conditional deletion of *Pcdh7* in VSNs mimicked the AOB defects seen in full knockouts, supporting its cell-autonomous role in axon targeting and partner selection. Together, these findings suggest that *Pcdh7* and *Pcdh9* are crucial for guiding VSN axons to the appropriate AOB region, influencing glomerular map organization and the social behaviors of rodents. All experiments preformed with N>3. 2R01HD097331(PEF)NICHD; R01HD114827(PEF)NICHD

S2-P25

Neural correlates of olfactory training-induced improvement in odor-hedonic perception and emotional behavior in mice subjected to early-life adversity

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Early-life adversity, such as childhood maltreatment, is a major risk factor for the later development of psychiatric disorders, particularly depression. Beyond emotional alterations, early life adversity also affects the perception of the pleasant odorant through morphological and functional alterations within the olfactory-reward circuitry. This olfactory deficit may further contribute to emotional disturbances and reduced quality of life. Interestingly, olfactory training has been shown to have a beneficial effect on olfactory capacity and emotional well-being. In this context, we tested whether this odor training can rescue olfactory hedonics, emotional behavior and restore altered activity within the olfactory-reward network. To address this question, we used a transgenic mouse model (TRAP2), which enables the tagging of neurons activated via c-Fos expression, allowing the assessment of both odor perception and neuronal responses to the same odorant at two distinct time points, before and after 40 days of olfactory training. We quantified c-fos expressing neurons across thirty brain regions involved in olfactory, limbic, and reward circuits and analyzed patterns of correlated activity within this network in relation to odor-evoked responses and behavioral measures. We revealed that olfactory training improved perception of pleasant odorants as well as emotional and motivational behavior in mice subjected to early-life adversity. We further examined how neural activity and connectivity within the olfactory-limbic and reward systems evolve in parallel with the recovery of odor hedonic value and identify brain regions contributing to these improvements. These findings provide insight into how sensory experience reshapes olfactory and affective circuits altered by early-life adversity and support the therapeutic potential of olfaction-based interventions. This project is funded by French ANR project “COBRA” (APAFIS #38354) and Lyon 1 University doctoral fellowship.

S2-P26

Flavour-specific mapping of food odours in the insular cortex

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Flavour perception arises from the integration of gustatory and olfactory signals, yet how learned taste-odour associations are represented in the brain and translated into appetitive behaviour remains poorly understood. Here, we combine olfactometry, gustometry, psychophysics and functional neuroimaging to examine whole-brain responses to an array of orthonasal food odours and their associated tastes. Healthy volunteers (N = 39) rate sweet and savoury liquids of a range of concentrations on their pleasantness and intensity. They then continue to rate food odours on their perceived sweetness and savouriness as well as wanting to eat. During the functional neuroimaging component, participants receive oral taste stimuli and orthonasal odours while they undergo fMRI scanning. Behavioural modelling indicates that perceived sweetness and savouriness of food odours exert orthogonal effects on resultant appetitive behaviour. Quality control of the fMRI data shows robust activation of the piriform cortex in response to odours compared to the clean-air explicit control condition. Multidimensional scaling of the consensus representational dissimilarity matrix of ensemble activation patterns in the agranular insular cortex shows flavour-specific clusters of patterns, with savoury odours displaying the most similar ensemble patterns. This flavour-specific mapping was not observed in the piriform cortex or lateral orbitofrontal cortex. Compromise quality of the agranular insula is significantly higher than that of the piriform and comparable to that of a higher-order processing system such as the orbitofrontal cortex. Taken together, these results provide evidence of the involvement of the agranular insula in retrieving the flavour identity of orthonasal odours, thereby providing a potential target for behavioural and neurostimulation interventions aimed at modulating eating behaviour.

S2-P27

Infrared thermography as a physiological indicator of the impact of human emotional odors on sheep (*ovis aries*).

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Positive human-animal relationships can elicit positive emotions and welfare in animals. Few animal species are able to perceive human emotional odors and assessments of this ability mainly relied on behavioural measures. This study aims to test infrared thermography as a non-invasive tool, complementing behavioural measures, to detect physiological responses in sheep exposed to human emotional odors. The experiments were conducted on French commercial sheep farms. Thermal images of the left eye of 64 ewes were taken before (baseline) and after exposure to human emotional odor. Three human emotional odors were used (joy, neutral, stress) and each odor was tested on a different day to avoid any olfactory contamination of the environment. These odors were collected from human sweat using cotton pads during emotional-eliciting films. Odors were presented to the ewes during 30 seconds. Stress odor induced an immediate and sustained increase in ocular temperature (Mean $\Delta = +0.58 \pm 0.56^\circ\text{C}$ at 15 seconds after stimulus exposition, Paired t-tests: $p < .001$, with significant differences across all phases up to 135 seconds following odor exposure, $p < .01$). The odors of joy, neutral, and control (cotton pad without emotional odour) produced no significant effect on ocular temperature. These results demonstrate that human stress odor, but not neutral or joy odors, triggers a rapid and sustained hyperthermic response in ewes, a pattern consistent with an activation of the sympathetic fight-or-flight response linked to threat perception. These findings suggest that ewes may have exhibited a form of emotional contagion induced by the exposure to human stress odor. More generally, the results indicated that infrared thermography can be a reliable physiological parameter to assess emotional responses to odors in sheep. This work was funded by a PhD fellowship.

S2-P28

The olfactory processing of social cues in females is highly influenced by prolactin

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Social behaviors result from neuroendocrine responses driven by olfactory system (OS) cue processing. Both OS, the accessory and the principal, express prolactin hormone (PRL) receptors (PRLR). To understand the role of PRL on this processing in female mice we aimed 1) to reveal the expression of PRLR within the olfactory bulb (OB) during sexual maturation (puberty onset -PO-, maturity -SM- and adulthood -A-), 2) to analyze the behavioral response when exposed to male stimuli after acute PRL, the OB activation, and the intracellular pathway involved; and 3) to determine critical periods of PRL actions in short- and long-term olfactory function by using PRL and dopamine receptor agonist (cabergoline -Cab-) and antagonist (sulpiride -Sulp-) treatments (tx) in juvenile and prepubertal periods. Our results show that in 1) PRLR expression is constant in MOB but decreases in A in AOB; 2) behaviorally, acute PRL augmented the exploration of male stimuli, and increased activation (cFos) of the main OB (MOB) mitral cells (MC), and the accessory OB (AOB)-MC. ERK pathway was enhanced after odor exposure only in MOB. In 3) tx with PRL and Cab during a juvenile period altered the OB cell response (cFos) after exposure to male stimuli, impairing the AOB granular cells (GC) and increasing the MOB-MC with both tx. Finally, all females at SM discriminated female vs male odors. However, at A, only the ones with Sulp and Cab tx maintained the discrimination capacity; olfactory preference for female, gonadectomized (GDX) males and intact males stimuli among all groups was similar during SM and A; when compared between stages, differences were evident in A only in Cab and Sulp, where Cab increased the preference for females vs GDX males, and Sulp preferred intact vs GDX males. Overall, our results confirm that PRL participates in the processing of social cues, probably acting during specific maturational periods by recruiting both OS. Support: UNAM-DGAPA-PAPIIT IN214822, IN220025.

S2-P29

Stimulus temperature changes the psychophysical dimensions of salt taste.

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Temperature is a fundamental yet understudied parameter of gustatory experience. In this study, we designed a set of experiments to examine how stimulus temperature affects the motivational and sensory-discriminative properties of sodium (NaCl) and non-sodium (KCl) salts. In brief-access tests, water-deprived mice consistently preferred cool (14 °C) over warm (36 °C) stimuli regardless of salt content. Progressive ratio tests confirmed cool stimuli possess greater appetitive efficacy, suggesting that, under our experimental conditions, water-deprivation may confound temperature preferences in brief-access procedures. In contrast, operant conditioning tasks revealed robust temperature effects on sensory-discriminative function. In a two-response taste detection task in which mice were trained to distinguish each salt from water, enhanced sensitivity to both NaCl and KCl was observed at 36 °C compared to 14 °C, with detection thresholds significantly lower at warm temperatures. Critically, in discrimination tasks where mice were required to distinguish between NaCl and KCl with stimulus intensity controlled by anchoring concentrations at their respective detection thresholds, mice showed significantly impaired discrimination performance at 36 °C compared to 14 °C, indicating that the qualitative signatures of the two salts become perceptually more similar at warmer temperatures. These findings demonstrate that temperature profoundly modulates the sensory-discriminative properties of salts: while warm temperatures enhance detection sensitivity, they simultaneously reduce qualitative discriminability between sodium and non-sodium salts. This temperature-induced perceptual similarity suggests that KCl may serve as a more effective NaCl substitute in foods served at warmer temperatures, with potentially important implications for sodium reduction strategies in human nutrition. This research was supported by NIH award number F31DC021630, T32DC000044 and R01DC019326.

S2-P30

The nasal cycle as a marker of central nervous system activity

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Nasal airflow is asymmetrical, due to obstruction by erectile tissue leading to greater airflow in one nostril over the other. The greater airflow nostril shifts periodically approximately every 2.5h, in an ultradian rhythm known as the "nasal cycle" (NC). Nasal airflow asymmetry influences olfactory perception and attention, but how it relates to brain activity beyond olfaction remains unclear. To assess this non-olfactory relation between the NC and brain activity, participants engaged in two sessions of a visuospatial decision task, followed by 8- minutes resting state. Electroencephalography, Electrocardiography and stereo nasal respiration were obtained during the experiment. A significant negative correlation emerged between the difference in NC amplitude, and P300 ERP in the visuospatial task: symmetrical nasal breathing was associated with a higher P300 peak ($r = -0.48$, $p = 0.006$), and a shorter latency of this component ($r = 0.37$, $P = 0.04$) ($N = 31$). Furthermore, symmetric breathing was associated with higher interhemispheric alpha (8-12 Hz) connectivity (wPLI debiased) during resting state. Cluster permutation analysis detected significant connectivity between parietal and frontal electrodes between hemispheres ($N = 33$). The results suggest that NC fluctuations are linked to alternating brain activity, and point to the NC as a potential marker of brain activity and interhemispheric connectivity. These results highlight the question of whether the NC is an olfactory adaptation that influences non-olfactory brain activity, or a non-olfactory brain process that influences olfaction. This question remains unanswered.

S2-P31

The lingual GLP-1 system modulates sweet taste detection and glucose metabolism.

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Glucose is an important nutrient for the body, and is mainly provided by diet. When glucose-containing foods are ingested, it induces in the gustatory system a sweet taste perception and a cephalic response of insulin release that has an important role in the regulation of glucose metabolism. These glucose-related gustatory responses are independent on the canonical sweet taste receptor, suggesting the involvement of an alternative glucose-sensing mechanism not well-described yet. The hormone glucagon-like peptide 1 (GLP-1), known to be secreted by the intestine in response to nutrient intake and to have a role in glucose homeostasis, has been located in taste cells, as well as its receptor GLP-1R. The aim of this study was to investigate the role of the lingual GLP-1 system in gustatory function related to glucose. A virogenetic approach was used to reduce the expression of GLP-1R specifically in the taste system of sugar naïve and sugar-exposed adult mice. Gustatory responses to sweet solutions were assessed using brief-access taste tests. The regulation of blood glucose and insulin levels was explored using oral and intraperitoneal glucose tolerance tests. The results show that lingual GLP-1R downregulation impairs the ability to perceive glucose, and induced a basal hyperglycemia in sugar naïve mice. In sugar-exposed mice, the first results indicate that the experience with the post-ingestive effects of glucose and fructose modifies the gustatory functions of GLP-1R. Overall, the lingual GLP-1 system seems to have a role in gustatory functions associated with glucose, that is dependent on the diet experience. This project is supported by the Human Nutrition and Food Safety division of INRAE and by the French National Research Agency (ANR).

S2-P32

Processing mechanisms of temporally complex odours across olfactory circuits**Shambhavi Phadnis¹, Tobias Ackels¹**¹ *Sensory Dynamics and Behaviour Laboratory, Institute of Experimental Epileptology and Cognition Research, University of Bonn Medical Center*

Olfactory stimuli in natural environments occur as turbulent plumes with complex temporal structure, including fluctuations in concentration, intermittency, and whiff duration shaped by a range of environmental factors, such as distance from the source and airflow dynamics. Recent evidence suggests that the mammalian olfactory system resolves such temporal features with high precision. Yet, the circuit mechanisms underlying their encoding, transformation, and refinement across processing stages remain poorly understood. To address this, we characterise how projection neurons in the olfactory bulb (OB) respond to odour plumes with precisely controlled temporal statistics, recreating sampling conditions encountered in naturalistic environments. Recordings are performed in head-fixed mice using multiple chemically distinct odorants, allowing us to assess whether representations generalise across chemical classes. We then dissect local circuit contributions by pairing extracellular recordings with optogenetic perturbations of OB inhibitory interneurons. We examine how this information is transformed in higher olfactory regions, including the piriform cortex and the lateral entorhinal cortex, using simultaneous multi-site extracellular recordings combined with selective silencing of feedback projections to the OB. Together, this approach allows us to identify which temporal features of odour plumes are conserved, reformatted, or filtered across different processing stages, and to generate predictions about which features drive behaviourally relevant computations, including odour source localisation and navigation in complex environments.

S2-P33

Sensory dysfunction in the 3xTg-AD mouse model: Investigating the degeneration of visual and olfactory systems in early Alzheimer's Disease

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Alzheimer's disease (AD) is the leading cause of dementia and the fifth leading cause of death worldwide. Its distinctive pathological features include beta-amyloid (A β) plaques and intracellular neurofibrillary tangles of hyperphosphorylated tau protein. AD is a progressive neurodegenerative disorder primarily characterized by cognitive deficits and memory loss. However, emerging evidence suggests that sensory systems are affected long before the onset of hallmark clinical symptoms. Among these, visual and olfactory impairments have been identified as potential early biomarkers of AD progression, reflecting underlying neuropathology within the brain's sensory circuits. Objectives of the study: To better understand the progression of AD-related neurodegeneration in the 3xTg-AD mouse model, that exhibits both A β and tau pathology, we examined the mechanisms that drive neurodegeneration in the individual sensory systems: the visual and olfactory systems. Methods: Initially, behavioral characterization, including fear-conditioning, odor discrimination, and multi-sensory tests, are performed on control mice at ages corresponding to the neurodegenerative stages of the 3xTg-AD mice, in which the accumulation of beta-amyloid has been observed. Results and conclusions: A decline in associative and working memory was found to correlate with sensory impairments in aged animals compared to young controls. Then, we will evaluate the sensory integration together with cognitive impairment in 3xTg-AD mice performing different behavioral tests in response to different stimuli. Indeed, 7-tesla magnetic resonance and in vivo electrophysiology, will be performed to study brain morphology and neural activity in controls compared to 3xTg-AD mice. Subsequently, to further dissect the sensory system impairments, we compared the retinal layers atrophy observed in AD patients using Optical Coherence Tomography, together with odor discrimination tests, with the data obtained from 3xTg-AD.

S2-P34

Data-driven exploration of functional domains and functions of human olfaction

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The functions of olfaction are commonly described through major behavioral roles such as ingestive behavior, hazard avoidance, and social communication. To move beyond a strictly adaptive perspective, we propose a framework distinguishing functional domains, broad spheres of activity in which olfaction is involved meaningfully, from functions, defined as the underlying mechanisms engaged in these domains. This study aims to test this literature-based framework using a large-scale approach. A survey was conducted in France with 435 participants (345 women, 88 men, aged 18-82), comprising 242 normosmics and 193 olfactory experts, combining Likert-scale ratings and open-ended responses on domains and functions of olfaction. Data were analyzed using multivariate statistical methods (Exploratory Factor Analysis) and topic modeling (BERTopic) enhanced with a large language model (LLM). Results confirm the existence of multiple latent dimensions underlying olfactory functions and functional domains. Beyond the classical domains, additional dimensions emerge encompassing medicine, navigation in the environment, spirituality and the experiential and aesthetic qualities of odors. These dimensions differ in their empirical prevalence, reflecting heterogeneous expression across individuals. Olfactory profiles are associated with specific dimensions, particularly those not directly linked to core adaptive functions. These findings provide initial support for the relevance of this framework in explaining the individual differences in olfactory perception and profiles. Future work will integrate intercultural data and further validate the model empirically. This study is a part of Typhaine de Simone's PhD project, funded by an ANRT CIFRE research program between Université Claude Bernard Lyon 1 and MANE.

S2-P35

Cross-modal associations between odors and intranasal sensations

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Odor perception results from the integration of odorants through the nasal cavity. This study aims to determine whether there are systematic correspondences between odorants and the intranasal sensations produced during sampling, and how these sensations are associated with perceived intensity and emotional responses. Thirty participants underwent an experiment involving 20 odorants. Each trial included perceptual ratings of five intranasal dimensions (temperature, tactility, duration, spatiality and temporality), together with intensity and hedonicity (5-point scales) and autonomic responses. A Principal Component Analysis separated perception of tactility, duration, spatiality and temporality (PC1, 63.5% variance) from temperature (PC2, 20.9%). Hierarchical clustering on these PCs identified 4 clusters: 1/ Low ratings (e.g., peach, raspberry) (pleasant, low intensity), 2/ Ratings close to the mean (e.g., lavender, coffee) (neutral), 3/ High perceptual ratings (e.g., garlic, onion) (unpleasant, high intensity), 4/ Low temperature, high duration (peppermint, eucalyptus) (freshness). Significant differences between clusters were found for all intranasal descriptors. Furthermore PC1 correlated positively with perceived intensity ($r=0.89$, $p<0.0001$) and negatively with hedonicity ($r=-0.76$, $p<0.0001$), while PC2 showed a negative correlation with hedonicity ($r=-0.48$, $p=0.03$). Finally, PC1 showed a negative correlation with expiratory respiratory volume. Overall, these results suggest that intranasal sensations can be associated with certain categories of odors, thereby paving the way for the use of such associations in paradigms involving the sensory substitution of smell with intranasal sensations produced, for example, by the trigeminal system. This Project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 964529

S2-P36

Olfaction: a key player in cognitive development and degeneration

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The olfactory system is anatomically interconnected with brain regions essential for cognition. In recent years, dysfunction in the sense of smell has been increasingly recognized as a hallmark feature shared among several neurological conditions, including both neurodevelopmental and neurodegenerative disorders, while the neural mechanism is largely unclear. We found that the Alzheimer's disease (AD) mouse models had impaired olfactory behaviors even in the absence of AD pathological marker beta-amyloid deposition in the brains or memory deficits. At the same stage, the number of mature olfactory sensory neurons on the olfactory epithelium decreased, indicating there is a reduced input from the olfactory periphery to the olfactory bulb (OB). Consequently, the decreased oscillatory synchronization between the OB and its downstream region piriform cortex further confirms a functional dysfunction of the olfactory neural circuit before AD onset. These findings suggest that the concomitant dysfunction of both olfactory behavior and gamma oscillations has important implications for early AD pathogenesis. When it comes to the very early stage of life, the olfactory ability of neonates is much more mature compared to the other senses, but the long-lasting impact of olfactory inputs on brain development and cognitive functions remains unknown. After experiencing a transient inhibition of the olfactory synaptic inputs shortly after birth, mice had declined performance in cognitive tasks later in life, accompanied by a persistent reduction in neural network activity and dendritic complexity in the limbic system along development, indicating that olfactory processing during a specific "critical window" is crucial for the development of the higher-order brain networks and the maturation of cognitive abilities. In summary, olfactory dysfunction is one of the common denominators predicting cognitive decline.

S2-P37

Gastric bitter taste receptors differentially regulate essential and toxic metal ion homeostasis

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Bitter taste receptors (TAS2Rs) are traditionally associated with oral chemosensation and food avoidance. Emerging evidence indicates that TAS2Rs expressed in the gastrointestinal tract function as integrative sensors linking luminal chemical cues to gastric physiology. This work reports novel insights into the receptor-specific roles of gastric TAS2Rs in regulating metal ion homeostasis. Using an immortalized human gastric cell model, we demonstrate that activation of distinct TAS2Rs elicits highly specific and divergent functional responses. TAS2R43, e.g., has been identified as a previously unrecognized co-regulator of gastric zinc homeostasis, revealing a dual role for this receptor in coordinating proton secretion as an essential part of gastric digestive function with essential trace element balance [1]. In contrast, activation of TAS2R8 and TAS2R10 differentially modulated cadmium uptake, indicating receptor-dependent control of gastric ion transport and toxicant handling [2]. Our findings challenge the traditional view of bitter taste receptors as redundant aversive sensors and instead establish gastric TAS2Rs as functionally specialized regulators of gastric metal handling. The differential regulation of essential versus toxic metal ions further suggests a role for TAS2Rs in adaptive gastric defense mechanisms. Collectively, this work defines a novel functional framework for gastric bitter taste receptors, with translational relevance for the development of strategies aimed at modulating micronutrient bioavailability, and protection against dietary metal-based toxicants.

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S2-P38

Shifted perceptual space across major datasets in olfaction research

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Computational efforts linking odorant structure to perception rely primarily on three large public datasets: Dravnieks (1985; 144 monomolecules, 146 descriptors), Keller/Vosshall (2017; 480 monomolecules, 23 descriptors), and Bierling/Croy (2021; 66 monomolecules, 22 descriptors). Whether these datasets reflect a shared olfactory perceptual space remains unknown. To address this, we applied principal component analysis (PCA) to each dataset. In all cases, the first component captured $\geq 30\%$ of the variance (Dravnieks = 30.8%, Keller/Vosshall = 37.9%, Bierling/Croy = 34.1%), and the first four components together captured over half (Dravnieks = 56.4%, Keller/Vosshall = 76.7%, Bierling/Croy = 80.2%). We then computed pairwise Euclidean distances in PCA space and cross-correlated them across datasets, using each pair's full intersection of shared molecules (28 of which were common to all three). Correlations were modest: Dravnieks $\leftrightarrow$ Keller $r=0.52$ ($N=62$), Keller $\leftrightarrow$ Bierling $r=0.44$ ($N=65$), Dravnieks $\leftrightarrow$ Bierling $r=0.26$ ($N=28$; all Mantel $p \leq 0.05$). Repeating the analysis in raw descriptor space yielded similar results (all Mantel $p \leq 0.01$). These relatively low correlations may reflect limitations of verbal descriptors or genuine differences across datasets. To distinguish between these possibilities, we split the Keller dataset and repeated the analysis across halves. This yielded substantially higher correlations ($\sim r=0.7$; 1000 splits), roughly double those observed between datasets. These findings suggest a consistent olfactory perceptual space within datasets, but a shifted space across datasets. Such shifts may arise from differences in participants (across decades and cultures), stimulus delivery, descriptor usage, or molecular composition. Regardless of origin, these differences highlight an important constraint for models trained and tested across datasets.

S2-P39

The natural statistics of human olfactory experience: A multi-national project

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How many distinct odors does a typical human encounter throughout a typical day, and what are these odors? What are the sources of variance underlying this question? In this work we present an ambitious multi-center project (Israel-Sweden-Germany), funded by an ERC Synergy grant. We built a cellphone app that, at random times, sends users a brief questionnaire. It opens with two questions - "Do you currently smell anything?" and "Were you aware of the odor before we asked?" - followed by rating scales for intensity and pleasantness, and an option to photograph the source. By probing 24,000 participants 4x/day we will obtain ~1 million ratings. So far, 3,200+ participants across nine countries (Nigeria, Sweden, Israel, Germany, USA, UK, Italy, Kenya, Canada) have produced 109,000+ ratings. Participants reported smelling something at 64% of probes and were consciously aware at 47%; given a smell, awareness reached 74% - most odors do reach consciousness. Reported smells skew strongly pleasant (mean ~70/100), and counter-intuitively, pleasantness rises with intensity ($\rho=+0.39$): the most intense odors are typically also the most pleasant. Food and food-adjacent terms (cooking, homemade food, meal) dominate descriptors, followed by coffee and perfume. Awareness peaks at midday and evening, plausibly entrained by meal times; outdoor moments yielded 1.4x higher awareness than indoor (62% vs 44%). Awareness varied 2.4-fold across countries (25% in Italy to 59% in Nigeria), with distinctive cultural fingerprints - gingerbread and candles in Sweden, markets and restaurants in Nigeria. Men reported more aware moments than women ($\chi^2(1)=1156$, $p<.001$) and people in their 30s peaked highest ($\chi^2(5)=548$, $p<.001$), though both effects are heavily affected by Nigeria's share of the sample. Olfactory experience is also temporally clustered: having just smelled something, the chances of consciously smelling rises from 47% to 76%. We are actively expanding to reach many more countries.

Session 3 — 17 September 2026, 12:30-14:00

S3-P1

Investigating the activation mechanism of insect odorant receptors

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Abstract: Insects detect odors through odorant-activated ion channel complexes that are composed of an odorant receptor (OR) subunit and an odorant receptor co-receptor (Orco) which are thought to exist in a 1:3 OR/Orco ratio in vivo^{1,2}. Recently published cryo-EM structures of OR/Orco complexes from mosquitoes with bound known agonists (AgamOR28 from *Anopheles gambiae* with 2,4,5-trimethylthiazole, and AegOR10 from *Aedes aegypti* with o-cresol^{1,3-4}), showed a displacement of the transmembrane helix 7b (TM7b) of the ORs which allows ions to flow through the ion channel. To elucidate the mechanism of this displacement, we utilized molecular dynamic (MD) simulations using Amber force fields⁵ and the MD code Gromacs⁶. These simulations were subsequently used as a basis to investigate an OR with unknown structure, namely SpicOR from the lily moth *Spodoptera picta*, revealing binding pocket dynamics that can be influenced by a single amino acid substitution, in link with change in OR sensitivity.

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S3-P2

From sensors to senses: a longitudinal study of personal pollution exposure, olfactory threshold, and well-being

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Air pollution exposure has been shown to reduce olfactory sensitivity and negatively affect mental health, yet most studies rely on outdoor stationary monitoring data that do not reflect individual exposure in real-world environments, nor do they account for indoor pollution. This longitudinal study assessed personal air pollution exposure, psychological well-being, and predicted olfactory sensitivity among participants during the winter and summer seasons. The participants constantly wore personal air quality sensors for 10 consecutive days, capturing exposure in both indoor and outdoor environments across two seasonal measurement waves (summer and winter, approximately six months apart). Olfactory sensitivity was measured using Sniffin Sticks threshold test, and psychological well-being was measured with validated questionnaires. The Atmotube PRO monitors two distinct classes of pollutants: particulate matter (PM1, PM2.5, PM10), representing airborne solid and liquid particles (e.g., dust, smoke), and total volatile organic compounds (TVOC), reflecting a broad, non-specific measure of gaseous organic pollutants (e.g., solvents, cleaning agents, formaldehyde, benzene), detected using a metal-oxide sensor. For the purposes of statistical analyses, PM2.5 was used as the primary particulate matter indicator due to its strongest model fit. Importantly, VOC and PM represent fundamentally different physical and chemical pollutant categories and are measured independently by the device. Preliminary mixed-effects analyses indicate that higher VOC exposure is associated with lower olfactory sensitivity, with seasonal variation in this relationship indicating that elevated levels of VOCs in the winter accelerate the decrease in olfactory sensitivity. No significant effects were observed for PM2.5. Olfactory thresholds decreased with age, while psychological well-being was not a significant predictor of olfactory sensitivity.

S3-P3

Sensory Drivers of Oral Nutritional Supplement Acceptability in Restrictive Anorexia Nervosa and Restrained Eaters

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Oral Nutritional Supplements (ONS) play an important role in supporting nutritional rehabilitation in restrictive anorexia nervosa (AN). Nevertheless, their clinical utility is frequently limited, as ONS can elicit feelings of disgust, leading to treatment compliance issues. Despite this, the impact of ONS sensory properties on their acceptability, especially taste and flavoring agents, remains largely unexplored. As AN is associated with food restriction, the investigation of ONS perception among individuals with restrictive eating behavior but without eating disorder is of interest to determine whether their perceptions and acceptability differ according to specific population. The aim of the present study was to identify the barriers and facilitators influencing ONS overall acceptability among individuals with AN and Restrained Eaters (RE) with a focus on sensory, cognitive, emotional, and motivational factors. Focus groups were conducted as semi-structured discussions, allowing participants to explore their expectations, needs, opinions, and motivations regarding ONS consumption. Sessions involved ONS samples and plant hydrosol samples to smell. The discussions were audio-recorded, verbatim transcribed and analyzed using qualitative thematic analysis. Preliminary results showed that RE reported disgust and unpleasant odors and tastes as major barriers, with a negative affective response toward “artificial” odors. For AN, the stigmatization due to medical packaging, perceived heaviness during consumption, and calories contents are the major barriers. Lack of aromas diversity and formats were mentioned. Both groups found that scents from hydrosol could be interesting in ONS juice form and can be used to suit dietary preferences and seasons. More “natural” colors were preferred. The findings provide in-depth insights into patient perceptions of ONS and may guide the development of patient-centered formulation strategies. Project funding: ANR AROMA n°20-CE21-004.

S3-P4

Representation and transformation of temporally complex odours in the mouse olfactory bulb

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In natural conditions, olfactory cues are often encountered as dynamic odour plumes whose temporal profile carries information about the environment. Recent studies have shown that the mouse olfactory system can access high-frequency information from such temporally complex odours (TCOs), despite limitations in sampling rate and signal transduction kinetics. However, the relevant temporal features, as well as the local and centrifugal circuit motifs involved in processing them, remain unclear. Here, we investigate how TCOs are represented and transformed in the mouse olfactory bulb (OB), the first odour-processing centre in the olfactory system. To this end, we perform two-photon Ca^{2+} imaging in OMP-GCaMP6f mice, allowing us to record input activity from olfactory sensory neurons at the glomerular layer of the OB. Because neural activity in the OB is linked to the breathing rhythm, we use a fast odour delivery device with millisecond precision to present stimuli at defined phases of the breathing cycle and probe neural correlates of specific temporal features. We find that responses to paired odour pulses separated by 10–250 ms, spanning a single sniff cycle, depend on the sniff phase of each pulse. Odour plumes often contain multiple odorants with distinct temporal dynamics. In this context, we show that the sequence of paired pulses of different odours can be predicted from glomerular population responses, even at pulse intervals as short as 10 ms. We also use binary odour mixtures to explore the interaction between stimulus composition and temporal features. We further compare these glomerular input responses with output activity recorded from OB projection neurons in Tbet-GCaMP6f mice to assess how feature representations change from input to output within the OB. Together, our results demonstrate that the OB encodes fine-scale temporal features of olfactory cues and provide insight into how these dynamic sensory inputs are transformed for downstream processing.

S3-P5

Investigating the effects of olfactory training in individuals with specific anosmia to R(-)-carvone

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Background and Objectives: Specific anosmia (SA) describes a markedly reduced sensitivity to an odorant despite otherwise normal olfactory function. While olfactory training (OT) has been shown to improve olfactory performance in individuals with olfactory loss, its efficacy in individuals with SA remains poorly understood. This study examines whether OT in individuals with SA to R(-)-carvone is associated with changes in sensitivity to the odor and general olfactory function. **Methods:** Odor thresholds for R(-)-carvone were assessed in adult healthy individuals (N = 102) to define SA as inability to detect the odorant at a concentration 100-times greater than the concentration perceived by 50% of the sample. Individuals considered to have SA (n = 18) underwent three months of OT, which consisted of twice-daily exposure (30 seconds each) to three suprathreshold odorants: R(-)-carvone, eucalyptol, and phenylethyl alcohol. Psychophysical testing (Sniffin' Sticks extended test and odor-specific thresholds) was performed before and after training. Compliance was measured using a modified four-item compliance score and the individual significance of olfaction was evaluated using a 20-item questionnaire. **Results:** There were no significant changes in specific threshold levels for R(-)-carvone, eucalyptol, and phenylethyl alcohol. There were no significant changes in TDI scores. **Conclusion:** In a sample of healthy individuals with SA to R(-)-carvone, three months of OT did not significantly improve the sensitivity to the trained odor or general olfactory performance. These findings suggest that individuals with SA and otherwise normal olfactory function may show a limited response to OT compared to individuals with general olfactory loss, where OT has shown to be effective. Future studies with larger sample sizes are needed to confirm these findings. **Funding:** This study was supported by a grant from the Deutsche Forschungsgemeinschaft (DFG HU441/29-1).

S3-P6

Sequence analysis and modeling of vomeronasal type-2 receptors (V2Rs) in mammals

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Semiochemicals are chemical signals perceived by animals in their environment, they play a key role in chemical communication. These signals include small molecules to larger proteins and can induce behavioural or physiological changes in animals. Semiochemicals are perceived by different chemosensory receptors (CRs), including vomeronasal type-2 receptors (V2Rs) in the vomeronasal organ (VNO), which is the focus of this study. V2Rs are thought to be specialized for the detection of non-volatile peptides and proteins. Despite their role in semiochemical communication, relatively few studies have been conducted on the V2Rs of various animal species to examine their structures. Here, we present the results of different sequence analysis on selected sequences of mouse, rat, rabbit and horse, in order to examine the conservation and characteristics of these sequences. We also present the structural modeling of these 25 selected sequences. Indeed, various computational tools have been used to estimate the secondary structures, the hydrophobicity, the conserved residues, the post-translational modifications and the orientation of residues. Comparison with evolutionary related receptors were also performed using calcium sensing receptors obtain from BLAST analysis. Finally, structural models have been obtained from TrRosetta and Chai-1 and studied dynamically. The analysis reveals the expected seven transmembrane helices (7TM) specific to G-protein coupled receptors (GPCRs) that show a high conservation. The Venus Flytrap (VFT) domain, including around the 500 first amino acids, is composed of series of helices and strand with a hydrophobic central region. Also, few amino acids have been identified as conserved across all V2R sequences but absent in evolutionary related receptors, they are probably involved in the specificity of V2Rs functioning.

S3-P7

Olfactory training shows no positive effect on olfactory working memory in older people**Barbara Żyżelewicz¹, Anna Oleszkiewicz^{1,2}, Michał Pieniak¹**¹ Institute of Psychology, University of Wrocław, Poland² Interdisciplinary Center Smell & Taste, Department of Otorhinolaryngology, Faculty of Medicine Carl Gustav Carus, Technische Universität Dresden, Dresden, Germany

Olfaction is closely linked to cognitive functioning, including processes such as working memory. In older adults, age-related decline in both olfactory and cognitive abilities raises the question of whether sensory-based interventions may support cognitive performance. While olfactory training (OT) is a well-established method for improving smell function, its potential cognitive benefits, particularly in relation to working memory in the olfactory domain, remain unclear. This study examined whether olfactory training may enhance olfactory working memory in older adults. The final sample included 101 healthy individuals aged 63 to 94 years (Mage = 72.6, SD = 5.44; 89% women). Participants in the experimental group completed a 12-week olfactory training twice daily using four odors, while the control group followed the same procedure with odorless stimuli. Olfactory working memory was assessed using tasks engaging memory processes within the olfactory modality, including odor concentration sorting and matching pairs of odors (smellory). The smellory task was completed by 84 participants aged 64 to 84 years (Mage = 72.1, SD = 4.89, women = 88.1%). For the sorting task, the omnibus model revealed no significant Time*Group*Odor interaction, $F(2,196) = 0.62$, $p = .537$, indicating no differential improvement across groups. Similarly, in the smellory task, the Time * Group interaction was not significant, $F(1,82) = 0.866$, $p = .355$, suggesting no effect of training on odor memory performance. These findings suggest that standard olfactory training may not be sufficient to enhance working memory processes related to olfaction in older adults. However, they contribute to ongoing research exploring new applications of olfactory training beyond sensory rehabilitation. Further studies are needed to determine whether extended or more intensive training protocols may yield cognitive benefits in aging populations.

S3-P8

Did Proust need to taste his madeleine? Oral cues offer no advantage over orthonasal olfactory cues in evoking autobiographical memories.

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When Marcel Proust (Swan's Way 1913) described his mental time-travel experience, he did not only sniff the madeleine cake after dipping it in the tea, but he also tasted it. Can the addition of oral cues, including gustation, somatosensation and retronasal olfaction evoke more Proustian memories than orthonasal olfactory cues alone? Here we used 4 different food objects (popcorn, tea biscuits, vanilla and peanut butter) and measured the change in evoked memories relative to a word cue for orthonasal olfactory cues (30 participants) compared to oral cues (30 participants) in an experimental design. The sample included 32 women and 28 men ($M = 23.81$, $SD = 4.37$, range = 18–40). The dependent variables included various experiential ratings regarding emotionality, vividness, age of the memory and memory content. We observed that using oral cues confers no advantage over orthonasal olfactory cues. If anything, orthonasal cues are slightly better in terms of evoking emotion at the time of recall ($F[1,173]=9.420$, $p = .002$, $\eta^2_p = 0.052$). Self-reported detailedness was also higher for orthonasal olfactory cues ($F[1,173] = 11.271$, $p < .001$, $\eta^2_p = 0.061$), but content analyses based on word counts did not confirm this. In daily life it may be pleasurable to share memories around the dinner table, but when it comes to investigating autobiographical memories in a laboratory, there is no advantage to using oral cues. Funding source: This project is funded by the TÜBİTAK 1002-A program.

S3-P9

Long COVID, Smell and Taste Dysfunction in Adults: the 2024 U.S. National Health Interview Survey

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Background: Smell and taste alterations (including distortions like phantosmia and dysgeusia) have been linked to long COVID. **Methods:** The 2024 National Health Interview Survey (NHIS) included 32,629 U.S. adults aged ≥ 18 years. We estimated the prevalence of smell and taste alterations and distortions and their associations with long COVID, defined as symptoms lasting ≥ 3 months, including neurologic symptoms of fatigue, brain fog, headache, dizziness, and smell and taste changes. Smell alterations consist of smell difficulties last 12 months, worsened ability compared with age ≤ 25 , altered perception of odors, or persistent phantom smells. Taste alterations consist of taste difficulties last 12 months, worsened taste compared with age ≤ 25 , or persistent unwanted taste. **Results:** Long COVID prevalence was 14.0% (males, 11.3%; females, 15.9%). Smell alterations were reported by 21.9% overall and 46.2% among those with long COVID; taste alterations were 12.6% overall and 31.0% with long COVID. The prevalence of any smell or taste alteration was 25.8% overall (8.6% smell and taste both, 13.2% smell only, and 4.0% taste only) and 51.1% among those with long COVID (26.0% both, 20.1% smell only, and 5.0% taste only). Distortions were reported by 6.1% (smell) and 3.2% (taste) overall, increasing to 17.6% and 9.9%, respectively, among those with long COVID. Sensory alterations were most prevalent among non-Hispanic Whites and individuals aged 55–64 years. The percentage of individuals who had fully recovered their sense of smell (61.3%) and taste (68.0%) was lower among those with long COVID compared to those without (smell, 89.2%; taste, 91.7%). **Conclusions:** Smell and taste alterations are common and more prevalent among individuals with long COVID, many of whom experience persistent dysfunction and incomplete recovery. These findings highlight the need for greater clinical awareness and targeted interventions to reduce the burden of long COVID-related sensory impairment.

S3-P10

Olfactory predictors of generalized anxiety in a non-clinical sample of young adults.

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While the relationship between olfaction and depression is well characterized. Similar link between anxiety is understudied. However, these two domains are linked by neuroanatomy as olfactory signal project directly to the limbic system (including amygdala), which is main processing area for anxiety reactions. Our study aimed to find out whether odor identification, olfactory sensitivity, and odor awareness can predict the level of the generalized anxiety in a non clinical sample of young adults. Participants were tested with the odor awareness scale (OAS-6), odor identification test (Sniffin' Sticks), and odor detection threshold (Sniffin' Sticks). Olfactory detection threshold was assessed for either pleasant odor (phenyl ethyl alcohol, PEA) or unpleasant, burnt-like odor (2-furfurylthiol, FFT). Generalized anxiety symptoms were assessed with GAD-7 questionnaire. We tested a total of 206 participants aged between 18 and 39 (Mage=26.1; SD=5.4; gender: 121 women, 82 men, and 3 non-binary or did not specify). Using linear regression analysis, we found that older age predicted lower anxiety levels ($b=-.44$, $p<.001$), while gender was not a significant predictor ($b=.87$, $p=.379$). Odor identification ability was not related to anxiety levels ($b=.20$, $p=.338$) whereas odor awareness was a trend-level negative predictor ($b=-1.21$, $p=.053$). In the two subsamples who underwent threshold testing with different odors we found a similar significant, small relationship between increased olfactory sensitivity and higher generalized anxiety symptoms (PEA: $r=.23$, $p=.020$; FFT: $r=.21$, $p=.033$). Altogether, we demonstrated that people who experience anxiety symptoms have also lower detection threshold to different odors, but this does not translate into increased ability to identify them. The link between odor awareness and anxiety awaits further in-depth investigation. Funding: National Science Centre in Poland (2022/45/N/HS6/00651)

S3-P11

Probing how the temporal structure of odour plumes informs mouse navigation behaviour

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Understanding how mammals use olfactory information for navigating their environment is key to uncovering their natural behaviours. Nocturnal rodents, in particular, rely heavily on odour cues for foraging, finding mates, and avoiding predators. In nature, odour plumes are highly dynamic, with rapid temporal fluctuations that carry spatial information, such as distance to the source. However, despite their importance, there is limited research on how mammals interpret these complex temporal odour signals. Our study investigates how mice use dynamic olfactory cues for navigation by combining behavioural, respiratory and neural measurements. For this, we designed a large open-field arena (150x100 cm) with a multichannel odour delivery system that simulates naturalistic airflow patterns and provides multiple odour sources. Mice are trained in a reward-conditioning paradigm in which they learn to initiate trials and navigate to the correct target among four odour sources, over long distances and without visual cues. In our first experiments, we observed that mice successfully learned the odour localisation task (n=4; accuracy: $80.64 \pm 5.68\%$). Additionally, both trial initiation and decision-making times decreased over sessions, indicating improved efficiency and a refined behavioural strategy. We quantify behavioural and physiological parameters at high spatiotemporal resolution. 2D and 3D tracking captures fine movement kinematics and exploratory strategies. Real-time odour encounters are recorded using a custom miniature sensor that captures fine-scale plume dynamics, enabling us to link odour events to the corresponding behavioural responses. In addition, we monitor respiratory dynamics using an implanted thermistor probe. We further aim to record neural activity in olfactory bulb and lateral entorhinal cortex using silicon probes. Together, this approach will identify which features of naturalistic odours drive behavioural decisions and guide navigation in mammals.

S3-P12

Role of a Glutathione Transferase in the olfactory system of *Drosophila melanogaster*

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Glutathione Transferases (GSTs) are key detoxification enzymes mainly studied in insects for their role in the biotransformation of plant defense chemicals and insecticide resistance. Although GSTs are primarily expressed in the detoxification organs of insects, such as the fat body and digestive tract. However, several studies have shown that many of them are also associated with the chemosensory organs, suggesting potential functions in chemical communication, beyond detoxification. In *Drosophila melanogaster*, 36 cytosolic GST genes have already been identified. Using transcriptomics analysis, we characterized and compared the expression profiles of these GSTs in chemosensory tissues (male and female antennae, proboscis + maxillary palps, legs, head of larvae and ovipositor). Then we focused on a GST gene that showed specific expression in antennae and a possible association with olfactory neurons, as suggested by single-cell RNAseq data analysis. We also showed that this GST was secreted, which is quite unusual for this enzyme family. In order to elucidate the physiological role of this GST in olfaction, we used a GST mutant strain and performed electrophysiological analysis to test the antenna response to different compounds of various chemical classes in this mutant strain compared to a wild-type strain. At the same time, we produced the recombinant enzyme and performed enzymatic activity tests to identify the catalytic efficiency of the enzyme with regards to various odorant molecules. Overall, our results support a role of this GST in olfactory processes, and the mechanisms potentially involved will be discussed.

S3-P13

Rethinking olfactory imagery as a multidimensional experience

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Although visual imagery has been investigated extensively, olfactory imagery has received relatively less attention. Most previous studies have measured the vividness of olfactory imagery by focusing on concrete objects (e.g., smell of coffee). We aim to go beyond this and explore the broader phenomenology of olfaction. To this end, we examined three different types of olfactory experiences including concrete object odors (e.g. lemon), ambient environmental odors (e.g. bakery), and socio-affective odors (e.g., partner). There were 15 items for each odor type, written in an immersive style using the second person singular to enhance imagery. Beyond expanding the range of olfactory experiences, we also aimed to characterize the phenomenology of olfactory imagery across multiple dimensions. In addition to vividness of imagery, we measured the onset, intensity, persistence, and ease with which olfactory images can be evoked. Furthermore, the familiarity of olfactory items was assessed. One hundred participants rated each item on all 5 imagery dimensions on a 7-point Likert scale. Initial analyses established the reliability of the new questionnaire. Subsequent factor analyses examined whether olfactory imagery is best characterized as a unified construct or as multiple distinct dimensions. Overall, this study provides a framework for examining olfactory imagery in its full complexity, highlighting its potential multidimensional nature. Eléonore Giraudet is supported by an aspirant FNRS fellowship granted by Fund for Scientific Research (F.R.S.-FNRS), as well as by a fellowship granted by SofinaBoël Fund for Education and Talent.

S3-P14

Inflammation modulates neurogenesis in the olfactory epithelium of adult zebrafish following olfactory bulb lesions

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Zebrafish (*Danio rerio*) robustly regenerate neurons throughout their lifespan, making them an ideal model for studying neural repair. Particularly, their olfactory system, comprising the olfactory bulb (OB) and olfactory epithelium (OE), is highly neurogenic and analogous to that of mammals. Although neuroinflammation is known to support recovery in the zebrafish brain, the role of inflammatory cells in OE repair is understudied. This study investigates the role of inflammation in OE repair following OB damage. For this, we induced retrograde OE degeneration via unilateral excitotoxic OB lesions using quinolinic acid (QA) in adult zebrafish of both sexes and examined recovery at 1, 6, 24, and 72 hours post-lesion (hpl). Immunohistochemistry was used to assess inflammatory markers, stem cell activation, cell proliferation, and neurogenesis. To evaluate the functional role of inflammation, we used pranlukast, an anti-inflammatory drug that works as a cysteinyl leukotriene receptor antagonist. Our results show in the OE, cell proliferation and neurogenesis are observed as soon as 6 hpl, while leukocytic density was observed as soon as 1 hpl. Anti-inflammatory treatment with pranlukast significantly reduced leukocytic recruitment and ablated OE cell proliferation to control levels. We also found differences in the proliferative and neurogenic response across the ipsilateral and contralateral OE. In the OB, QA lesions induced leukocyte migration and astroglial activation in the ipsilateral OB; while pranlukast reduced this neuroinflammatory response, indicating that pranlukast is effective at ablating post-lesion inflammatory responses. This work provides a comprehensive time-course characterization of post-lesion neurogenesis in the OE of adult zebrafish. Our findings demonstrate that inflammation is essential for OE repair and highlight their contribution to neural repair in adult vertebrates. Funding: National Science Foundation (NSF) grant BRC-BIO 2313303

S3-P15

Newly identified amino acid substitutions in the C-terminal domain further improve functional expression of difficult-to-express ORs

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Humans have 391 odorant receptor (OR) genes. Due to their poor functional expression in heterologous cell systems, reproducible receptor-ligand pairs are known for only a minority and the pharmacology of human olfaction is still poorly understood. Using a well-expressed OR as template, the C-terminal domains of poorly expressed class I and class II OR were engineered by several rounds of mutations leading to dramatically increased sensitivity paralleled by increased cell surface expression. These optimized C-terminal domains facilitated high surface expression and strongly increased sensitivity of the assay for many ORs. However, even with this improvement, not all human ORs can be functionally expressed in vitro. Thus, we sought to find additional C-terminal domain variants that can further boost the cell surface expression of ORs. A total of more than 300 single amino acid substitutions in the C-terminal domain were screened for increased sensitivity of OR7C1. Different combinations of the most beneficial substitutions were tested with OR7A10 which we previously identified as the OR for molecules with a peppery odor character. For the best variants we observed an additional 30-fold increase in sensitivity which was paralleled by increased cell surface expression. This newly identified C-terminal variant did also increase the sensitivity of OR7C2 and OR10G3 towards Ambermax and ethyl vanillin, respectively and hence can be used to further increase expression of multiple difficult-to-express receptors.

S3-P16

Taste receptor–inflammation crosstalk predominates in gustatory tissue in a cafeteria diet model

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Introduction: In addition to their role in chemosensory perception, taste receptors serve as metabolic sensors in extra-orally. Obesity-linked inflammation has been associated with reduced taste receptor expression in gustatory tissue, yet whether these interactions extend beyond the oral cavity remains unclear. **Methods:** We investigated this question using a cafeteria diet model of obesity-associated metabolic dysfunction in C57BL/6 mice (18 weeks), with standard diet-fed animals as controls. Gene expression of Tas1r2, Tas1r3, CD36, Ffar1, GPR84, and Ffar4 across circumvallate papillae, jejunum, stomach and spleen was assessed using RT-qPCR. Morphological changes in fungiform papillae were assessed by quantifying papillae number and area. Tissue-specific correlation networks between taste receptor expression, fungiform papillae markers, inflammatory markers and cardiometabolic markers were determined. **Results & Conclusion:** Cafeteria diet resulted in obesity (92.1±2.4% higher body weight, $p<0.0001$), metabolic dysfunction, and systemic inflammation, accompanied by reductions in fungiform papillae number (−24%, $p<0.01$) and area (−36%, $p<0.05$). In circumvallate papillae, Tas1r2 was reduced by 85% ($p<0.05$), alongside CD36, Ffar1, GPR84, and Ffar4, while in jejunum, downregulation was restricted to CD36, Ffar1, and GPR84. Correlations between taste receptor expression and dietary intake patterns supported the functional relevance of these molecular changes. Network analysis identified tissue-specific correlation structures, with taste receptor-inflammation interactions confined predominantly to gustatory tissue, indicating that obesity-induced inflammation most strongly impacts gustatory tissue, while extra-oral taste receptor regulation occurs independently of systemic inflammatory signals. This work was supported by the Slovak Research and Development Agency (APVV-21-0355) and the Ministry of Education, Science, Research and Sport of the Slovak Republic (VEGA1/0341/23).

S3-P17

Functional characterization of juxtaglomerular cells in the mouse accessory olfactory bulb

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The accessory olfactory bulb (AOB) serves as the primary relay for vomeronasal sensory input to the brain. Here, conspecific chemical information - such as social or reproductive status cues - is encoded and conveyed to downstream neural circuits. Despite its central role in social chemosignaling, however, the cellular and circuit mechanisms underlying stimulus representation in the vomeronasal system remain poorly understood. Using large-scale transcriptomics along with morphological analysis in male and female mice, we identified at least twelve distinct AOB cell types. Among these, we characterized a population of juxtaglomerular interneurons within the AOB glomerular layer. These juxtaglomerular cells have been proposed to mediate inhibition within glomerular modules, suggesting a role in sensory processing and stimulus discrimination within the AOB. However, their precise functional contribution to circuit dynamics remains elusive. To investigate juxtaglomerular cells, we employ whole-cell patch-clamp electrophysiology and assess their electrical characteristics, including passive membrane properties, excitability, ionic currents, and synaptic inputs. By integrating physiological, morphological, and transcriptomic data, we aim to define the role of JGCs within AOB circuitry and thereby advance our understanding of vomeronasal information processing.

S3-P18

Identification of key pheromones and neural mechanisms involved in bumblebee reproduction

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Sex pheromone communication is crucial in insects for attracting mating partners. Despite its fundamental role in insect ecology and its implications for conservation, it remains poorly understood. Current research on the detection and neural processing of species-specific sex pheromones, and how they initiate copulation, has primarily focused on a few species of Lepidoptera and Diptera. In contrast, the mating systems of social insects have been strongly overlooked, despite their complex mating behaviours. This is particularly true for the buff-tailed bumblebee (*Bombus terrestris*), a widespread pollinator and influential model for studying perception, learning and memory. Even if we know that *B. terrestris* males establish flight paths along which they deposit pheromones from the cephalic labial glands (CLG), attracting both virgin queens and other males, the compounds involved are utterly unknown. Although extracts of male CLG are behaviourally attractive for queens, and some CLG compounds elicit electrophysiological responses in the queen antenna, it is still unclear which are sexually attractive to males and queens and how they are processed in the neural systems of both castes. In our project, we aim to obtain a complete picture of *B. terrestris* sex pheromone communication by i) investigating which pheromonal CLG compounds attract males and queens (gas chromatography-electroantennographic detection, GC-Mass Spectrometry and 4-quadrant olfactometer), ii) describing possible adaptations in their olfactory systems (neuroanatomical staining); and iii) recording neural activity by means of in vivo calcium imaging to reveal how sex pheromones are processed in the brain. This project is funded by the Structure et dynamique des systèmes vivants (SDSV) doctoral school of University Paris-Saclay, Agence nationale de la recherche (ANR, BumbleDate project), and the Procope-Mobility grant program.

S3-P19

Chemical Intolerance and Smell: Meta-analysis Reveals Altered Response Bias Rather Than Enhanced Sensitivity

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Objective. Chemosensory dysfunction is frequently implicated in multiple chemical sensitivity (MCS) and chemical intolerance, yet empirical evidence for altered olfactory function remains inconsistent. We conducted a pre-registered systematic review and meta-analysis to characterize olfactory-related outcomes across psychophysical, rating-based, and physiological domains. **Methods.** Searches in PubMed and Web of Science identified studies published 1991–2025 that: (i) applied explicit criteria for MCS, chemical intolerance, sick building syndrome, or building-related symptoms; (ii) included an olfactory or trigeminal challenge; and (iii) reported structured sensory, symptom, or physiological outcomes. Following screening, 34 studies provided extractable data (MCS: $n = 663$, $\text{Mage} = 43.0 \pm 9.7$; controls: $n = 924$, $\text{Mage} = 39.9 \pm 8.4$). Multilevel meta-regression was used to estimate pooled effects and test moderators (stimulus standardization, outcome domain). **Results.** The full-model pooled effect was non-significant ($b = 0.28$, $\text{SE} = 0.18$, $p = .116$). After removal of 29 outliers ($k = 84$), a significant effect emerged ($b = 0.32$, $\text{SE} = 0.12$, $p = .008$), indicating elevated olfactory-related scores in MCS as compared to non-MCS groups. Moderator analyses revealed that this effect was driven by non-standardized odor tasks ($b = 0.43$, $\text{SE} = 0.16$, $p = .009$), and reported symptoms ($b = 0.50$, $\text{SE} = 0.22$, $p = .02$) and, to a lesser extent, odor perceptual ratings ($p = .06$). Olfactory ability measured via standardized tests did not differ between groups ($p = .79$). **Conclusion.** MCS does not seem to be characterized by enhanced olfactory ability but by heightened symptom and perceptual responses, particularly to non-standardized odor stimuli. These findings underscore the importance of stimulus standardization and implicate central evaluative and interoceptive mechanisms rather than amplified sensory input in the MCS phenotype. **Funding:** This work was supported by Asahi Kasei Homes Corporation.

S3-P20

Gustatory ERP-Based Classification of Taste Dysfunction and Normogeusia Using Machine Learning

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Taste dysfunction is typically assessed using subjective psychophysical tests, while objective neurophysiological measures remain limited. This study aimed to investigate whether gustatory event-related potentials (ERPs) can be used to differentiate patients with taste dysfunction from individuals with normal taste function (normogeusia) using machine learning approaches. A total of 67 participants were included, comprising patients with taste dysfunction ($n = 43$; mean age: 57.7 ± 11.9 years; 23 females) and individuals with normogeusia ($n = 24$; mean age: 55.5 ± 13.0 years; 13 females). Sweet and salty taste stimuli were presented, and EEG signals were recorded. Machine learning models, including support vector machines, random forest, and convolutional neural networks, were applied to classify the two groups. The models achieved a classification performance of approximately 80%, with the best performance obtained using the CNN model. These findings suggest that gustatory ERPs contain meaningful information related to taste function and may provide a potential objective approach for distinguishing taste dysfunction. This approach could contribute to improving the clinical assessment of gustatory disorders.

S3-P21

Deciphering the molecular basis for activation and inhibition of olfactory receptors.

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Olfactory receptors (ORs) are part of the largest family of G protein-coupled receptors and constitute the molecular basis for the detection and discrimination of a vast chemical space of volatile compounds. ORs function through a combinatorial mechanism, meaning that a single odorant can activate multiple receptors, while a single receptor can respond to structurally diverse ligands. Beyond their sensory role, select ORs are ectopically expressed in non-olfactory tissues; Olfr412, for instance, is expressed in spermatozoa and participates in chemotaxis, highlighting the broader physiological relevance of this receptor family. Despite significant advances in the field regarding the molecular mechanisms of odorant recognition and receptor activation, receptor inhibition remains poorly understood. This study integrates computational modeling with experimental validation to decipher and further understand the molecular determinants of OR activation and inhibition. Olfr412, the ectopically expressed OR in mice, bound with different types of odorants served as the selected model systems. We combine molecular docking and simulations to characterize agonist, and inverse agonist binding at the atomic level. Comparative analysis of OR-agonist and OR-inverse agonist complexes reveals differences in receptor-ligand interactions highlighting the differences between activation and inhibition. Additionally, virtual screening was performed on a large dataset of odorants collected from the PyrFume database to identify other potential agonists and inverse agonists. The selected odorants were subsequently tested experimentally to validate their effect on OR activity. Furthermore, Umbrella Sampling was employed to map the translocation pathways of odorants through the Olfr412. Collectively, this work deepens our understanding of OR activation and inhibition at the molecular level, with implications for both their sensory and ectopic function, enabling the design of modulators.

S3-P22

Multi-dimensional study on sex-specific patterns of taste deficits, α -synuclein, and Supervised Learning for diagnosis and disease stratification in Parkinson's disease

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Parkinson's disease (PD) is a complex neurological disorder. α -synuclein is described as a PD biomarker, with genetic mutations linked to susceptibility. Taste deficit is an underexplored PD non-motor feature with diagnostic potential and sex patterns remain unclear. We assessed taste perception for five stimuli, salivary α -synuclein (α syn) profiling, four α -synuclein gene (SNCA) polymorphisms, demographic and clinical features in 99 PD patients and 60 healthy controls (HCs) within a unified, sex-aware analytical framework. A Supervised Learning (SL) model classified PD versus HC and moderate (MSPD) versus high severity PD (HSPD), determining the impact of each feature. Taste identification was markedly reduced in PD, with modality and sex-specific deficits. Saltiness and astringency were the most affected and were modulated by SNCA variations. Females preserved sour recognition, while males showed marked citric acid misidentification. SL models achieved high classification accuracy for PD versus HCs, identifying NaCl under threshold, age, and incorrect identification of tannic acid as the most influential features for PD prediction. Tremor-dominant PD and shorter disease duration were associated with MSPD, while longer duration and akinetic-rigid type predicted HSPD. The rs2583988 CC genotype was associated with MSPD, while the rs2583988 TT genotype and lower taste acuity predicted HSPD. In sex-stratified analyses, SL modeling revealed that the inability to perceive saltiness was the most informative feature; astringency misidentification strongly predicted female PD, whereas sour misidentification characterized male PD. Salivary oligomeric α syn was elevated only in PD females who failed to identify astringency. These findings highlight sex-specific interplay among taste dysfunction, salivary oligomeric α syn, and SNCA genetics in PD, supporting the integration of sensory and genetic markers into non-invasive, sex-aware diagnostic and stratification strategies.

S3-P23

Assessing olive oil adulteration: a combined study of human olfaction and nanostructured gas sensor responses

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Olive oil (OO) is a cornerstone of the Mediterranean diet, valued for its sensory qualities and health benefits; however, its adulteration with cheaper seed oils (SO) is a significant issue, reducing quality and potentially posing health risks for consumers. Although effective, current detection methods rely on complex laboratory techniques and specialized personnel. Olfaction plays a crucial role in the food sensory evaluation, including OO; however, the ability of untrained individuals to distinguish pure OO from adulterated samples is generally considered limited. This study aimed to evaluate whether human olfaction can discriminate among 100% pure OO (100), 25% (25), and 50% (50) adulterated samples with SO, and to compare these results with nanostructured gas sensors performances. The study included 25 subjects (F = 16, M = 9), aged 20–40 years. All participants were preliminarily assessed using Sniffin' Sticks ODOFIN test (Threshold, Discrimination, and Identification), resulting in normosmic olfactory function with a mean score of (cutoff = 16). Participants were then asked to discriminate 100 vs 50, 25 vs 50, and 100 vs 25 OO samples by using a triangle-test-approach: the average scores were 2.5/6, 2/6, and 1.8/6, respectively. In parallel, eight different nanostructured gas sensors were tested on the same OO samples: Tin-Titanium-Based (STN and ST10) and Tungsten-Based (W11) sensors demonstrated the best performance, exhibiting clear progressively decreasing responses with increasing levels of adulteration. For STx sensors were observed variations of 12% and 22% by comparing 100 sample vs 25 and 50 ones respectively, and these variations were approximately 11% and 25% for W11 one. These percentages were obtained by averaging all measurements, while the S.E. was always negligible. Finally, these findings underline the strong capability of nanostructured gas sensors to outperform human olfaction in detecting OO adulteration.

S3-P24

Prenatal development of the rabbit vomeronasal and olfactory systems at embryonic day 20: early differentiation within a 30-day gestation

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The mammalian vomeronasal system plays a crucial role in the detection of pheromones and other chemical cues involved in intraspecific communication. Despite its functional relevance, information on the prenatal development of this system in mammals remains scarce, particularly beyond conventional laboratory rodents. This study aims to address this gap by analysing the prenatal development of the vomeronasal and olfactory systems in the rabbit. Understanding this process is especially relevant because it may provide insight into the possible existence of maternal-foetal chemical communication during prenatal development, with potentially important practical implications for early postnatal behaviour and survival. To this end, we employed immunohistochemical and lectin-histochemical approaches, including antibodies against G proteins, GnRH, GAP-43 and other neural markers, as well as a panel of lectins including UEA, VVA, LEA and STA. At embryonic day 20 in the rabbit, the vomeronasal system already displays a notably advanced degree of morphological development, particularly at the level of the vomeronasal organ and nerves. However, the accessory olfactory bulb shows a lower degree of differentiation and lamination at this stage. Overall, the olfactory system appears to be more advanced than the vomeronasal system during this prenatal period. These findings raise relevant questions regarding the spatial and temporal patterns of differentiation and functional maturation of both chemosensory systems during prenatal development, as well as their possible implications for chemical communication and behavioural adaptation during the first days of life. This study was funded by the Spanish Ministry of Science and Innovation, grant PID2021-127814OB-I00.

S3-P25

A wide sensory window between odor and irritation thresholds enables both safety and consumer acceptability of fragrances

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Objectives: Odor and sensory irritation thresholds of chemicals have broad relevance across scientific, regulatory, and industrial contexts, including safety assessment, product development and quality control, and environmental air quality monitoring. In the cosmetics, personal care, and household product industries, odor perception plays a critical role in fragrance formulation, inhalation safety evaluation, and overall product acceptability. We undertook this study to measure the thresholds for odor and irritation in the same individuals for 8 fragrance materials. These materials were selected according to the availability or lack of existing relevant target data (human or animal) and their volume of use, categorized as >1,000 or 100–1,000 metric tons. **Methods:** 30 males and females, aged 18-38, provided odor and nasal irritation (lateralization) thresholds for phenylethyl alcohol acetic acid, linalool, eucalyptol, menthol, styrene, terpineol and cinnamic aldehyde. **Results:** As previously shown for other chemicals, irritation thresholds for all but one fragrance chemical were significantly higher than corresponding odor thresholds-- in some cases an order of magnitude higher. The ratio of the irritation threshold to the odor threshold indicates how effectively a chemical's smell acts as a warning signal for its potential toxic or irritating effects and ranged from approximately 6-fold to over 240-fold across the tested chemicals. **Conclusion:** This work demonstrates that, for a range of widely used fragrance ingredients, odor detection occurs at concentrations far below those that elicit trigeminal-mediated chemesthetic responses. The resulting sensory window between odor and irritation thresholds provides a robust margin that is directly relevant to inhalation safety assessment and consumer acceptability. Importantly, the results reinforce that normal fragrance use occurs well within a range that does not trigger adverse sensory effects.

S3-P26

Exploring the mechanisms and causes of VNO inflammation: differences between domestic pigs and sheep

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The role of the vomeronasal organ (VNO) in animals has been well documented. In recent years, inflammation of this organ has been described and associated with aggressive behaviours in pigs and cats. The causes of VNO inflammation have been investigated in farm pigs and in a murine model, suggesting that the living environment may represent an important factor in the onset of this pathological condition. To further explore the mechanisms underlying vomeronasalitis, we review and discuss our previous work in pigs and sheep. In pigs, vomeronasalitis appears to be highly prevalent in animals reared under both intensive and free-range conditions. The non-sensory epithelium (NSE) is always affected, while the sensory epithelium (VNSE) is involved in 83% and 85% of VNOs, respectively. Preliminary results in farm sheep show that the VNSE is inflamed in 53% of VNOs, whereas the NSE is affected in 47% of cases. Overall, our studies demonstrate that vomeronasalitis affects both domestic pigs and sheep, although with some differences. Pigs appear to be more frequently affected than sheep, regardless of the rearing system. This may be related to species-specific behaviours absent in sheep, such as rooting and digging, which could increase the presence of microparticles in the oral cavity and facilitate their entry into the vomeronasal ducts, thereby triggering an inflammatory response. This hypothesis is supported by observations in a murine model, where environments with higher levels of microparticles were associated with an increased frequency of vomeronasalitis. Interestingly, in pigs the NSE is always inflamed, suggesting a potential primary defensive role against external insults, whereas in sheep the NSE is affected in less than 50% of cases. The reasons underlying these interspecies differences require further investigation to better characterise the causes and mechanisms of vomeronasalitis in domestic species and to develop strategies for its prevention.

S3-P27

Loss of smell and taste among recipients of hematopoietic stem cell transplantation

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Hematopoietic stem cell transplantation (HSCT) is a potentially curative treatment for patients with selected hematologic disorders. However, both HSCT and the pretreatment prior to HSCT, may affect olfactory and gustatory senses. The aim of this case-controlled, prospective study was to assess olfactory and gustatory functions in HSCT patients at three timepoints: prior to HSCT (T1), 10-15 days after HSCT (T2), and 3 months after HSCT (T3), and compare these functions with those for healthy controls. This ongoing study reports data from 30 patients at T1 (14 F, 16 M, age (mean±SD) 56.5 ± 10.0 yrs), 20 patients at T2 (9 F, 11 M, age (mean±SD) 52.5 ± 10.9 yrs), 11 patients at T3 (3 F, 8 M, age (mean ± SD) 52.5 ± 9.6 yrs) and 29 controls (19 F, 10 M, age (mean±SD) 45.9 ± 9.6 yrs). Olfactory function was assessed using Sniffin' Sticks test battery (TDI score: threshold + discrimination + identification, score 1-48) and gustatory function using Taste Strips (score 0-16). Results showed that TDI scores were lower in patients than controls at all timepoints (mean±SD): T1 = 26.9 ± 4.7, T2 = 26.7 ± 6.1, T3 = 28.2 ± 3.1 vs controls = 31.0 ± 3.1 (p < 0.01). Taste Strips scores (mean ± SD) were also lower in patients than in controls: T1 = 8.9 ± 4.2, T2 = 9.1 ± 3.6, T3 = 12.0 ± 2.2 vs controls = 13.5 ± 1.9 (p < 0.001). These results indicated that HSCT patients have significantly reduced olfactory and gustatory functions compared with healthy controls both prior to transplantation and in the early posttransplant period, with evidence of partial improvement by 3 months. These findings highlight the need for prospective monitoring of chemosensory outcomes in HSCT recipients and underscore that both patients and medical professionals would benefit from targeted information about these complications.

S3-P28

Depicting Evoked field potentials to Olfactory Stimuli using on-scalp MEG: Preliminary Results for a Smell Oddball Task

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During recent decades, substantial progress has been made in delineating the olfactory functional neuroanatomy. Its temporal dynamics, however, remain poorly understood as methodological limitations restrict the brain areas that can be accessed (Han, 2021; Regemorter et al., 2022; Wang et al., 2025). On-scalp magnetoencephalography (MEG) is a new methodology that measures magnetic evoked response fields through sensors placed directly on the head. This allows for temporally precise measurements of deeper brain areas with for higher sensitivity and spatial precision. The potential of the method to capture the evoked response to olfactory perceptual stimuli, given its better resolution for deeper structures like the insula, remains to be explored. This study investigates the evoked response to odors using an OPM-MEG system (from FieldLine Medical, with 136 dual-axis sensors) using an olfactometer (Lundström et al., 2011). To maintain the participants' attention throughout the long scans, we implemented an odd-ball task utilizing two different odors (watermelon and coconut, both associated with the sweet taste quality but distinct enough to be distinguished easily). The experiment is divided into 64 blocks separated by 5 min breaks to prevent habituation to the presented odors, and a smell intensity rating is included in between blocks to monitor habituation is avoided. Preliminary results show evoked responses at both sides of the central line, which source localization via MRI places in proximity to perceptual areas of interest. These findings show the potential for osMEG for contributing to the characterization of the temporal dynamics of the olfactory response and its integration with cortical signaling pathways such as those involved in motivational and metabolic processing.

S3-P29

Macrophages are dispensable for taste bud regeneration upon nerve injury repairSuyoung Ki^{1,2}, Yongtaek Jeong^{1,2}¹ Department of Pharmacology, Korea University College of Medicine, Seoul 02841, Republic of Korea² BK21 Graduate Program, Department of Biomedical Sciences, Korea University College of Medicine, Seoul 02841, Republic of Korea

Peripheral sensory nerve injury triggers Wallerian degeneration, a process characterized by axonal fragmentation, myelin breakdown, and robust macrophage infiltration. While macrophages are widely implicated in debris clearance and repair within the injured nerve trunk, their functional role in the remodeling of sensory end organs remains poorly defined. Here, we investigated whether macrophages are required for taste bud degeneration and regeneration using a glossopharyngeal nerve transection model in mice. We first observed a marked accumulation of Iba1-positive macrophages in the circumvallate papilla after nerve injury, peaking at 2 weeks and coinciding with maximal taste bud loss. To test the functional requirement for these cells, we used both pharmacological and genetic depletion models. Despite near-complete depletion of infiltrating macrophages, the extent of taste bud loss at 2 weeks was indistinguishable from controls. Likewise, macrophage deficiency did not alter structural recovery at 4 weeks, with taste bud number, size, and the relative composition of different taste cell types remaining unchanged. These findings indicate that the major structural phases of taste bud degeneration and regeneration proceed despite macrophage depletion. Thus, although macrophage infiltration is a prominent feature of peripheral sensory denervation, it is not an essential driver of structural end-organ remodeling in the taste system.

S3-P30

ItypOR41 utilizes universal verbenoid anchors, whereas the stereochemical structure of (S)-cis-verbenol induces unique receptor activation.

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The European spruce bark beetle, *Ips typographus*, is a destructive forest pest whose outbreaks cause catastrophic spruce dieback across Eurasia. Its mass-aggregation behavior is coordinated by the pheromone (S)-cis-verbenol, which is detected by ItypOR41, the second most highly expressed odorant receptor in the beetle's antennae. The objective of this study was to identify the specific residues responsible for binding verbenols and verbenone and to elucidate the molecular mechanism behind the unique potency of (S)-cis-verbenol. We integrated molecular dynamics (MD) simulations (Amber, CPPTRAJ, MMGBSA) with HEK293 cell-based functional calcium assays. Functional assays confirmed that (S)-cis-verbenol exhibits significantly higher potency than trans-verbenol. Crucially, we identified universal ligand-binding anchors; mutations at these sites eliminated the receptor's response to all verbenols and verbenone. Regarding the secondary objective, MD simulations identified an "upward" posture for (S)-cis-verbenol involving higher electrostatic interactions with D151 and S201. However, alanine substitutions at these sites, including a double mutation, did not reduce potency dramatically or abolish the cis-preference. These findings demonstrate that ItypOR41 employs a two-layered recognition system: common anchor points essential for general verbenoid recognition, and a stereochemical "induced fit" mechanism where the cis-geometry itself, rather than specific bonds with residues like D151 or S201, drives superior receptor activation. We conclude that the cis-structure more effectively lowers the energetic barrier to the active state, facilitating a stronger conformational change. This research provides a critical foundation for the rational design of eco-friendly, species-specific semiochemicals for bark beetle population regulation. Fund; Korea National Research Foundation (RS-2024-00414882) and Carl Trygger Foundation (CTS25:4199)

S3-P31

Quantifying the cross-linguistic translation appropriateness of odor descriptors using large-scale corpus analysis

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Verbal descriptors are widely used to pinpoint the perceptual qualities of odorants. Standardized sets, such as Dravnieks' Atlas of Odor Character Profiles (Dravnieks, 1972), provide a common language to compare odorants and are frequently utilized in psychological rating studies. However, translating these English scientific terms directly into other languages risks overlooking language-specific cultural contexts that intrinsically link a term to the olfactory modality. This project aims to quantify the appropriateness of directly translating commonly used verbal descriptors from Dravnieks' Atlas into diverse target languages. Our methodology involves machine-translating these English descriptors into various languages. We then assess the relationship of these translated terms to the olfactory modality across languages using the Olfactory Association Index (OAI) and Olfactory Specificity Index (OSI; Iatropoulos et al., 2018). To compute the OAI and OSI, we utilize the multilingual "Colossal Clean Crawled Corpus" (mc4; Raffel et al., 2019), a large dataset of web-scraped documents spanning 108 languages. While this is an early-stage project, preliminary workflow validation confirms the feasibility of extracting these indices from the mc4 dataset to evaluate translation efficacy. We will present our first statistical results detailing the cross-linguistic OAI and OSI distributions, alongside further project outlooks. Ultimately, this large-scale text analysis will highlight potential cultural and linguistic pitfalls in the direct translation of odor descriptors, offering a more culturally robust framework for international olfactory research. This work is funded by the Volkswagen foundation within project 'Olfactorial Perceptrics' (No. 9B396).

S3-P32

The time-dependent effects of benign taste experience on neural plasticity, learning, and taste discrimination.

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In associative learning, we study how future behaviors are influenced by past sensory experiences. For example, in conditioned taste aversion (CTA), animals readily learn to avoid tastes paired with malaise. Our research has repeatedly replicated a Latent Enhancement (LE) of CTA learning towards a novel taste when it is prefaced by benign taste experiences. We have also shown that these experiences are processed in gustatory cortex (GC) through immediate early gene activity and optogenetics. Recent in-vivo electrophysiological data suggest that TE increases neural plasticity in GC such that neural representations of tastes become more discriminable. We argue that more discriminable tastes would be more associable in CTA learning which can lead to stronger aversions. Here, we test the hypothesis that TE enhances discriminability by examining novel taste thresholds and CTA discrimination within a brief access task using female Long Evans Rats (N=85). Our results (using two-way ANOVAs) reveal lower taste thresholds and reduced CTA generalization to other sweet tastes following TE. We also test the hypothesis that TE induced plasticity is time dependent by inducing delays between TE, CTA, and CTA testing. TE occurring within 1 week of CTA learning still produced a LE of CTA that persists for 2 weeks. Lastly, we also examine neural activity (c-FOS) in the basolateral amygdala (BLA) and nucleus accumbens (NAc) — areas involved in taste processing and learning — to characterize the neural correlates of TE induced plasticity. These findings begin to characterize the impact of TE on behavior and neural dynamics of taste learning. Research reported in this abstract was supported by the National Institute of General Medical Sciences of the National Institutes of Health under Award Number P20GM103499. The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

S3-P33

Stimulus-driven modulation of sensory sensitivity in mouse olfactory sensory neurons

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Fast, short-term sensory adaptation to prolonged or repetitive stimuli is essential for maintaining dynamic sensitivity in sensory systems. In olfactory sensory neurons (OSNs), odorant detection relies on odorant receptor (OR) activation, which triggers a G-protein-mediated cAMP cascade. Responses (to prolonged and strong stimulation) are counterbalanced by Ca²⁺/calmodulin, resulting in adaptation. Given the high ligand affinity of ORs, many OSNs exhibit very low odorant detection thresholds, resulting in sensitivity ranges spanning several orders of magnitude. However, it is unclear whether mechanisms of peripheral response modulation extend beyond adaptation and whether such modulation depends on stimulus concentration. Here, using patch-clamp recordings and Ca²⁺ imaging in mouse OSNs, we investigate how varying interstimulus intervals and stimulus intensities modulate transduction currents, Ca²⁺ responses, and action potential generation. We dissect (i) which and (ii) how concentrations are de- and encoded in the olfactory periphery, (iii) which types of modulation occur, and (iv) which signaling cascade stages play important roles in modulation. Together, we seek to understand how OSNs fine-tune their sensitivity and responsiveness to represent a wide range of odorant concentrations.

S3-P34

Orexin neurons contribute to state dependent control of olfaction.Willem Koppenol^{1,2}, Janice Bulk^{1,2}, Sophie Steculorum¹¹ Max-Planck-Institute for Metabolism Research, Cologne, Germany² Graduate School for Biological Sciences, Faculty of Mathematics and Natural Sciences, University of Cologne, Germany

Food intake relies on the integration of internal metabolic cues and external sensory signals to drive feeding-related behaviors. Increasing evidence indicates that the nutritional status dynamically modulates olfactory processing, biasing behavior toward food seeking over other olfactory-driven tasks. Yet, the mechanisms underlying this state-dependent modulation of olfaction remain poorly understood. Here, we investigate the role of orexin (Ox) neurons from the lateral hypothalamic area (LHA), a key hypothalamic hub regulating homeostatic control, in these processes. Virus-based tracing confirmed orexinergic projections to several olfactory regions and sn-RNAseq and RNAscope revealed the expression of Ox-receptors 1 & 2 throughout olfactory regions. To functionally assess the role of Ox-neurons in state-dependent modulation of olfaction, we employed chemogenetics to selectively activate or inhibit these Ox-neurons, thereby mimicking or suppressing their endogenous fasting-induced activity across a range of olfactory tasks. Chemogenetic activation of Ox-neurons in ad libitum-fed mice increased food odor sniffing behavior compared to social and neutral odors, mimicking the fasted state. In contrast, chemogenetic inhibition of Ox-neurons in fasted mice decreased food odor sniffing behavior compared to social and neutral odors. Total time of sniffing behavior during all odor sniffing and investigation experiments was significantly increased upon activation and decreased upon inhibition of Ox-neurons, confirming a broad bidirectional modulatory role in olfactory engagement. During dis-/habituation assays, chemogenetic inhibition of Ox-neurons disrupts odor memory while no effects were observed upon chemogenetic activation. Collectively, our work reveals that LHA Ox-neurons function as modulators for metabolic state-dependent olfactory processing. This project is funded by the modulation in olfaction research grant (FOR5424) from the German Research Association (DFG).

S3-P35

Application of an Electronic Nose in the Noninvasive Identification of Parkinson's Disease

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Background. Parkinson's disease (PD) is a common neurodegenerative disorder. Electronic nose technology detects volatile organic compounds (VOC) patterns and offers a promising non-invasive diagnostic tool. **Objective** To evaluate whether an electronic nose can distinguish VOC profiles from the skin of patients with Parkinson's disease and healthy controls. **Methods.** In this cross-sectional study, 25 clinically diagnosed patients with PD (age: 68.2 ± 9.2 years) and 27 healthy controls (age: 71.4 ± 8.6 years) were enrolled. Sebum samples were collected using cotton pads and analyzed using an electronic nose from Universidad de Extremadura. Data were analyzed using logistic regression and support vector machine (SVM) models. Model performance was evaluated using cross-validation, calibration analysis, and permutation testing. **Results.** Logistic Regression Model 3 demonstrated the most stable performance in distinguishing patients with PD from healthy controls, AUC=0.91, accuracy=0.808. Calibration analysis indicated acceptable model fit (Brier score = 0.125). Permutation testing confirmed that the model performance was significantly better than random classification, with a cross-validated AUC of 0.83 ($p = 0.0099$). Similarly, SVM Model 2 showed strong classification performance, correctly identifying 84.8% of PD patients and 89.1% of healthy controls. The model achieved a cross-validated AUC of 0.94 and an F1-score of 0.87. Permutation testing further confirmed that the model performance was significantly superior to chance (AUC = 0.94, $p = 0.005$). **Conclusion.** Electronic nose detection of sebum-derived odor signatures, analyzed using logistic regression and SVM models, can effectively distinguish patients with Parkinson's disease from healthy controls.

S3-P36

Proof-of-concept trial of pre-meal olfactory nudging versus habituation in women with overweight or obesity

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Objective. Odor-based behavioral strategies are understudied in ingestive behavior research, despite the role of olfaction in food choice. We hypothesized that daily brief olfactory interventions would be feasible and accepted in women with overweight or obesity and produce short-term changes in diet and olfactory awareness. **Methods.** In a 2-week, proof-of-concept, pilot randomized controlled feasibility trial, females (N=45; age 18-55 years; 60% white) were randomized to one of three interventions: olfactory nudging (ON; brief pre-meal exposure to “healthy” banana/strawberry fruit odors); olfactory habituation (OH; 10-min pre-meal exposure to “unhealthy” banana pudding/strawberry cake odors); or control (10-minute pre-meal no-odor exposure). **Results.** ON intervention was rated as more liked and acceptable than OH and control (overall $p=0.006$; pairwise $p=0.04$ and $p=0.007$). Baseline BMI and smell identification ability did not differ by group, whereas autonomous motivation was higher in ON and OH than control (6.1 ± 0.9 and 6.1 ± 0.7 vs 5.2 ± 1.1 ; $p=0.02$). At 2 weeks, body weight change did not differ across groups (ON $+0.3\pm2.5$ lbs, OH $+0.1\pm1.3$ lbs, control $+0.4\pm2.4$ lbs; $p=0.44$). Olfactory awareness showed a nonsignificant increase in OH ($\Delta= +2.0\pm8.4$) vs ON ($\Delta= -2.1\pm5.0$) and control ($\Delta= -1.7\pm8.9$). Exploratory dietary patterns were mixed, with directional shifts in OH (lower protein and %protein, higher %carbohydrate) and ON (lower %fat). In adjusted analyses, higher baseline autonomous motivation (control: 5.2 ± 1.1 vs. ON: 6.1 ± 0.9 and OH: 6.1 ± 0.7 ; $p=0.02$) predicted higher protein intake ($b=7.56$, $p=0.046$) and lower alcohol intake ($b=-3.06$, $p=0.03$) at 2 weeks. **Conclusion.** These early findings suggest that brief olfactory interventions are feasible and acceptable, particularly ON, and may result in short-term diet-related changes. Larger studies are needed to better understand efficacy of odor-based interventions and influence on ingestive behavior.

S3-P37

How chemistry and temporal perception shape acidity language in wine

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Acidity is a key structural component of wine, yet little is known about how experts describe the quality of acidity. This study investigated how acidity descriptors, such as crisp and zesty, relate to chemical composition, temporal perception, and retronasal aroma access in Riesling and Chenin Blanc, two high-acid varieties with distinct acidity profiles. Six dry white wines were selected: Riesling from Alsace, Clare Valley, and Marlborough, and Chenin Blanc from Vouvray, Swartland, and Marlborough. Wines were analysed chemically using WineScan. A trained sensory panel (N = 13) evaluated acidity using Time Intensity analysis, extracting area under the curve, maximum intensity, and time to maximum intensity. An online questionnaire with Masters of Wine (MW) and MW students (N = 67) identified common descriptors, and an in-person expert blind tasting (N = 44) assessed how well eight descriptors matched each wine under nose-clip and free-smelling conditions. Riesling showed higher total perceived acidity and maximum intensity than Chenin Blanc, whereas Chenin Blanc took longer to reach peak acidity (all $p < .001$). Online responses revealed style-specific associations, including zesty for Riesling and bracing for Chenin Blanc. In blind tasting, however, most descriptors except broad matched Riesling more strongly. Descriptor use was predicted by acid-related and non-acid components. Higher total acidity, lower pH, and greater temporal intensity were associated with racy, crisp, sharp, and bracing, whereas broad showed the opposite pattern. Retronasal aroma access increased the use of racy and bracing, indicating that acidity language is not purely taste-based. These findings show that wine acidity is a dynamic multisensory percept shaped by chemistry, temporal dynamics, and retronasal aroma. Understanding acidity descriptors as perceptual rather than conventional terms may improve sensory science approaches to wine and professional acidity communication.

S3-P38

Beyond OR51E2: systematic profiling reveals novel dysregulated olfactory receptors in prostate cancer

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Olfactory receptors (ORs), a family of G protein-coupled receptors, are increasingly recognized as being ectopically expressed in non-olfactory tissues. Emerging evidence links specific ORs to cancer pathophysiology, highlighting their potential as biomarkers and modulators of tumour behaviour. Although OR51E2 is well established as upregulated in prostate cancer, the broader landscape of OR expression in this disease remains insufficiently characterised. In this study, we systematically profiled the expression of 18 ORs at mRNA and 3 ORs protein levels using RT-PCR, real-time RT-PCR, and immunofluorescence. Analyses were conducted in normal prostate epithelial cells (PNT1A) and compared with androgen-dependent (LNCaP, 22Rv1) and androgen-independent (DU145, PC3) prostate cancer lines. Our results reveal a more complex OR expression profile than previously identified. Beyond OR51E2 and OR51E1, multiple ORs were detected in both normal and cancer prostate cells. Notably, OR8B8 and OR8D1 were significantly downregulated in prostate cancer cell lines in comparison to normal prostate cells, suggesting a potential loss of function associated with tumourigenesis. These findings expand the current understanding of OR dysregulation in prostate cancer and uncover novel candidates warranting further investigation. Future functional studies are needed to determine whether these receptors contribute to disease progression and whether they may represent viable diagnostic markers or therapeutic targets. Funding: This work was funded by Programa Operacional Regional do Centro (project grants CENTRO-08-5864-FSE-000039); Oncology Research Fellowship Centro Académico Clínico das Beiras (Núcleo Regional do Centro da Liga Portuguesa Contra o Cancro, LPCC).

S3-P39

Kokumi peptide RHE under a low-salt condition enhances salt perception and modulates EEG responses in humansChuang Yu¹, Piao Yi¹, Tao Jiang²¹ Faculty of Flavour Fragrance and Cosmetics, Shanghai Institute of Technology, Shanghai, China² Team of Neuro-Éthologie Sensorielle, CRNL, INSERM U1028, CNRS UMR5292, University Lyon 1, Lyon, France

Objectives: Kokumi compounds enhance taste perception, but their effects on cortical activities of salt stimuli remain unclear. This study examined whether the kokumi peptide RHE (p3), identified from *Agaricus bisporus*, modulates human salt perception and salt-evoked EEG responses. **Methods:** Peptides were identified from aqueous extracts of *Agaricus bisporus* by LC-MS/MS and screened using a calcium-sensing receptor (CaSR)-based approach. The ten trained participants (5 females, 5 males; 20-28 years) completed sensory evaluation and EEG recording. Saltiness was assessed by quantitative descriptive analysis (QDA) and time-intensity (TI) methods. Salivary aldosterone was measured by ELISA. EEG was recorded during oral stimulation with low-salt solution (S3, 3 mg/mL NaCl) and S3 supplemented with RHE (S3-p3). **Results:** In QDA, saltiness intensity increased averagely from 5.1 for S3 to 7.9 for S3-p3 ($p < 0.05$). TI analysis showed that saltiness duration increased from approximately 45s for S3 to 120s for S3-p3. Salivary aldosterone increased by 46.47% following S3-p3 stimulation compared with S3. EEG topography showed that S3-p3 elicited more persistent frontal and left temporal activities than S3 alone. Power spectral analysis showed dominant low-frequency responses, with greater theta (4-8Hz) and alpha (8-13Hz) activities under S3-p3 than under S3 ($p < 0.05$). These findings suggest that S3 and S3-p3 could evoke different perceptual and cerebral responses. Further studies are required to determine how these EEG differences relate to perceptual significance and subjective saltiness experience. **Conclusions:** RHE enhanced the intensity and persistence of saltiness perception under low-salt conditions. This effect was accompanied by changes in theta and alpha activity and sustained frontal-temporal cortical responses, providing preliminary neurophysiological evidence for kokumi-mediated modulation of salt perception.

Session 4 — 17 September 2026, 17:30-19:00

S4-P1

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A hindbrain neuronal population mediates water perception and consumption

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Drinking water is essential for maintaining body fluid homeostasis. Animals can identify aqueous liquids from available options through ingestion and selectively consume them, indicating the presence of sensory mechanisms that detect ingested water. However, the underlying neural mechanisms remain unknown. Here, in mice, FOS expression-based whole-brain neuronal activity mapping and single-neuron Ca^{2+} imaging identified the waist area of the parabrachial nucleus (wPBN) as responsive to water drinking. Moreover, optogenetic inhibition and caspase-mediated ablation of the wPBN neurons impaired water-drinking but not feeding behavior, demonstrating the essential role of the wPBN in water-drinking behavior. Using in situ hybridization and Ca^{2+} imaging, we further found that, within this region, a neuronal population expressing the oxytocin receptor (Oxtr) is enriched for neurons selectively responsive to ingested water. Closed-loop optogenetic inhibition of these neurons at the onset of drinking led to immediate cessation of water intake. Finally, we found that Oxtr-expressing wPBN neurons project to the thalamus and cerebral cortex, and that optogenetic silencing of this population impaired the ability to discriminate water from non-aqueous liquids, but not water-seeking behavior, suggesting their involvement in the perception of ingested water. Together, this study provides evidence for a neuronal population that encodes sensory information underlying the perception of water and its consumption, advancing our understanding of the neural basis of hydration. This work was supported by Japan Science and Technology agency CREST (JPMJCR21P3), Japan Society for the Promotion of Science KAKENHI (23H00400), and the Lotte Research Promotion Grant, and was reported in part in *Current Biology* (36:1787-1799, 2026).

S4-P2

Transcutaneous vagus nerve stimulation at olfactory bulb-relevant frequencies modulates human olfactory function

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Transcutaneous vagus nerve stimulation (tVNS) is a non-invasive technique that engages central neuromodulatory pathways and has been shown to modulate sensory processing across multiple modalities. Here, we investigated whether tVNS influences human olfactory perception. Sixty-seven healthy adults were assigned to one of two stimulation frequency groups, 25 Hz (beta) or 55 Hz (gamma), and each received 20 minutes of tVNS applied to the left cymba concha as well as a control condition with identical stimulation delivered to the left earlobe. Odor discrimination, odor intensity, and visual attention were assessed before and after stimulation. Gamma frequency tVNS significantly decreased odor discrimination performance, whereas beta frequency did not. Odor intensity ratings increased modestly following stimulation in both groups, with a stronger effect observed after gamma-frequency tVNS. There were no changes in visual attention, and the stimulation intensity, which was selected by the participant, was not associated with olfactory outcomes. These findings demonstrate that tVNS can modulate human olfactory processing in a frequency-dependent manner. The selective effect of gamma frequency, i.e. the known processing frequency of the human olfactory bulb, implies inhibitory interneuron engagement in the olfactory bulb. These findings demonstrate that non-invasive tVNS can modulate human olfactory processing in a frequency-specific manner and highlight stimulation frequency as a critical parameter for future translational applications.

S4-P3

Semantic learning outpaces perceptual change during perfumery training

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Olfactory expertise is often defined by perceptual acuity, but professional training may first reorganize how odors are categorized and described. We tested this hypothesis in a longitudinal study of 39 perfumery students assessed before training, after 6 months and after 18 months. Students completed free odor description, odor-evoked association, odor discrimination, odor recognition and free odor sorting with group labeling using 92 named olfactory materials, including monomolecular compounds, essential oils, perfumes and compositions. Text responses were analyzed with a French NLP pipeline using part-of-speech tagging, lemmatization, FastText embeddings and cosine-similarity measures of within-student consistency, between-student convergence and alignment with an expert teacher. Longitudinal effects were tested with linear mixed-effects models and random forest classifiers predicted training stage from semantic and perceptual features. Semantic change exceeded perceptual change overall (beta = 0.17, SE = 0.06, 95% CI [0.05, 0.29], p = .0045). Semantic improvement was significant from T0 to T1 (beta = 0.42, SE = 0.05, p < .001) and from T1 to T2 (beta = 0.34, SE = 0.05, p < .001). Perceptual improvements were weaker: odor discrimination improved from T0 to T1 (beta = 15.39, SE = 5.70, p-adj = .021) but not from T0 to T2, whereas odor recognition sensitivity increased from T0 to T2 (d-prime beta = 0.74, SE = 0.29, p-adj = .033), mainly through correct rejections. Random forest models classified training stage above chance using all variables (accuracy = 0.80 +/- 0.12) or semantic variables alone (0.79 +/- 0.12), both outperforming perceptual variables alone (0.39 +/- 0.12; adjusted p < 10⁻⁴⁰). These findings suggest that early olfactory expertise is marked less by perceptual refinement than by the emergence of a shared, expert-aligned odor semantics. Funding: [none].

S4-P4

A microbiome-derived olfactory signal regulates inter-male aggression and social dominance in mice

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Many species use microbiome-derived metabolites as chemosensory cues, yet the chemicals involved and the sensory pathways that detect and process them remain poorly understood. Trimethylamine (TMA) is a volatile metabolite that is produced by the gut microbiome and selectively accumulated in the urine of sexually mature male mice. Here, we show that TMA regulates inter-male aggression and social dominance by activating trace amine-associated receptor 5 (TAAR5) in the main olfactory system. In wild-type mice, early aggressive behavior during male-male encounters strongly predicts eventual social status: dominant males initiate more attacks, whereas subordinate males display more defensive behaviors. Deletion of TAAR5 eliminated this asymmetry, with dominant and subordinate mice showing similar levels of aggressive and defensive behaviors. Strikingly, restoring TAAR5 expression in olfactory sensory neurons (OSNs) rescued the behavioral asymmetry, indicating that this effect is mediated by the main olfactory system and arguing against contributions from proposed TAAR5 expression in the brain. Finally, pharmacological suppression of microbial TMA production reduced inter-male aggression, and this effect was reversed by painting treated males with TMA, showing that microbiome-derived TMA is the key volatile ligand for TAAR5 in this context. Taken together, our findings identify TMA as a critical olfactory cue that signals the presence of sexually mature males and facilitates social hierarchy formation. More broadly, our results demonstrate that a microbiome-derived metabolite can shape mammalian social interactions through the main olfactory system and uncover a previously unrecognized role for the TAAR family in regulating social behavior.

S4-P5

Olfactory sensitivity and division of labor in the honey bee *Apis mellifera*

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Division of labor is found in many animal societies, with individuals specializing in specific tasks, and thereby enhancing the efficiency of the group. Task specialization is especially present in honey bees (*Apis mellifera*), in which it depends on the animal's age, but remains also flexible as it is influenced by the social context and the particular needs of the colony. Various models have attempted to explain division of labor, with the widely accepted response threshold model, which postulates that individuals differ in their response threshold to task-associated stimuli. Division of labor in honey bees relies on an efficient communication system among colony members, with the use of many pheromones involved in almost all aspects of their life, making olfaction a crucial sensory modality for these species. Even if the response threshold model has been extensively studied in the honey bees, especially on the gustatory modality, the role of olfactory sensitivity has remained greatly overlooked. We are thus testing the hypothesis that differences in olfactory perception, especially to pheromones, lead to task specialization in the honey bee colony. Behavioral assessments using an olfactometer as well as associative appetitive conditioning will allow to determine olfactory response thresholds to social pheromone compounds (9-ODA, geraniol, methyl oleate and beta-ocimene) in different worker task groups (nurses, guards and foragers). Complementary neurophysiological recordings using electroantennogram will allow to determine olfactory detection thresholds to the same olfactory stimuli. Using both behavioral and neurophysiological approaches, we expect to identify a variation of olfactory detection specific to each worker task groups, in particular to social pheromones.

S4-P6

GLP-1 signaling in the olfactory system modulates glucose metabolism in obese mice: effects of olfactory bulb versus intranasal exendin-4 administration

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Objectives: Glucagon-like peptide-1 (GLP-1) has a key role in glucose homeostasis. Emerging evidence highlights central GLP-1 signaling as an important regulator of metabolic function. In particular, the olfactory bulb (OB) has been identified as a structure linking sensory inputs to energy balance. In this context, intranasal delivery represents a non-invasive strategy to target central GLP-1 receptors and modulate metabolic processes. **Methods:** In lean and diet-induced obese mice, we performed intranasal administration of Exendin-4 (Ex-4) prior to metabolic testing. These results were compared with in situ Ex-4 injections in the OB that we have previously reported (see reference). **Results:** First, after intranasal administration, Ex-4 lowered basal glycemia as soon as 30min post-injection. This effect reached a plateau at 1h and was maintained for up to 5h in both males and females. Second, we performed Oral Glucose Tolerance Tests (OGTT) in mice fed either a standard or a high-calorie diet for 3 months. Following intranasal saline administration and glucose gavage, obese mice exhibited dysregulated glycemia. This profile was restored to control levels by intranasal Ex-4, mimicking the effects observed with in situ OB injection. In addition, like intra-OB injection, intranasal Ex-4 induced an increase in the insulin peak at 15min in the OGTT. Insulin sensitivity was also improved by intranasal Ex-4 in obese mice during an Insulin Tolerance Test. Finally, even in lean mice, intranasal Ex-4 regulated glycemic profiles. This effect was weaker using intra-OB injection. Mechanisms involving olfactory versus alternative pathways that could account for this metabolic rescue are currently under investigation. **Conclusions:** Intranasal administration of incretinomimetics has strong potential to normalize dysregulated glucose metabolism in obese mice. **Funding:** ANR grant Smellins ANR-24-CE17-7944

Ref: Montaner et al. (2024) Nat Commun. PMID 39138162

S4-P7

Modulating the modulators: chemogenetic dissection of bulbar dopaminergic contributions to olfactory behaviour.

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Dopaminergic neurons constitute a distinct class of interneurons in the olfactory bulb (OB), most of which retain lifelong regenerative capacity. Although dopamine modulates olfactory-related behaviours, the specific contribution of OB dopaminergic neurons to odour detection and discrimination remains unclear. Previous studies often lack anatomical specificity, are confounded by midbrain dopaminergic activity, and use diverse behavioural paradigms, resulting in inconsistent conclusions. To investigate the causal role of this bulbar dopaminergic population, we stereotactically injected inhibitory or excitatory DREADD viruses into the OB of dopamine transporter (DAT)-Cre adult mice. After one month of DREADD expression, clozapine-N-oxide (CNO) was administered intraperitoneally to transiently inhibit or activate these neurons, and mice were assessed in a battery of olfactory tasks, including odour habituation, sensitivity and discrimination tests, as well as responses to appetitive and aversive stimuli. Odours were delivered using a custom olfactometer and behaviour was analysed with DeepLabCut and SimBA to measure time spent near the odour port and locomotion, but also to assess sniffing and rearing. Silencing OB DA neurons altered different aspects of behaviour, particularly reducing attraction to appetitive odours (non-significant increase of odour exploration for the CNO group, increase of 33%, n=12, versus significant increase of 111%, n=11, and 140%, n=12, for the control groups), and increased sensitivity to specific odours, enabling detection at lower concentrations (significant increase of the odour exploration at low concentration only for CNO group, 8 sec for odour vs 3 sec for solvent, n=12, p=0.04). These findings highlight a specific role for OB dopaminergic interneurons in odour-guided behaviour. This research is supported by the UKRI Biotechnology and Biological Research Council, the Wellcome Trust and a Homerton College Junior Research Fellowship.

S4-P8

Non-monotonic psychometric functions in human olfaction

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An important question in human psychophysics concerns the limits of human perception. Typically, detection thresholds are measured using a staircase procedure. The assumption is that the function underlying threshold is a sigmoidal one. Yet, although rarely measured and/or reported, the psychometric functions for a number of odors have been shown to be non-monotonic, exhibiting discontinuities or “notches” in performance as a function of concentration. For example, non-monotonic functions have been observed for alpha-ionone (Cameron & Doty, 2025; Marshall & Moulton, 1981) and n-pentanoic acid (Marshall, Blumer & Moulton, 1981). The existence of notches provides a challenge for the threshold measurements, but also begs the question as to the neural underpinnings of this phenomenon. Here we report non-monotonic psychometric functions for both phenyl ethyl alcohol (PEA) and amyl acetate (AA). On each trial in these experiments, participants were presented with two stimuli either in Sonosonics’ Snap and Sniff wands or in glass jars. One stimulus was odorless mineral oil and the other was mineral oil plus one of a range of concentrations of odorant (PEA or AA), presented in random order (i.e., method of constant stimuli). We either tested a large number of participants ($n \sim 30$) in only one session that lasted about 30 minutes or we tested a smaller number of participants ($n \sim 10$) in multiple (5-10) 30-minute test sessions. In all of these conditions we found demonstrable notches in the psychometric functions although the precise concentration at which the notch appeared was somewhat variable across participants and odors. We will discuss neurophysiological evidence that the non-monotonic functions may be the result of two classes of receptors whose responses are concentration-dependent and/or cells that themselves have non-monotonic responses as a function of concentration due to inhibitory mechanisms in the olfactory bulb.

S4-P9

Axon-bearing, GABAergic/dopaminergic interneurons in the rat olfactory bulb: a cellular substrate for directed lateral inhibition between glomeruli?

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In the vertebrate and insect olfactory systems, odors are encoded as combinatorial patterns of activity across glomeruli, the brain's first olfactory processing units. In the insect antennal lobe, local interneurons showing oligo-glomerular, patchy innervation patterns implement a directed, glomerulus-specific form of lateral inhibition. Similarly, functional directed interglomerular inhibition exists in the vertebrate olfactory bulb (OB), but its cellular substrate remains elusive. Recently, glomerular dopaminergic/GABAergic interneurons have emerged as potential convergent analogues of the insect's oligo-glomerular neurons. Using acute, horizontal OB slices from transgenic rats expressing GFP under the tyrosine hydroxylase (TH) promoter, we performed in vitro whole-cell recordings from 43 GFP⁺ neurons, monitored spontaneous and evoked excitatory postsynaptic currents (EPSCs), and reconstructed their morphology using biocytin-labelled samples. Our preliminary data analysis shows that glomerular TH/GFP⁺ neurons (i) fire at rates that increase linearly with depolarization, showing weak adaptation; their key firing properties were reliably reproduced by a data constrained point neuron model (Hodgkin-Huxley), based on 4 membrane conductances: gNa⁺, fast inactivating; gK⁺, delayed rectifier; gCa²⁺, high-threshold, and gSK⁺, Ca²⁺-gated; (ii) present one or multiple axons that extend up to ~1 mm beyond the juxtaglomerular space of their "home" glomerulus, showing a clearly directed anisotropic glomerular innervation pattern (reconstruction of both glomeruli and axons for N = 3 cells); (iii) receive spontaneous and evoked EPSC bursts, likely originating from ensembles of bursting external tufted cells. Our results are in line with the hypothesis that long-range dopaminergic/GABAergic interneurons implement directed interglomerular lateral inhibition in the vertebrate OB that could contribute to configural mixture perception. Funding: DFG 430157889-SPP2205 Priority Program.

S4-P10

Auditory and visual object processing in olfactory cortex of individuals with life-long olfactory deprivation

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The traditional view of modality-specific brain organization has been challenged by demonstrations of cross-modal processing, i.e. the activation of specialized sensory cortex by input from other senses. An alternative theory suggests that rather than specializing in different sensory modalities, different parts of the cortex can specialize in different types of tasks. We recently addressed this question in the olfactory domain, showing that in normosmic individuals, the primary olfactory cortex (piriform cortex) responds to unimodal auditory and visual objects, regardless of how strongly they are associated with an odor. Here, we assess whether previous olfactory experience is a prerequisite for this cross-modal processing, using a unique group of individuals born without a sense of smell (congenital anosmia; CA; n = 30). First, we confirmed the presence of clear visual and auditory activations in the piriform cortex of the sensory deprived individuals. As compared to normosmic controls (n = 30), these activations and associated functional connectivity, both within the piriform cortices and between piriform cortex and other regions, were not only present but altered in a modality dependent way. Our results show that life-long absence of olfactory input does not impede cross-modal activations by visual and auditory objects in the piriform cortex. Further research will be needed to assess whether this activation is specific for objects, in line with the idea of a task-based rather than modality-specific organization of this brain region.

S4-P11

A flexible, semi-naturalistic arena for quantification of complex mouse behavior

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Animals continuously acquire sensory information from their environment and dynamically adapt their behavior in response to changing conditions. In mammals, such as mice, behavioral output arises from the integration of multiple factors, including resource availability, social context, and potential threats. Reproducing this complexity under controlled laboratory conditions remains a major challenge in behavioral neuroscience. Here, we introduce a semi-naturalistic and highly modular experimental platform designed to combine experimental control with ethological relevance. The system enables automated, high-throughput behavioral experiments with minimal experimenter interaction and supports continuous acquisition of high-resolution video data. In combination with pose-estimation-based analysis using DeepLabCut, this framework allows for scalable and quantitative characterization of behavior. A key feature of the platform is its modular architecture, which allows rapid reconfiguration of experimental conditions without the need for new hardware setups. We validate the system across three chemosensory proof-of-concept paradigms: (i) detection of olfactory-driven behavioral modulation in single animals, (ii) integration of a classical three-chamber assay within the same environment to demonstrate configurational flexibility, and (iii) extension to multi-animal settings for the analysis of social interactions. Overall, this approach provides a versatile and robust tool for studying complex mouse behavior under controlled yet ecologically meaningful conditions, enabling a closer alignment between laboratory experiments and naturalistic behavior.

S4-P12

Chemosensory function and ingestive behavior in adults using GLP-1 receptor agonistsGiorgia Rutigliani¹, Richard D. Mattes¹¹ *Department of Nutrition Science, Purdue University, West Lafayette, IN, USA.*

Glucagon-like peptide-1 receptor agonists (GLP-1RAs), initially developed for type 2 diabetes, are now approved for chronic weight management. In addition to improving glycemic control, GLP-1RAs reduce energy intake by ~20–35% and body weight by ~8–20%, depending on agent and dose. Commonly cited mechanisms include delayed gastric emptying, reduced appetite, and shifts in food preferences. However, changes in appetite ratings are often modest, suggesting that other behavioral mechanisms, including changes in cravings, food reward, and responses to external food cues, may also contribute to reduced energy intake. Emerging evidence suggests that GLP-1RAs may alter chemosensory function, although human findings remain inconsistent. This study examined associations between GLP-1RA use, chemosensory function, and ingestive behavior across three groups differing in duration of use. Adults were classified as pre-use, ~1-month users, or ~6-month users of GLP-1RAs. Outcomes included taste detection thresholds, suprathreshold taste intensity ratings, fat taste sensitivity, olfactory function, attentional bias toward food, and self-reported food cue responsiveness, cravings, food reward, and food-related intrusive thoughts. Sweet detection thresholds were higher in 1-month ($p = 0.008$) and 6-month users ($p = 0.002$) than pre-use individuals. Salty thresholds were also elevated in 6-month users ($p = 0.040$). Suprathreshold intensity ratings and olfactory function did not differ among groups. Food cue responsiveness, trait food cravings, food reward, and food-related intrusive thoughts were lower in 6-month users (all $p < 0.05$). Attentional bias toward food was also reduced at 6 months ($p = 0.046$). GLP-1RA use was associated with selective changes in taste sensitivity and reduced food reward, cravings, and food-cue reactivity, supporting a role for external food cues alongside interoceptive signals in GLP-1RA-related eating behavior. Sources of funding: Indiana CTSI

S4-P13

Desensitization, Adaptation and Habituation to Odors: Reconciling Terminology, Mechanisms, and Methods in Human Olfactory Research

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A core phenomenon in olfactory research is the decline in odor-related responsiveness during repeated or sustained exposure. Yet the field lacks a consistent mapping between phenomenon labels and mechanistic claims, because habituation, adaptation, and desensitization are used inconsistently across studies, species, and outcomes. A major source of confusion is that classic habituation frameworks define habituation as a response decrement that excludes sensory adaptation or fatigue and is diagnosed by criteria such as stimulus specificity, spontaneous recovery, and dishabituation, whereas much of human olfaction research quantifies the phenomenon through perceptual, behavioral, and physiological indices that do not always allow unambiguous locus attribution. In this perspective piece, we propose an explicit operational reconciliation: we use olfactory desensitization as a holistic phenomenon label for the measurable decrease in odor-related responsiveness, always anchored to the specific index used in a given study (e.g., desensitization in terms of perceived intensity, electrophysiological responses, or neuroimaging signals). We reserve adaptation for receptor-level or peripheral processes and habituation for central or cortical processes, while recognizing that peripheral and central processes can unfold in parallel and jointly shape the observed change. We then synthesize how key design features, including but not limited to stimulation schedule, short versus longer persistence windows, recovery characterization, and olfactory-trigeminal and attentional influences, determine what can and cannot be inferred about peripheral versus central contributions. Our goal is to provide a defensible conceptual framework and a practical methodological guide that offers an accurate classification of studies, improves interpretability and comparability across experimental paradigms and supports more precise hypothesis formation and inference for future human olfactory research.

S4-P14

Emotional and sensory processing, but not resting blood pressure, are associated with variations in olfactory perception

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Nonclinical elevations of resting blood pressure are associated with emotional dampening and reduced empathy. Given that empathic and emotional responses can also be triggered by olfactory stimuli, we examined whether resting blood pressure is associated with olfactory performance. We tested blood pressure and olfactory threshold, identification, and hedonic evaluations of Sniffin' Sticks odors (n=139). We also collected measures of sensory processing, anxiety, and alexithymia (difficulty in identifying emotions), all of which are known to vary with olfactory performance and emotional response. We did not find a relationship between resting blood pressure or anxiety and any measure of olfactory performance. However, results did suggest a link between sensory processing as measured by the Adolescent/Adult Sensory Profile (AASP) and odor identification scores, perceived odor intensity, and familiarity. Those with higher scores on AASP subscales of sensory sensitivity and sensation avoiding performed significantly worse in the olfactory identification portion of the Sniffin' Sticks TDI task ($r(135) = -.187$, $p = .029$ and $r(135) = -.22$, $p = .01$ respectively). Individuals with higher AASP scores in sensory sensitivity and sensation seeking perceived odors to be higher in intensity ($r(137) = .26$, $p = .002$ and $r(137) = .178$, $p = .036$), while those scoring higher in the externally oriented thinking (EOT) subscale of the Toronto Alexithymia Scale (TAS-20), perceived odors to be of lower intensity ($r(139) = -.17$, $p = .046$). Finally, those higher in the AASP sensation seeking subscale perceived odors as more familiar ($r(137) = .227$, $p = .007$). We did not compare food and non-food odors or investigate perception of social odors, which might have revealed additional differences in olfactory perception. Our results demonstrate the complex interactions among psychological, behavioral, and physiological factors that influence our everyday sensory evaluations of the world around us.

S4-P15

Dwarfing glutamic acid: Discovery of glutamic acid derivatives as highly potent umami receptor (TAS1R1/TAS1R3) modulators

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L-Glutamic acid is a common (e.g., the most abundant among free amino acids in human milk) endogenous amino acid and a ligand of the umami taste receptor TAS1R1/TAS1R3. Its monosodium salt, MSG, is widely used to deliver umami taste. Although regulatory agencies including the U.S. FDA and EFSA consider MSG safe at normal intake levels, it has been the subject of persistent public safety controversy due to anecdotal reports of transient symptoms after consumption. Thus, there is a need to lower the level of or provide alternatives to MSG as an umami flavor. Unfortunately, although many alternatives have been described, the most potent and effective are often synthetic, whereas consumer preference is shifting toward natural solutions. In this work we sought to identify glutamic acid derivatives that outperform L-glutamate at the human umami receptor TAS1R1/TAS1R3 and impart significantly improved taste profiles. Preference was given to natural or nature-inspired analogues. Although previously reported glutamate derivatives show reduced umami activity (vs glutamic acid), we believed that the opportunities to improve were not exhausted. Thus, we designed and synthesized a series of benzoyl- and cinnamoyl-linked glutamic acid chemotypes. Functional screening in TAS1R1/TAS1R3 cell-based in vitro assays identified multiple active compounds, including Cpd-A, with EC50 values in the low-micromolar range, compared with over 2000 μ M reported and confirmed under the same assay conditions for MSG. Originally nature-inspired, Cpd-A was later validated as a natural product. Human sensory evaluation showed that Cpd-A has a substantially stronger umami taste than MSG and that it also significantly enhances the umami intensity of MSG beyond additivity when used in combination. Funding: This work was fully funded by DSM-FIRMENICH

S4-P16

Obesity selectively impairs palatable taste-guided licking behaviour and is reversed by GLP-1 receptor agonists

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Obesity is associated with altered taste-guided feeding behaviour in humans, yet the underlying neural mechanisms remain unclear due to environmental and lifestyle variability. Here, we evaluated the effects of obesity on taste-guided licking behaviour and whether such changes can be selectively reversed by GLP-1 receptor agonists (GLP-1 RAs). Male C57BL/6J mice were maintained on either standard chow or a high-fat diet (60% kcal from fat) until the development of obesity. Semaglutide (41 µl/kg/day) was administered daily for at least 11 days before behavioural testing. Brief-access taste tests were performed across multiple conditions, including lean, diet-induced obesity (DIO), GLP-1 RAs administration, pair-feeding (PF), and 50% dietary restriction (DR). Licking behaviour was assessed across multiple taste modalities in separate 10-second trials. Obesity selectively reduced licking responses to palatable taste stimuli, including sweet, fat, and umami, with minimal changes observed to aversive taste stimuli, such as bitter and sour, indicating that obesity impaired taste-guided licking behaviours in a valence-specific manner. By contrast, GLP-1 RA administration but not PF or DR robustly restored this impairment. Since obesity is known to decrease the cell-surface expression of Gai-coupled dopamine D2 receptors (DRD2) in the striatum, thereby altering various reward-related behaviours, we next investigated a causal link between DRD2 and impairments in taste-guided licking behaviours. Chemogenetic rescue experiments of Gai signalling by introduction of inhibitory DREADDs (hM4Di) into DRD2-expressing neurons of Adora2a-Cre mice showed that the caudate putamen (CPu) in the striatum is important for appetitive taste-guided licking behaviours. These results suggest that GLP-1 RAs can restore altered taste preferences in obese mice. We are currently investigating whether GLP-1 RAs increase the cell-surface expression of DRD2 in CPu during obesity.

S4-P17

Olfactory communication and biocontrol potential in the European water vole

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In many rodent species, olfaction plays a central role in social and sexual communication through chemical signals. In France, outbreaks of the European water vole (*Arvicola terrestris*) cause severe agricultural damage and raise public health concerns due to their role as reservoirs of zoonotic pathogens. Conventional control based on anticoagulant rodenticides has led to environmental contamination, non-target effects, and resistance, resulting in increasing restrictions. Therefore, alternative environmentally friendly strategies such as biocontrol based on olfactory cues are being explored to improve trapping efficiency. The European water vole is a monogamous burrowing rodent that produces urine signals and lateral scent gland secretions potentially involved in social interactions. Previous studies showed attraction of both males and females to male urine, whereas no response to lateral gland secretions was detected under experimental conditions. Chemical analyses identified numerous volatile compounds in both sources, including sex-specific molecules, but their behavioral relevance remains largely unknown. This project aims to further investigate the attractive effects of selected olfactory compounds using behavioral preference tests conducted under controlled laboratory conditions and under field conditions through capture trials. In laboratory assays, individual responses to defined odor stimuli are precisely quantified in standardized experimental setups. In parallel, field experiments evaluate the same candidate compounds in real-world conditions by measuring their ability to enhance trapping success. This parallel approach enables direct comparison between controlled behavioral responses and ecological effectiveness, providing a robust assessment of both attractiveness and potential applicability of olfactory cues for biocontrol purposes.

S4-P18

Initial Evaluation of Generative AI Responses within the Hub4Smell Smell Test Data Repository Compared to Human Expert Analyses

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Objectives: Access to olfactory datasets remains fragmented across institutions, limiting large-scale analysis and discovery. Hub4Smell is a domain-specific repository designed to enable integration and scalable analysis of anonymized smell test data. We evaluated a generative artificial intelligence (GenAI) interface embedded within Hub4Smell to determine whether its responses to user queries align with expert human analyses, against a target agreement threshold of $\geq 95\%$. **Methods:** A GenAI system was integrated into Hub4Smell to support conversational querying of curated smell test datasets from human participants. A structured protocol was implemented in which two human experts independently answered a standardized set of prompts using only the repository data accessible to the AI. AI outputs were compared to expert responses using predefined accuracy metrics, including concordance scoring and qualitative error-type classification. **Results:** The GenAI interface produced coherent and contextually relevant responses across multiple query domains, including descriptive statistics, demographic trends, population-level patterns, and data quality assessments. Agreement with human expert responses was observed in 27 of 30 queries (90%). Identified discrepancies were primarily attributable to misinterpretation of query intent, inconsistent terminology alignment, and incomplete response generation (e.g., omission of result sets). These findings highlight key areas for improvement, including prompt optimization and implementation of guardrails to manage uncertainty. **Conclusions:** The Hub4Smell GenAI interface demonstrates substantial alignment with expert human reasoning in the analysis of olfactory datasets. Iterative refinement is expected to improve performance toward and beyond the 95% agreement benchmark, supporting scalable and user-friendly access to complex olfactory data and accelerating research in human olfactory science. **Funding:** NIDCD: R44DC022498 and P50DC022549.

S4-P19

Flavor-flavor and flavor-nutrient conditioning with odors: competition or amplification?

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You are hungry, and the sweet scent of vanilla draws you in to buy a cake. Two forms of conditioning explain this behavior: flavor-flavor (FFc) and flavor-nutrient (FNc), where associations with a rewarding taste or with postprandial metabolic effects are formed. Whether these two processes amplify, inhibit, or remain independent of each other remains unclear. To answer this question, fasted participants completed two sessions during which FFc and FNc were implemented and disguised via triangle tests across four blocks of five trials each. Each trial consisted of tasting three solutions and indicating whether one cup differed from the others. Of the five trials (randomized), two were conditioning trials, with three cups of either 1) an unfamiliar odor mixed with sucralose (sweet-paired), and 2) another unfamiliar odor left unpaired. The remaining trials served as distractors and included one odd stimulus among two others (coconut/almond odor, saline, or water). Each block was preceded by a caloric beverage (nutrient-paired) or water (control), depending on the session (FNc). Associative learning and metabolic responses were assessed via odor ratings (wanting, sweetness, liking). Blood glucose measures collected throughout the session. Results showed that FFc selectively increased the perceived sweetness of the sweet-paired odor, while other evaluations remained unchanged. Inter-individual analyses revealed a modulatory effect of sucralose liking: the more participants liked the sucralose solution, the larger the increase in both liking and wanting for the sweet-paired odor. FNc prevented a decrease in liking and wanting. Only the nutrient non-paired odors showed reduced liking and wanting after conditioning. The changes in wanting were related to individual differences in glucose processing, highlighting a selective link between metabolic responsiveness and motivational value. This study was supported by the ERC under the H2020 program (Grant No. 947886).

S4-P20

Evolution of pheromone detection in grain weevils

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Weevils of the genus *Sitophilus* are among the most destructive pests of stored cereals and cereal products worldwide. In particular, the rice weevil (*S. oryzae*), the corn weevil (*S. zeamais*) and the granary weevil (*S. granarius*) cause significant grain losses. Like many weevils, these species rely on male-produced aggregation pheromones to locate and colonize new grain resources. *S. oryzae* and *S. zeamais* produce (4S,5R)-5-hydroxy-4-methylheptan-3-one (sitophinone), whereas *S. granarius* emits (2S,3R)-1-ethylpropyl 2-methyl-3-hydroxypentanoate (sitophilate). In insects, pheromones are detected by olfactory sensory neurons that express odorant receptors (ORs). Analysis of the *S. oryzae* genome recently identified 100 OR genes, but their functions remain unknown and the pheromone receptors have not been identified. Here, we first used transcriptomics to identify ORs and quantify their expression levels in the antennae of the three *Sitophilus* species, hypothesizing that pheromone receptors would exhibit elevated expression levels. The OR repertoire was largely conserved, except for one OR whose sequence was more variable and which was highly expressed in all three species. We then studied its response spectrum in each of the three species using heterologous expression in *Drosophila* olfactory neurons. This revealed that this OR is indeed an aggregation pheromone receptor, tuned to sitophinone in *S. oryzae* and *S. zeamais*, and to sitophilate in *S. granarius*. These results demonstrate that a single OR adapted to detect two structurally distinct pheromones.

S4-P21

Environmental pollution in a dish: Modelling pollution-induced effects on taste and nutrient sensing using taste bud organoids

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The human taste system remains mechanistically unresolved and highly vulnerable to environmental toxicants, yet lacks physiologically relevant human models. Human induced pluripotent stem cells (hiPSCs) provide a diploid, karyotypically normal and tractable platform to generate complex 3D organoids. Here, we establish an hiPSC-derived taste bud organoid model by temporally modulating Wnt, cAMP, SHH, and JAK-STAT signaling while suppressing BMP and TGF β pathways. This approach yields stable, proliferative 3D cultures expressing key taste lineage markers. Immunocytochemistry and qPCR confirm expression of taste receptors (TAS1R3, TAS2R38), transduction molecules (GNAT3, SNAP25, CA4, PLC β 2, TRPM5), and epithelial markers (KRT14, KRT8). Organoids further express POU2F3, a master regulator of chemosensory cell identity, and LGR5, a stem/progenitor marker associated with epithelial and lingual stem cell populations. Thus, we generated a POU2F3-knockout hiPSC line, which will enable mechanistic interrogation of taste cell differentiation in combination with functional studies, including calcium imaging in response to sweet and bitter compounds and complementary protein and transcriptomic analyses. This work establishes a human organoid model to study taste biology and its disruption, opening new avenues in sensory research, toxicology, and nutrition. The project is funded by the Leibniz Association and part of the EpiDish project.

S4-P22

Decoding sensory memories of smells and sounds in a Virtual Reality environment.

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Our brains create internal representations of environments that we move through, interact with, and memorize. These representations are encoded as sets of cognitive maps in the hippocampal and entorhinal cortical networks. Successful restructuring of the maps is crucial for separating and accurately retrieving spatial memories. In humans, research has focused on visual cognitive maps, but little is known about the formation and restructuring of these maps based on other sensory modalities. We hypothesized that sensory-specific cognitive maps emerge for olfactory and auditory spatial memory representations in a constant visual environment. Participants (n=27) memorized locations of smells and sounds (olfactory and auditory conditions) linked to four objects: coffee, lawnmower, bonfire, and popcorn, within a Virtual Reality garden environment. Then, participants were presented with the smells and sounds and asked to navigate and place them in the correct location as precisely as possible, while their brain activity was recorded in an fMRI scanner. We assumed that successful remapping would enable above-chance condition decoding of hippocampal activity during navigation to the recalled locations of olfactory versus auditory cues. We thus performed multivariate pattern classification of recall episodes in spatial and sensory brain regions. Results showed better than chance condition classification from all selected regions of interest, suggesting multiple widespread sensory-specific maps. Additionally, we explored neural decoding of individual sensory objects in the same brain regions. Results showed that individual sounds (e.g., sound of a lawnmower) could be successfully classified only in the primary auditory sensory cortex. In contrast, smells could be classified in a wider network, including sensory regions, hippocampus and entorhinal cortex. These results support the notion of strong olfactory integration with the spatial memory systems in the human brain.

S4-P23

GraphNOSE: a graph transformer in olfaction for single molecule and mixture prediction

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Predicting olfactory qualities from molecular structure is an open problem. Although linear models can link molecular features to human-rated odor descriptors and show strong predictive power for well characterized regions of the chemical space, they often fail when extrapolating to novel chemical scaffolds, extreme molecular weights, or complex odor mixtures. Existing databases range from small, richly annotated datasets to larger but sparsely labeled corpora, leaving broad regions of chemical space without perceptual annotation. Whether non-linear modeling is needed across this range remains unclear. To address this gap, we introduce GraphNOSE, an open-source graph transformer framework that predicts multi-label odor descriptors from SMILES strings for single molecules and binary mixtures. GraphNOSE was trained on a curated dataset (GoodScents and Leffingwell). By integrating positional and structural encodings within a transformer-based graph architecture, GraphNOSE achieves strong performance with 6× fewer parameter than standard Graph Neural Network (GNN) baselines while consistently outperforming molecular language model embeddings, molecular fingerprints, and baseline GNNs by an average AUROC margin of 4.7% ($p < 0.001$). On novel scaffolds or molecular weight ranges outside training data, GraphNOSE achieves an AUROC of 84% on out-of-distribution (OOD) compounds. This exceeds the current state-of-the-art GNN for OOD in olfaction (OpenPOM: 81%, $p < 0.001$), and identifies conditions under which linear models empirically fail. Finally, we apply Explainable AI (XAI) methods that identify which substructures and molecular features drive odor predictions, providing mechanistic insight grounded in the model's learned representations. Together, these results suggest GraphNOSE as a scalable and interpretable architecture for olfactory prediction that generalizes to structurally novel compounds underrepresented in current perceptual databases.

S4-P24

Genomic and behavioral evidence for social olfaction in white-faced capuchins

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Olfaction is a key sense for many mammals, but its role in the social lives of anthropoid primates (monkeys, apes, humans) is still poorly known. Studies of chemosensory receptor genes and olfactory behaviors can provide insight into the evolution and current uses of olfaction in social contexts. Vomeronasal receptor (VR) genes are especially relevant for social olfaction as they are specialized for the detection of pheromones and other cues involved in social signaling, reproductive behavior, and individual recognition in mammals. However, they are of uncertain significance in anthropoid primates, who have reduced olfactory morphology and chemosensory gene repertoires relative to many mammals. Here, we examined the VR gene repertoires and genetic diversity in populations of wild white-faced capuchin monkeys (*Cebus imitator*, parvorder Platyrrhini) from Costa Rica and Panama using DNA sourced through fecal FACS (fluorescence-activated cell sorting from noninvasively collected fecal samples) or tissue. Using neutral loci, we also estimated background levels of variation and accounted for the effects of demography. We annotated 2 intact, 4 disrupted, and 1 truncated VR gene. Tajima's D was significantly lower for intact VR genes than neutral loci, consistent with potential purifying selection. For both intact VR genes, intact haplotype frequencies were $\geq 80\%$. We complement the genomic analyses with a detailed observational study of wild capuchins, documenting a rich and varied ethogram of olfactory behaviors with a clear potential for social communication in this species. Overall, we

provide evidence that VR genes are maintained by natural selection in a wild platyrrhine monkey. Our study provides compelling evidence of a role of olfaction in the rich social world of capuchins and a basis for future research to further explore the factors shaping contributions of olfaction to social communication and evolution in primates and other mammals.

S4-P25

Trajectory of retronasal olfactory function in patients with chronic rhinosinusitis with nasal polyps treated with biologics: a prospective longitudinal study

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Background: Olfactory dysfunction is a key feature of chronic rhinosinusitis with nasal polyps (CRSwNP). While biologics improve orthonasal olfaction, longitudinal data on retronasal function and differences between inflammatory targets remain limited. This study evaluated retronasal olfactory recovery over time during biologic therapy and compared IL-4/IL-13 versus IL-5 pathway blockade. **Methods:** In this prospective longitudinal study, adults with severe CRSwNP starting biologics were assessed at baseline and after 4, 12, and 24 weeks. Orthonasal olfaction was measured using Sniffin' Sticks (TDI), and retronasal function with a validated 20-item identification test. Clinical outcomes included nasal polyp score (NPS), peak nasal inspiratory flow, and SNOT-22. Repeated-measures ANOVA was used to evaluate longitudinal changes and time-by-treatment interactions. **Results:** Sixty-nine patients were included (44 dupilumab, 25 mepolizumab). At baseline, median TDI was 15.0 and retronasal score 7.2, indicating severe impairment. All outcomes improved significantly over time ($p < 0.001$). Retronasal function improved rapidly within 4 weeks (7.2 to 10.4). TDI increased progressively (15.0 to 26.0 at 24 weeks). Significant time-by-treatment interactions were observed for retronasal identification, TDI, SNOT-22, and NPS, indicating non-parallel recovery patterns. Dupilumab was associated with greater improvements in retronasal function and symptoms compared with mepolizumab. **Conclusions:** Biologic therapy leads to substantial recovery of both retronasal and orthonasal olfaction in CRSwNP, with distinct temporal dynamics. Retronasal function shows rapid improvement and may be particularly sensitive to IL-4/IL-13 blockade, supporting its role as a functional endpoint.

S4-P26

Olfactory discrimination in four year old children using a simplified two alternative forced choice task - preliminary results

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The sense of smell is essential for personal safety, enabling detection of threats such as gas leaks, smoke, or spoiled food. Olfactory discrimination is a key olfactory ability to navigate in the environment and distinguish threatening odors from the harmless olfactory background. Women are known to better discriminate odors, but the origin of this advantage is unclear. The aim of the present study is to verify if this difference is innate or learned throughout development and compare boys and girls as young as 4 years old to compare their ability to discriminate odors. Odor discrimination is typically measured using the Sniffin' Sticks three-alternative forced-choice (3AFC) paradigm, which places substantial cognitive demands. Therefore, we implemented a simplified two-alternative forced-choice (2AFC) procedure based on Cameron's (2024) approach, using fewer trials (8 vs. 16), and involving odors familiar to children. The study included 229 children (114 boys, 115 girls). Mean performance of the group was 5.24 out of 8 points scale. Girls ($M = 5.31$, $SD = 1.54$) achieved similar scores to boys ($M = 5.16$, $SD = 1.52$), ($t(227) = 0.77$, $p = .443$). These findings lend credit to the hypothesis that the advantage of women over men is not present in early childhood but arises in later development. These results are baseline measurements for the ongoing longitudinal study that will allow to address this hypothesis.

S4-P27

Associative learning between intranasal electrical stimulation patterns and olfactory concepts

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Objective: We aim to investigate the feasibility of associative learning between eight intranasal electrical stimulation patterns and eight olfactory concepts in healthy individuals. Demonstrating such associative learning to substitute olfactory sensations may provide a conceptual basis for the future development of olfactory implants. **Methods:** The study was conducted as a three-session clinical trial with healthy participants (n=44), who underwent either associative learning (n=24) or a perceptual control condition. The associative learning group completed a training sequence in which they learned to associate the intranasal trigeminal stimuli to the olfactory concepts, followed by a testing sequence in each session to assess recognition performance. The stimulation parameters were derived from prior pilot experiments to maximize perceptual separability. The perceptual group was exposed to the same sequences in an equivalent order but without any association between electrical stimulation patterns and olfactory concepts. The third and final session included an identical testing sequence for both groups to compare effects of associative and perceptual learning. **Results:** Preliminary results suggest that all participants in the associative learning group (n = 24) improved in associating electrical stimulation patterns with olfactory concepts from the first session (26%) to the third session (40%) ($F[2, 44]=25.1$, $p<0.001$; all group comparisons $p<0.05$). In contrast, the perceptual learning group (n = 20) reached significantly lower scores in the third testing (15%, $p<0.001$) compared to the associative learning group. **Conclusions:** These findings indicate that associating electrical trigeminal stimulation patterns with olfactory concepts is feasible. This approach may inform the development of an olfactory prosthesis based on trigeminal substitution.

S4-P28

Machine Learning-Driven Discovery of Behaviorally Active Odorants Across Insect Species and Humans

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Insect-borne diseases including malaria and Dengue fever pose devastating global health burdens, with mosquitoes alone responsible for millions of deaths annually. Chemical repellents represent a frontline defense, yet rational discovery of effective odorants is profoundly challenging — not only due to the complexity of insect olfactory systems, but because repellents must simultaneously possess pleasant scent profiles to ensure human compliance. Synthetic repellents like DEET fall short on both fronts, driving urgent need for better alternatives. We developed chemical structure-based machine learning (ML) models to predict odor valence across species with evolutionarily unrelated olfactory receptors: humans (~400 GPCRs), *Drosophila* (~100 ion-channels), honey bees (>180 receptors), and mosquitoes. For humans, models predict across 146 distinct odor character categories, enabling systematic exploration of how molecular structure governs scent perception and hedonic valence. Screening chemical spaces exceeding 10 million compounds, we identified odorants that are powerfully repulsive to mosquitoes while scoring as pleasant and fragrant to humans — outperforming DEET in both repellency and sensory appeal, directly addressing the challenge of developing repellents people will actually use. We further extended this framework to honey bees (*Apis mellifera*) to protect pollinators from pesticide-treated crops. Models refined through iterative species-specific behavioral assays were applied to screen >50 million compounds, yielding >130 repellent candidates confirmed through laboratory and open-field validation. Together, these studies establish iterative ML modeling as a scalable framework for rational discovery of behaviorally active volatiles, with transformative applications in vector control, fragrance design, and pollinator protection.

S4-P29

The impact of Sex Hormones on Nasal Respiratory Patterns in Women

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During the menstrual cycle, women experience fluctuations in Estrogen and Progesterone. During pregnancy, these sex hormones surge, but are suppressed when using hormonal contraceptives (HC). Progesterone is a known respiratory stimulant in pregnancy, whereas Estrogen's role remains debated. Given potential links between sex hormones and nasal respiration, we hypothesized that respiration varies systematically across the menstrual cycle, but less so with HC use. To test this, we recorded nasal airflow using the Nasal Holter, a novel wearable for continuous 24h respiratory monitoring. Naturally cycling (NC) women were measured in three 24h sessions: menses, pre-ovulation, and luteal phase; HC users were measured twice on-HC and once off-HC. Standardized nasal flow features were extracted separately for wake and sleep and compared across phases. Thus far (n=13 NC women), Duty Cycle Inhale (proportion of the breath cycle spent inhaling) during sleep—but not wake—varied across the cycle: longest during menses (0.35 ± 0.02), shorter at ovulation (0.34 ± 0.02), and shortest in the luteal phase (0.33 ± 0.04) ($F(2,24)=4.02$, $p=0.03$, $\eta^2=0.25$). The Coefficient of Inhale Duration showed an opposite trend ($F(2,24)=3.31$, $p=0.054$). A Linear Discriminant Classifier using respiratory features identified cycle phase with 79.5% accuracy (chance=33.33%, $p<0.0001$). These findings support systematic variation in nasal airflow across the menstrual cycle. The observed changes reflect timing rather than volume, suggesting a central (brain) mechanism of respiratory pace-making rather than peripheral musculoskeletal effects. This previously unidentified nasal airflow shift may relate to concurrent olfactory shifts previously described along the menstrual cycle. Given the ease of measurement, it may have practical implications for women's health and fertility. This work was funded by ISF grant 232/21 awarded to Noam Sobel.

S4-P30
=====**A diet experience with two metabolically distinct sugars reprograms the oral glucosensing.****Sandrine Chometton¹, Lindsey A. Schier²**¹ *Université Bourgogne Europe, Institut Agro, CNRS, INRAE, UMR CSGA, 21000 Dijon, France*² *Department of Biological Sciences, University of Southern California, Los Angeles, CA, USA*

Glucose is an essential source of energy for all living organisms. Because this nutrient is mainly provided by the diet, it is necessary for the body to rapidly detect and motivate the ingestion of glucose-containing substances. The oral taste system is critical for recognizing nutrients in the environment and initiating ingestion. Chemically- and metabolically-diverse compounds including simple sugars, like glucose, low-calorie sweeteners, and D-amino acids engage a common “sweet taste” receptor (TAS1R2+TAS1R3). Recent work demonstrates that experience with the post-ingestive effects of two metabolically distinct sugars, glucose and fructose, enables animals to subsequently discriminate these two initially similar-tasting compounds based on orosensory information, and generates a preference for glucose over fructose. This result has also been observed in mice lacking the canonical sweet taste receptor, showing that a TAS1R-independent pathway is involved. We thus aim to understand the gustatory signal necessary for the taste discrimination of glucose and modulated by a post-ingestive experience with sugars. We showed that sweet-sensitive sugar-exposed mice have a higher glucokinase mRNA expression in the taste system compared to sweet-sensitive sugar naïve mice, and that this enzyme plays a role in glucose discrimination in brief-access taste tests. Furthermore, we demonstrated that the glucose transporter SGLT1, but not the enzyme glucokinase, plays a role in glucosensing in sweet-insensitive sugar-exposed TAS1R2+TAS1R3 knock-out mice. These results show that the post-ingestive effects of sugar reprogram metabolism-dependent and -independent glucosensing pathways in the taste bud cells. Overall, these findings expand our understanding of orosensory mechanisms underlying glucose appetite. Sources of funding: National Institutes of Health (NIH) and Institutional Start-Up funds from the University of Southern California (USC).

S4-P31

Exploring barriers to seeking support for Smell and Taste Disorders (SATDs) amongst ethnic minority communities in the UK.

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Objectives of the study. The objective of the study was to identify the challenges that need to be addressed to improve the support received by people from ethnic minority communities in the UK, with a focus on Smell and Taste Disorders (SATDs). **Methods used.** A qualitative study was undertaken, using semi-structured focus group discussions and interviews, augmented by an online survey. 9 focus group discussions and 2 interviews were conducted with a total of 39 participants. The survey received 89 responses. Participants ranged from the following ethnic groups: African, Indian, Caribbean, Pakistani, Chinese, Arab, Bangladeshi, and mixed ethnic groups. The responses were thematically coded to identify the following: - Key challenges faced by minority communities in seeking and receiving support for SATDs - Strategies that they feel will help them seek and receive support for SATDs. **Essential results.** Five challenges were identified which hindered the participants from receiving and seeking support for SATDs. These were: discrimination, cultural differences, communication barriers, other responsibilities, and unfamiliar systems. The strategies identified to address these challenges were: stakeholder involvement, training and education of staff, diversification of staff, improvement of language accessibility, awareness campaigns about SATDs, support groups, and improvement of accountability in healthcare settings. **Conclusions.** The identified challenges, and existing healthcare strategies and relevant literature were reviewed to create strategies to improve SATD support and healthcare for people from ethnic minority communities in the UK. Future work needs to be done to explore the accommodation of intersecting ethnic identities as well as to explore challenges in support for SATDs throughout the SATD process, from diagnosis to support. **Identification of sources of funding-** Norfolk & Waveney ICB Research Capability Funding- UEA MRes Grant- ENT UK Grant

S4-P32

Object or cake? Selective processes in the human brain for the recently discovered food area in the ventral visual stream

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The human ventral visual cortex shows category-selective organization to recognize and discriminate between ecologically important categories like faces, words, body parts, and places. The food-image-selective region in the ventral stream has been identified by recent data-driven studies, which is called the ventral food stream. The aim of this study is to investigate the ventral food stream by employing a novel object-food illusion: a hyper-realistic cake. In hyper-realistic cake videos, the initial object perception shifts to a perception of cake throughout the video. We conducted two different experiments. In Experiment 1, we collected behavioral data from 22 participants (mean age=21.24 ±1.33; age range: 18-24; 12 female) to validate the hyper-realistic cake videos. In Experiment 2, we collected data from 25 participants (mean age=24.48 ±1.33; age range: 18-55; 13 female). In the first fMRI session, we applied a functional localizer one-back task to localize food-selective regions for each participant. Then we showed validated object-food illusion videos to participants during the second fMRI session. Consistent with the recent findings of food selectivity, we identified the food-selective response in the ventral food stream by using a functional localizer task. During the video session, the ventral food stream exhibits a significant increase in activity when the object perception shifts to perception of a cake. Despite showing spatial variability between participants, there is a selective response to visual food cues in the fusiform cortex. This study shows the food-selective activity in the ventral visual cortex in response to illusory visual stimuli in a hypothesis-driven manner. Further analyses are ongoing, including multivariate pattern analysis to characterize activity patterns in the ventral visual cortex, specifically for food images. This study is funded by TÜBİTAK 1001 project under award number 223K638.

S4-P33

High-frequency transcutaneous auricular vagus nerve stimulation improves olfactory function in patients with idiopathic and post-infectious olfactory dysfunctionSusanne Weise¹, Sarah Elsner¹, Thomas Hummel¹¹ Smell & Taste Clinic, Department of Otorhinolaryngology, University Hospital Carl Gustav Carus, Technische Universität Dresden, Dresden, Germany

Objective: We aimed to evaluate the effects of high-frequency transcutaneous auricular vagus nerve stimulation (taVNS) on olfactory function in patients with olfactory dysfunction (OD) of different etiologies in a placebo-controlled design. **Methods:** This prospective, randomized, sham-controlled clinical trial included 60 patients with OD (30 idiopathic, 30 post-infectious). Olfactory function was measured using the “Sniffin’ Sticks”, gustatory function with “taste strips”, and cognitive function using the “Regensburg Word Fluency Test”. Odor intensity ratings were additionally obtained. All measures were performed before and after 10 minutes of taVNS (80Hz) or sham stimulation. **Results:** TaVNS led to significant improvements in odor threshold ($p < .001$), discrimination ($p = .002$), and overall olfactory scores ($p < .001$) compared to sham treatment in both OD groups. It also enhanced odor sorting ability for phenyl ethyl alcohol ($p = .01$), increased phonemic verbal fluency ($p < .001$), and improved changes in phonemic fluency ($p = .02$). Olfactory performance was related to verbal fluency measures (for taVNS: the change of discrimination, semantic verbal fluency $r=0.38$, $p=0.04$), although no consistent association was observed across all parameters between changes in verbal fluency and olfactory function. Lower initial discrimination scores ($p = .05$) were linked to clinically meaningful gains in smell function. **Conclusions:** The results suggest a functional interaction between the olfactory system and afferent vagal pathways, supporting taVNS as a promising potential treatment option for OD. Additional research is required to better understand the mechanisms involved and to assess long-term effectiveness in larger populations.

S4-P34

How glia shape adaptive plasticity in the mouse olfactory bulb**Tess Stanley¹**, Elisa Galliano¹¹ *Department of Physiology, Development and Neuroscience, University of Cambridge, United Kingdom*

Over the past two decades, glial cells have emerged as key contributors to neuronal plasticity. However, these discoveries are often confined to a narrow subset of limbic structures and sensory cortices, limiting our understanding of how glia shape adaptive plasticity in first-order sensory areas exposed to rapidly changing chemosensory environments. The adult mouse olfactory bulb (OB) provides an ideal model to address this gap, given its location one synapse from the nose and capacity for rapid adaptation. Using brief (1 day) or sustained (7 days) sensory deprivation via unilateral naris occlusion, or sensory enrichment via passive odour exposure, this project investigates how altered sensory input modulates microglial and astrocytic activation in the OB glomerular layer in adult mice of both sexes. Immunohistochemistry, confocal microscopy, and semi-automated morphological analysis were used to assess the density and morphology of microglia and astrocytes stained for IBA1 and GFAP, respectively. Preliminary results suggest that the duration of naris occlusion differentially impacts glia. One day of deprivation produced no significant effects (mixed-model nested one-way ANOVA, all $p > 0.9$) while 7 days of deprivation increased astrocytic ($p < 0.01$) immunoreactivity, alongside reduced microglial branching ($p < 0.0001$), indicating a more activated state. In contrast, enrichment selectively increased astrocytic immunoreactivity after both 1 day and 7 days (all $p < 0.01$). This pilot work lays the groundwork for unravelling how astrocytes and microglia differentially contribute, alongside neurons, to coordinate dynamic olfactory bulb responses to changing sensory input. Future work will adopt longitudinal imaging to characterize changes in glia-neuron interactions during altered sensory input.

S4-P35
=====**Decoding semantic complexity in natural vs. non-natural odor descriptions**

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The linguistic representation of odors is frequently characterized as deficient in Western languages, primarily relying on source-based analogies or subjective hedonic judgments. Despite this apparent limitation, the structural organization of olfactory descriptive strategies remains an underexplored frontier in cognitive science. This study investigates the semantic divergence between descriptions of "natural" and "non-natural" odors by analyzing a large-scale corpus (N = 411) through the lens of computational linguistics. To quantify these descriptive architectures, we developed a Natural Language Processing pipeline focused on lemmatization and data filtering. Cognitive complexity was evaluated using Moving-Average Type-Token Ratio (MATTR) and Shannon entropy, while semantic organization was modeled through co-occurrence networks to analyze graph density and lemma interconnectivity. The results reveal a fundamental shift in descriptive logic depending on the odor's nature. Descriptions of synthetic odors exhibit higher lexical diversity, suggesting a reliance on specialized, technical terminology. Conversely, descriptions of natural odors display significantly higher Shannon entropy and network density. These findings indicate that while natural olfactory language utilizes a more focused vocabulary, it is anchored in a highly integrated and balanced cognitive structure. In contrast, synthetic descriptions appear semantically fragmented, relying on isolated categorical labels rather than interconnected associations.

S4-P36

Does your compound stimulate GLP-1 release? A new screening tool.

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Enteroendocrine cells, comprising about 1% of the intestinal epithelium, are specialized cells that sense nutrients and mediate hormone secretion. Among them, enteroendocrine L-cells are responsible for secreting Glucagon-like Peptide-1 (GLP-1), which is essential for regulating appetite and glucose metabolism. GLP-1 promotes insulin secretion from the pancreas, suppresses appetite and contributes to weight loss. To detect GLP-1 secretion from enteroendocrine cell supernatants, we optimized and validated a GLP-1 secretion assay. We employed NCI-H716 and STC-1 cell lines, well-characterized models for human and murine intestinal L-cells, respectively, along with a reporter cell line that expresses the recombinant GLP-1 Receptor in combination with the chAMPion reporter system. Our high-throughput screening (HTS)-grade GLP-1 secretion assay was functionally assessed using carbohydrate stimuli, including fructose. Additionally, by testing specific receptor agonists, we confirmed a role for Bitter Taste Receptor 38 (TAS2R38) and Free Fatty Acid Receptor 4 (FFAR4, also known as GPR120) in stimulating GLP-1 release from NCI-H716 and STC-1 cells, respectively. In conclusion, we developed a powerful tool for identifying and pharmacologically profiling novel GLP-1 stimuli, demonstrating its applicability for the identification of new therapeutic strategies in the treatment of diabetes and obesity.

S4-P37

Selecting wine stimuli with consistent sensory descriptions across expertise levels

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Wine knowledge involves linking wine identities (defined as a grape variety × region of origin, e.g. Riesling from Mosel) to expected olfactory and gustatory attributes. Little empirical evidence identifies which identities are reliably recognized and consistently described across expertise levels. Stimulus selection in wine cognition and expertise research currently depends largely on convenience or expert intuition. This study aims to provide an empirically grounded alternative. In a cross-sectional online study (target N = 400), wine drinkers in France, spanning the expertise gradient from regular consumers to certified professionals, complete a knowledge task based on the WSET Level 3 recommended sample (70 grape × terroir). Participants indicate which identities they recognize and, for each, report familiarity, prior tasting experience, expected color, six odor descriptor families (fruity, floral, herbal/green, spicy, oak/toasty, earthy/mineral), and three structural attributes (sweetness, acidity, body). Expertise is indexed by training, professional involvement, years of experience, and a validated 10-item wine knowledge test (Majid & Croijmans, 2016). Two complementary metrics drive stimulus selection. Recognition coverage: proportion of participants per expertise level recognizing a given identity. Descriptive agreement: distance between participant ratings and a reference profile derived from expert tasting references and WSET notes, combining categorical and continuous components. Identities combining high recognition with stable, expert-aligned descriptions across expertise levels are retained. The output is a standardized stimulus set indexed by recognition and descriptive agreement across the expertise gradient. Future chemosensory and cognitive studies can draw subsets matched to the target population's knowledge level, supporting wine identification, memory, and expertise-assessment tasks. Funded by Institut Lyfe and the Lyfe Foundation PhD fellowship.

S4-P38

Deconstructing the olfactory crosstalk between acetylation and phosphorylation in Alzheimer's disease

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Post-translational modifications (PTMs) are essential for the correct dynamism of biological processes and their crosstalk adds complexity to protein regulation. In Alzheimer's disease (AD), PTMs research has focused on the neuropathological Tau protein, showing their crucial role in its regulation. Here, we integrated the proteome, phosphoproteome, and acetylome of AD-affected olfactory bulbs (OB) to characterize the PTM landscape and key protein regulation across AD progression. Protein extracts were obtained from 50 post-mortem OB of controls and AD-Braak stage I-IV subjects. Digested proteins were enriched in phosphorylated and acetylated peptides and LC-MS/MS data were acquired from Orbitrap Exploris 480. Spectronaut/SpectroMine, Perseus, Metascape, and Ingenuity Pathway Analysis softwares were used for data analysis. We observed a progressive proteomic impairment across Braak stages affecting mitochondrial metabolism and neuropathological pathways. Moreover, we quantified more than 1500 and 250 differentially phosphorylated and acetylated residues in AD-affected OBs that impacted on protein regulation. To study PTM crosstalk, we selected proteins with significant correlations between acetylated and phosphorylated sites. Most of them were known interactors of Tau, APP, or A β plaques, and played key roles in ion exchange channels or microtubule assembly. Interestingly, CKMT1A, implicated in energy transduction in high-demand tissues, presented hypo-phosphorylation at S147 and Y153 that positively correlated with acetylation at K148 and K337 at Braak III-IV stages, indicating a possible crosstalk between them and highlighting these sites as promising therapeutic targets for future studies. Overall, this multiomics profiling elucidates the contribution of PTMs to disease onset and progression and sheds light the PTM regulation of key targetable candidates in AD. This study is supported by the grant PID2023-152593OB-I00 (MCIU/AEI/ 10.13039/501100011033 / FEDER, UE).

Session 5 — 18 September 2026, 12:30-14:00

S5-P1

Effects of physical exercise and olfactory training with pleasant and unpleasant odors on verbal fluency and depression

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Objectives of the study: Mechanisms of olfactory training (OT) in older adults require a more thorough investigation, considering its potential beneficial effects in age-related olfactory, cognitive, and affective decline. To address this question, we examined the role of OT odor hedonics in 128 participants aged between 50 and 85 yr (Mage = 68.2 ± 7.4; 105 women and 23 men). Additionally, we explored the role of physical activity in olfactory function, depression scores, and verbal fluency. Experimental methods used: Participants were randomly assigned to 1 of 4 groups, of which 2 performed OT with either (i) pleasant or (ii) unpleasant odors, or (iii) performed physical exercises and no OT or (iv) a control group without OT or physical exercises. Essential results: Results indicate a beneficial effect of OT, both with pleasant and unpleasant odors, on one key domain of cognitive function, namely verbal fluency. Furthermore, physical exercise alone also caused an increase in verbal fluency. Finally, a significant beneficial effect of OT on depression score was noted, but only in the group that displayed relatively robust depression symptoms in the pre-training session. Conclusions: OT may be of particular use for older people due to their risk of age-related cognitive decline and depression. Identification of sources of funding: This study was funded by AbScent (AbScent grant 2022 to AS). AO was supported by the National Science Center (OPUS 2020/37/B/HS6/00288).

S5-P2

Processing history influences bitterness and astringency in rapeseed protein

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Rapeseed protein isolates (RPI) are characterized by bitter taste and astringency, limiting their use in food applications. This study investigated how pH shifts and heat treatment affect chemosensory perception in RPI solutions. A factorial design used three pH conditions (7.0, 5.0, and 5.0 raised to 7.0) and two treatments (control or treatment at 90 °C for 30 min followed by centrifugation at 5000 g for 30 min). Samples contained up to 4% protein (cruciferin and napin). A trained panel (n = 10) evaluated samples in triplicate over three sessions using a 15 cm unstructured scale for bitterness, astringency, sweetness, and overall flavor. Data were analyzed using ANOVA simultaneous component analysis (ASCA). Heat treatment, pH, and their interaction, significantly affected the sensory outcome ($p < 0.01$). Lower pH and thermal processing were associated with higher bitterness and astringency. Interestingly, the sample with pH shift (5.0 to 7.0) behaved differently from the other two and was scored markedly lower in bitterness and astringency, indicating an effect of the processing history beyond final pH. Sweetness was not prominently influenced, suggesting that it may be driven by more stable components. Both processes may induce protein unfolding and alter the protonation state of the molecules in the system, modifying non covalent interactions. This influences the availability of taste-active compounds (e.g., bitter polyphenols like kaempferols) for receptor binding and for oral mechanisms involved in astringency. Further analysis will help identify compounds and interactions underlying the observed changes. These results show that sensory perception in RPI systems is shaped not only by final physicochemical conditions, but also by processing history. Understanding food matrix transformations can help explain and improve sensory quality in food applications. Funded by Novo Nordisk Foundation: NNF21OC0065495. UCPH Research Ethics Committee approval: 504-0347/26-7000.

S5-P3

Bitter taste signalling as a chemosensory gatekeeper at the blood-brain barrier

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Brain barriers – the blood-brain (BBB) and the blood-cerebrospinal fluid barrier (BCSFB) – are essential for central nervous system protection, tightly regulating brain homeostasis while restricting drug entry. This dual role remains a major challenge for the treatment of neurological disorders. We previously demonstrated that human BCSFB cells express a functional taste signalling machinery, including bitter, sweet and umami taste receptors. However, evidence for a similar chemosensory role at the BBB is still limited. Here, we aimed to characterise the taste signalling pathway and bitter taste receptors (TAS2Rs) in the human BBB using the hCMEC/D3 cell line. RT-PCR analysis revealed the expression of key signalling components and 21 of the 26 known human TAS2Rs, which was confirmed by Sanger sequencing (>90% similarity). Immunocytochemistry further demonstrated the presence of GNAT3, PLC β 2 and TRPM5, along with TAS2R4, TAS2R5, TAS2R10, TAS2R14, TAS2R39 and TAS2R43 proteins. Functional relevance was assessed through intracellular calcium mobilization assays using TAS2R ligands already employed in the treatment of central nervous system disorders, such as glioblastoma, meningitis and other infections, Alzheimer's, Parkinson's and cognitive decline, and stroke. These ligands elicited calcium responses that were significantly reduced upon inhibition with probenecid, indicating that activation is at least partially mediated by the canonical taste signalling pathway. Altogether, our findings demonstrate that BBB endothelial cells harbour a functional and diverse TAS2R repertoire, supporting a chemosensory role for this barrier. Importantly, these results highlight the potential of targeting taste signalling pathways to modulate BBB function, offering new strategies to overcome drug resistance and enhance brain drug delivery.

S5-P4

Experience-dependent plasticity shapes odorant mixture representations in the antennal lobe via local interneurons

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Odorants are detected by olfactory receptor neurons (ORNs) that project to the antennal lobe (AL), the first olfactory neuropil in the insect brain. In the AL, ORNs make synaptic contacts with: i) projection neurons (PNs), which in turn send olfactory information to other brain areas; and ii) local interneurons (LNs) that form a dense network of lateral inhibitory and excitatory interactions within the AL. Functional and computational studies indicate that this local network transforms sensory input, presumably to enhance perception of meaningful odors. Here, we investigate the role of GABAergic LNs in both learning-dependent plasticity in the AL and the ability of flies to perceive meaningful odorants within a complex mixture. For that aim, we performed aversive olfactory conditioning using a single odorant as a conditioned stimulus and tested olfactory preference in a T-maze by exposing the flies to the conditioned odorant, or to the conditioned odor embedded in a binary mixture of different proportions with a novel odorant (distractor). We determined the proportion threshold needed for flies to detect the learned odorant immersed in the mixture. Next, we demonstrated that blocking the activity of different groups of LNs in the AL impairs the ability of animals to detect the learnt odorant embedded in mixtures. We then asked whether aversive conditioning affects the neural representation of mixtures in the AL. We recorded odor-evoked responses in PNs using in vivo calcium imaging of pure odorants and mixtures while concomitantly training animals with the same protocol used in the T-maze experiments. This approach allowed us to study the neural representation of stimuli before and after training. We found that the representation of a mixture in the AL changed after the learning. This experience-dependent plasticity suggests that the local interneuron network enhances the salience of the learned odorant, facilitating its detection within complex environments.

S5-P5

Understanding the attractiveness of human body odors: the link between perception and chemical composition

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In humans, the selection of a sexual partner is partly based on olfactory criteria. While some factors that cause variation in body odor have been known for some time to modulate perceived attractiveness, the underlying mechanisms are still poorly understood. In particular, a better understanding of the chemical basis of olfactory attractiveness could help us to better understand this form of nonverbal communication. In this study, 115 men and women of different ages (children, young adults, older adults) provided body odor samples taken from the neck and armpits. These samples were analyzed by both experts and non-experts for their perceptual qualities, and were chemically analyzed by GC-MS after collection using a specific device called ABOV (Analysis of Body Odor Volatiles). Inter- and intra-individual differences were observed at the perceptual and chemical levels, particularly in relation to age, gender, hormonal status (menstrual cycle phase in women and testosterone levels in men), and odor source. In addition, we examined the relationship between attractiveness of body odors and their chemical characteristics. Overall, the results of this study provide further insight into human body odor, and an understanding of olfactory attractiveness in humans.

S5-P6

Grounding olfactory perception in language: Benchmarks and models for generating natural language odor descriptions from molecular structure

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Recent advances in deep learning have enabled prediction of odorant perception from molecular structure, opening new avenues for odor classification. However, most existing models are limited to predicting percepts from fixed vocabularies and fail to capture the full richness of olfactory experience. Progress is further limited by the scarcity of large-scale olfactory datasets and the lack of standardized metrics for evaluating free-form natural-language odor descriptions. To address these challenges, we introduce Odor Description and Inference Evaluation Understudy (ODIEU), a benchmark which includes perceptual descriptions of over 10,000 molecules paired with a model-based metric for evaluating free-form text odor descriptions. The model-based metric uses Sentence-BERT (SBERT) models which are fine-tuned on olfactory descriptions to allow better evaluation of human-generated odor texts. Using the fine-tuned SBERT models, we show that free-form odor descriptions contain additional information in their syntactic structure compared to semantic labels. We further introduce CIRANO (Chemical Information Recognition and Annotation Network for Odors), a transformer-based large-language model that generates free-form odor descriptions directly from molecular structure, thus implementing the molecular structure-to-text (S2T) prediction. CIRANO achieves performance comparable to humans. Finally, we generate human-like descriptions from mouse olfactory bulb neural data using an invertible SBERT model, yielding neural-to-text (N2T) predictions highly aligned with human descriptions. Together, CIRANO and ODIEU establish a standardized framework for generating natural language olfactory descriptions and evaluating their alignment with human perception.

S5-P7

Perception of Caloric-Richness Influences the Metabolic Dose-Response Relationship of Sugar-Sweetened Beverages

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Sugar-sweetened beverages are consistently linked to weight gain and metabolic syndrome; however, the mechanisms behind these effects are still not fully understood. In particular, the extent to which perceived caloric dose influences metabolic responses to liquid carbohydrate intake has received limited attention. Sweetness is thought to serve as a predictive cue for energy content, and disruptions in the relationship between sweetness and caloric content may alter physiological responses to food. The current study investigated how perceived caloric-richness affects the metabolic dose-response relationship to sweeteners. Ten healthy participants completed a series of experimental sessions in which they consumed beverages spanning a range of carbohydrate doses (0-150 kcal), adjusted relative to individual metabolic requirements. In one condition (4 participants), only sucrose was used, while in another condition (6 participants), a constant dose of the non-nutritive sweetener sucralose was added. Metabolic responses were assessed using indirect calorimetry before and after drink consumption. Participants also rated their perception of the first sip of the drink. We observed that caloric dose and sweetener type interacted to affect resting energy expenditure (REE). Perceived caloric-richness moderated the interaction (generalized linear mixed model $\chi^2 = 19.390$, $p = 0.002$), such that in high caloric doses with the added non-nutritive sweetener, energy expenditure was lower when participants perceived it to be low in calories. This may impact consumers when they wrongly believe that beverages with added artificial sweeteners are also low in calories.

S5-P8

Sodium chloride and sour taste in humans – a pilot study

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Salty and sour tastes are among frequently confused tastes at low concentrations. We were interested whether the interaction of salty with sour at low concentrations would enhance sour taste. We analysed the effect of adding NaCl to citric acid (CA) (both compounds at low concentrations) on sour taste. 34 subjects, men and women aged 23-70, were recruited. The following tastants were prepared: standard CA dilutions - 1CA (0.13 g/l), 2CA (0.16 g/l), 3CA (0.25 g/l), and their mixtures with standard NaCl dilutions - 1NaCl (0.16 g/l), 2NaCl (0.24 g/l), 3NaCl (0.34 g/l), 4NaCl (0.48 g/l). Three series of tastants samples (15 ml) were prepared: (1) 1CA, 1CA+1NaCl, 1CA+2NaCl, water, 1CA+3NaCl, 1CA+4NaCl, (2) 2CA, 2CA+1NaCl, 2CA+2NaCl, 2CA+3NaCl, water, 2CA+4NaCl, (3) 3CA, 3CA+1NaCl, water, 3CA+2NaCl, 3CA+3NaCl, 3CA+4NaCl. The subjects were asked to identify samples' tastes with the "sip and spit" method. Series were analysed separately. A contingency table with two variables: sample (CA or CA+NaCl) and answer (sour identified or unidentified) was used in statistical analyses. In each series, a number of subjects did not recognise sour taste in the pure CA solutions (1CA, 2CA, 3CA). Yet, some of them identified sour taste when NaCl was added to CA. In the series 3, 41 % participants did not recognise sour taste in 3CA sample. Among them, 92 % detected sour taste in CA+NaCl samples. The contingency table showed a relationship between sour taste recognition and the sample ($p = 0.04$). The results indicated that the addition of NaCl significantly affected the recognition of sour taste in this series. In most cases, the two lowest concentrations of NaCl helped to identify sour taste (0.16 g/l and 0.24 g/l). The results suggest that NaCl has the potential to improve the recognition of sour taste under certain conditions. Further studies are required to fully understand the mechanisms of sour-salty interaction in the context of sour taste.

S5-P9

A low-dimensional code for perceptual similarity in olfaction.

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Olfaction still lacks a unifying circuit-level framework linking odorant receptor (OR) activity to olfactory percepts. To investigate how OR responses shape perceived stimulus similarity, we developed a novel approach that exploits the anatomical convergence of ORs onto individual glomeruli to control sensory input at the level of specific OR types. Using two-photon and widefield imaging, we identified numerous glomeruli and measured their responses to 121 odorants. We then generated synthetic olfactory stimuli by optogenetically activating selected glomerular combinations. To assess perceptual similarity, we trained mice in a novel Go/No-Go task to report perceived differences between test stimuli and a reference activation pattern. Using this approach, we found that glomeruli within the reference set differ in perceptual relevance: activation of some glomeruli elicited substantially larger perceptual changes than others. Glomerular perceptual impact correlated with similarity in glomerular odorant responses, consistent with a low-dimensional organization of olfactory representations. Using only glomerular responses to odorants, we inferred a latent perceptual space with an autoencoder objective and found that ~12 dimensions were sufficient to capture odor representations. Projections of synthetic activity patterns into this space accurately predicted behavioral responses to novel stimuli. Moreover, the low dimensionality of the latent space predicts the existence of non-overlapping glomerular activity patterns that generate indistinguishable percepts. These glomerular metamers were confirmed experimentally within the resolution of our assays. Our results indicate that olfactory perception is low-dimensional and inherently structured for efficient odor representation. More broadly, they support a model in which olfactory circuits transform high-dimensional OR inputs into a structured low-dimensional representation that links glomerular activity to perception.

S5-P10

Neural Integration of Peripheral Taste and Internal State in *Drosophila*Youngseok Lee¹¹ *Department of Integrative Biotechnology, Kookmin University*

Sodium is essential for physiological homeostasis, yet excessive intake is harmful, requiring adaptive regulation of salt consumption according to internal sodium status. Using *Drosophila melanogaster*, we identified an integrated internal commander that coordinates innate high-salt avoidance with homeostatic salt appetite. At the peripheral level, a specialized pharyngeal gustatory pathway functions as a primary gatekeeper against excessive salt intake. The ionotropic receptor IR60b, together with its coreceptors IR25a and IR76b, is selectively expressed in a single pair of gustatory receptor neurons in the labral sense organ and is specifically activated by high Na⁺, generating a robust rejection signal that limits ingestion of harmful salt concentrations. In parallel, we uncovered a central neuroendocrine circuit that dynamically adjusts salt preference according to sodium need. Brain-wide functional screening identified insulin-producing median neurosecretory cells (Ilp2/MNCs) and the leucokinin (LK)–leucokinin receptor (LKR) pathway as key regulators of state-dependent salt appetite. Under sodium deprivation, these neurons function as direct central sodium sensors, exhibiting dose-dependent Na⁺-evoked activity even when synaptic transmission is blocked, indicating intrinsic sodium sensitivity. Anterior leucokinin neurons further provide feedback modulation to fine-tune circuit output. Mechanistically, cAMP–PKA signaling gates sodium-evoked neuronal activity, and constitutive activation of this pathway drives persistent salt-seeking regardless of internal state. Together, our findings establish a multilayered regulatory framework in which peripheral high-salt sensing provides innate aversion, while central sodium-sensing neuropeptidergic circuits override this barrier to promote adaptive salt appetite and maintain ionic homeostasis.

S5-P11

Olfactory epithelial remodeling in eosinophilic chronic rhinosinusitis: a murine model study

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Chronic rhinosinusitis (CRS) affects approximately 11% of the Western population and is frequently associated with olfactory dysfunction (OD), significantly impairing patients quality of life. Clinically, CRS is classified into two phenotypes, with eosinophilic CRS (eCRS), characterized by type 2 (T2) inflammation, associated with more severe OD. Despite its prevalence, the mechanisms underlying OD in eCRS remain poorly understood, partly due to limited investigation of the olfactory epithelium in murine models. Emerging evidence suggests that OD may result from epithelial remodeling or barrier dysfunction, but the mechanisms driving epithelial damage remain unclear. This study aims to characterize the impact of eCRS on the olfactory epithelium and to determine whether epithelial alterations are associated with metaplasia or epithelial-mesenchymal transition (EMT). Methods: Male and female mice were administered ovalbumin (OVA) combined with *Aspergillus* protease (AP) three times per week for 12 weeks. Control mice were treated with PBS following the same protocol. Olfactory function was assessed using the buried food test and habituation/dishabituation test. Coronal skull sections were processed for immunohistochemical analysis of olfactory epithelial markers (OMP, KRT18, SOX2, KRT5), respiratory markers (TUBB4), mucus production (MUC5B), EMT markers (SNAIL, SLUG), and tight junction proteins (ZO-1, E-cadherin). The OVA-AP model induced OD and reduced olfactory epithelial area, with increased mucus production and weakened tight junction integrity. No increase in ciliated respiratory cell marker expression was detected in affected regions. These findings do not support respiratory metaplasia as a major mechanism of epithelial loss. Ongoing quantitative analyses of epithelial cell populations and EMT markers will further elucidate inflammation-driven epithelial remodeling in eCRS. Final results will be presented at the conference. Funding: F.R.S-FNRS

S5-P12

Synthesized human fear odor and linguistic processing: Cognitive and physiological evidence from an eye-tracking study

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Humans primarily communicate through language, but also rely on non-verbal cues such as auditory signals (prosody, pitch, volume) and visual signals (facial expressions, gestures) to shape the meaning of the verbal language. These cues convey affective states and can bias receivers' behavior in both written and oral communication. A large body of research evidences that human body odors also transmit non-verbal emotional information, with specific molecules in sweat signaling fear. Prior studies show that these fear-related molecules influence the interpretation of emotionally ambiguous faces and emotional processing in the brain. However, it remains unclear whether these chemosignals have any effect on the interpretation of linguistic communication, as has been found for other non-verbal communication channels. In a fully within-subjects, pre-registered experiment, participants completed two lab sessions differing only in the odor present: fear-related (molecules + solvent) or neutral (solvent only). In each session, participants read sentences containing either neutral or negative emotional words in short blocks, while eye movements were recorded. After each block, they rated subjective emotions and task engagement. Physiological measures (heart rate variability and skin conductance) were also recorded to assess emotional responses. At the end of each session, recognition memory for the words was tested. We expected interactions between odor condition and sentence emotionality, demonstrating that olfactory cues influence emotional language processing at multiple levels. This study explores how fear odors, as contextual cues, impact awareness, attention, and cognition during linguistic tasks. It is among the first to investigate a role for smell in linguistic communication, expanding chemosensory research into ecologically valid areas of verbal and non-verbal interaction, and opening new avenues for multisensory approaches in affective cognitive neuroscience.

S5-P13

What ligand binding reveals about mammalian odorant-binding protein function.

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Mammalian odorant-binding proteins (OBPs) are members of the lipocalin family that are abundantly expressed in the olfactory epithelium and commonly associated with olfaction. Despite the availability of high-resolution structural data, their precise physiological role remains unclear, particularly whether they directly participate in odor recognition or instead contribute more indirectly to odorant transport and processing. To investigate the molecular determinants and functional consequences of ligand binding in OBPs, we combined molecular modelling, network analysis, site-directed mutagenesis, isothermal titration calorimetry (ITC), and synchrotron circular dichroism (CD) spectroscopy. Our results show that ligand entry and binding occur without major conformational rearrangements or destabilization of the protein scaffold, both in wild-type and mutant OBPs[1]. Although allosteric pathways influence ligand access and binding affinity, the overall fold remains remarkably stable throughout the interaction process. This pronounced structural resilience argues against a direct role of OBPs in odor recognition, which generally involves dynamic conformational transitions. Instead, our findings support a model in which OBPs primarily function as robust scavenger and buffering proteins that capture, transport, and regulate the local availability of odorant molecules. Importantly, this intrinsic stability, together with conserved ligand-entry mechanisms across the lipocalin family, highlights OBPs as promising biological sensing elements for bio-nose technologies, where durability, reproducibility, and sustained ligand binding are critical properties.

[1] M. Lalis, et al. How Allosteric Mutations Control Ligand Binding in Lipocalin Protein: Odorant Binding Protein as a Test Case, *Cell Mol Life Sci*, 2025.

S5-P14

Koku compounds modulate fat and viscosity perception in milk: a temporal sensory study

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Objective: CaSR-binding peptides have been linked to the flavor-enhancing sensation koku, yet their role in modulating fat oral sensation in humans remains unclear. To test whether koku-related peptides act as positive allosteric modulators of fat perception, as demonstrated in animal models, this study investigated the sensory effects of glutathione (GSH) and γ -Glu-Val-Gly (EVG) in milk of varying fat content. **Methods:** Seven trained panelists (57% female, age: 32.9; range 21-66) evaluated six treatments in triplicate using Temporal Check-All-That-Apply (TCATA, 60 s per sample), in a 2x2 design crossing matrix (skim 0.1% or semi-skim 1.5% fat milk) with treatment (reference, 1 g/L GSH, or 0.01 g/L EVG). Following each TCATA evaluation, panelists rated viscosity, sweetness, and animalic flavor intensity on line scales. **Results:** Linear mixed models with Tukey-corrected post-hoc revealed significant viscosity increases for 1.5% fat milk enriched with GSH ($\beta = +16.6$, $p = 0.011$) and EVG ($\beta = +15.3$, $p = 0.019$) relative to unenriched skim milk, but no significant effect when compounds were added to skim milk alone, indicating a fat-dependent mechanism. Pairwise contrasts were directionally consistent but did not survive correction, likely reflecting limited panel size. TCATA showed that all enriched treatments exceeded unenriched skim milk in fatty/greasy character; notably, EVG-enriched skim milk was rated as equally fatty/greasy as unenriched 1.5% fat milk. Sweetness was influenced by fat content only, with no compound effect of enrichment. **Conclusion:** These preliminary results support a fat-dependent amplification of viscosity and fat perception by koku peptides and suggest EVG as a candidate for a fat mimetic in reduced-fat dairy formulations.

S5-P15

Representations of odor mixtures in the antennal lobe of *Aedes aegypti*

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The sense of smell plays in mosquitoes an important role in many behaviors such as foraging and host seeking. Olfactory cues in nature usually include blends of odorants instead of single monomolecular odorants. Different odor components activate their cognate sensory neurons at the peripheral olfactory organs. While the coding of odor mixtures by the sensory neurons has been examined in many studies, very little is known about how odor mixtures are represented by the central nervous systems of mosquitoes. To investigate how odor mixtures are represented in the *Aedes aegypti* brain, we observed the responses for binary odor mixtures and their components in the primary olfactory center, i.e., antennal lobe. Using in-vivo patch clamp electrophysiology in immobilized mosquitoes, we targeted single neurons (cell bodies) in the antennal lobe and delivered ecologically relevant odors to their antennae. The order of stimuli (different mixtures and their components) was randomized from experiment to experiment to avoid bias. Brief (1-second) pulses of odor were presented to the antennae, and the spiking responses of the recorded neurons were determined by counting spikes in the response window. The experiments showed diverse response patterns to the odor mixtures. In general, our results indicate that the responses to the binary mixtures are often different from the responses to the individual components and suggest a trend towards nonlinear summation of the component activities rather than a simple linear summation. These nonlinear effects point to extensive interactions of the input channels at the level of the antennal lobe, likely mediated by the local neurons. Our results highlight the complexity of olfactory processing even at the early stages of mosquito olfaction. Funding: The work has been funded by the Council of Scientific and Industrial Research, the Science and Engineering Research Board, and the Department of Biotechnology, India.

S5-P16

Attentional modulation of human single-neuron odour representations

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The same odour can be perceived differently depending on whether we notice it, ignore it, or deliberately search for it. Olfaction is fundamentally context-dependent, yet the neuronal coding mechanisms by which attention reshapes human odour processing remain unknown. Our recent single-neuron recordings revealed that neurons in the human piriform cortex and medial temporal lobe encode odour identity, revealing principles of cellular odour coding in the human brain. However, it is unclear how this code is modulated by higher-order cognitive demands and at which stage of the olfactory pathway attention exerts its influence. To address this question, we recorded single-neuron activity in patients undergoing intracranial monitoring while they performed an odour-visual attention task. On each trial, participants received identical, precisely controlled odour and visual inputs, but were instructed to attend either to the odour or to the visual cue. This design allowed us to isolate attentional modulation from sensory stimulation. We found that odour identity was encoded fastest and most precisely by neurons in the human piriform cortex, whereas odour coding emerged later and less prominently in medial temporal lobe regions. In contrast, medial temporal lobe activity more accurately predicted the current attentional focus, suggesting that these regions integrate olfactory information with task demands. Together, these findings reveal a functional division within the human olfactory system. Piriform cortex provides a rapid and precise representation of chemical odour identity, while medial temporal lobe circuits flexibly link odour information to attentional focus. Our results show how cognition dynamically reshapes sensory coding in the human brain.

S5-P17

OSRNs and Resource Editing through Subcellular Cluster Mapping in Fish**Subrata Kumar De¹, Gour Maity¹, Swaraj Kumar Sarkar²**¹ Department of Zoology, Vidyasagar University, Midnapore, West -721 102, West Bengal, India² Brainware University, Barasat-700125, West-Bengal, India

Olfactory sensory receptor neurons (OSRN's) within the olfactory neuroepithelium are a unique cellular architecture in fish, *Pseudapocryptes lanceolatus* (Bloch and Schneider, 1820), belonging to the Order: Perciformes (Teleostean: Gobiidae) of South East Asia (IUCN Status: Least Concern, Ver. 3.1). Olfactory structure examined under light and transmission electron microscope using the 2.5 % glutaraldehyde in 0.1 M phosphate buffer at 4 °C for 2 hrs. (primary fixation). and 1 % osmium tetroxide (secondary fixation) for 1 hrs. respectively. The olfactory neuroepithelium is a pseudostratified structure of distinct cellular diversity with definite functions. OSRN are designated for searching for food, selection of mating, avoiding the prey, etc (Hara, 1971). In a particular habitat, the subcellular components show a distinct subcellular topology and a sharp pattern of subcellular accumulation close to the perinuclear space and definite protoplasmic vesicular morphs within the OSRN. The mitochondrial structures with cristae are well focused within the cytoplasmic spaces of Olfactory sensory receptor neurons (OSRN's). The subcellular structural orientation within the protoplasm indicates the probable affinity to food preference and/or the functional motive (?) of the animal to characterise the preference of food editing geometry of the particular species of a particular ecological habitat, so that the animal can easily survive in their preferred ecological niche.

Reference:

1. Hara T J (1975) Olfaction in fish. Progress Neurobiology, 5: 271-335.

S5-P18

Olfactory experience and behavioural processing as a human factors frontier in spaceflight.

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Human factors research overlooks the dimensions of olfactory experience and behavioural processing in spaceflight. In microgravity, the perception of smell changes dramatically. Bodily fluids accumulate in the cranial and nasal cavity, constrict the nasal passages, and reduce olfactory sensitivity. Additionally, astronauts contend with stinky matters, carrying out inevitable needs in confined, shared environments with minimal privacy: vomiting, menstruation, and waste. The result is an environment characterised by sensory monotony, persistent odours, olfactory fatigue, and altered perception. Our approach systematically reviews the literature to identify gaps in the understanding of olfactory experience in spaceflight. We validate this with expert accounts across aerospace medicine, life-support engineering, astronaut training, and human factors research. Design recommendations are grounded in astronauts' lived experience. Olfactory stimuli in confined spacecraft contribute to psychosocial distress. We examine how sensory perception and olfactory processing inform novel solutions for health and comfort in orbit, with the human operator at the centre of mission-critical design. Promising interventions include aromatherapeutic, transdermal, and inhalation systems for alleviating nausea; odour-sequestering containment products; and smart garments with e-textile sensors. Multisensory technology systems and olfactory perception training offer scalable tools for resilience. The changes to the olfactory system in reduced gravity represent a critical frontier in spaceflight human factors research. Beyond containment engineering, the multidisciplinary integration of sensory design, technology development, and materials science is essential to address adaptation, fatigue, and psychosocial tolerance and advance human well-being beyond Earth.

S5-P19

Impact of olfactory and threat priming on human odor detection thresholds

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Human experimental studies have demonstrated that inducing a specific emotional state (e.g., disgust) might temporarily influence odor detection threshold. In the present study we aimed to verify whether similar effects can be obtained by shifting one's attention towards odors using cognitive priming. To this end, we created four newspaper-like stories based on possible interactions of Theme (olfactory, non-olfactory) and Threat (threatening, non-threatening) factors. Participants' detection threshold was measured using the Sniffin' Sticks test, twice – before and after reading the stories, towards one of two odorants assigned randomly (non-threatening, pleasant odor, phenyl ethyl alcohol, PEA; unpleasant, threatening odor, 2-furfurylthiol, FFT). A total of 195 young adults aged 18 to 39 (Mage=26.1years, SD= 5.3; 117 women) completed the study. We hypothesized that olfactory, threatening prime would temporarily increase odor detection threshold for FFT. Using linear mixed-effects model, we found the four-way interaction between Theme, Threat, Odor, and Time was not statistically significant ($F=.11$, $p=.736$, $\eta^2p<.01$). However, we found a significant Theme*Odor*Time interaction ($F=4.72$, $p=.031$, $\eta^2p=.02$). Upon unpacking this interaction with Holm-corrected post hoc tests, we found that participants who read non-olfactory related stories (both threatening and non-threatening) showed lower detection threshold for PEA (but not FFT) post-reading, as compared to baseline ($p=.013$). Other groups did not show any changes in odor detection thresholds over time. Our findings demonstrate that compared to emotional state induction, olfactory-threatening priming does not influence olfactory sensitivity. Funding: National Science Centre in Poland (2022/45/N/HS6/00651)

S5-P20

Crossmodal Correspondences between Sound and Smell Mediated by Affect

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Multisensory perception is a fundamental aspect of everyday experience, yet the mechanisms linking different sensory modalities remain unevenly understood. Olfaction occupies an important position within the multisensory system because odors often evoke affect and memory, but describing odor quality using a consensual set of perceptual dimensions remains difficult. How smell is associated with stimuli from other senses therefore raises key questions about crossmodal correspondences [Deroy, Crisinel, & Spence, 2013]. Despite metaphorical parallels between music and perfumery, empirical evidence for sound–smell correspondences remains limited. The present study investigates whether affect may function as an indirect link between odors and sounds. Three experiments are conducted. In Experiment 1 (already completed; N = 57), participants sorted 83 musical excerpts, created by sound designers at Ircam Amplify specifically for this experiment, into 8 affective categories (Apathy, Depression, Happiness, Irritation, Relaxation, Sensuality, Stimulation, and Stress). Results showed clear affective induction for some categories (e.g., Relaxation), whereas others elicited weaker reactions (e.g., Apathy). In Experiment 2, participants will sort 24 odors into the same affective categories. In Experiment 3, participants will directly match sounds with odors selected from the previous experiments. By comparing direct sound–smell associations with indirect affect-based matches, this work aims to assess if affect mediates these correspondences. Finally, musical audio features (e.g., melody, tempo, dynamics) will be analyzed to identify dimensions underlying the observed affective profiles and to determine which features predict crossmodal sound–smell associations.

S5-P21

Beyond smell loss: cognitive and affective alterations in post-COVID olfactory dysfunction and implications for digital rehabilitation

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Millions of individuals continue to experience persistent symptoms following SARS-CoV-2 infection, with olfactory dysfunction representing one of the most disruptive and long-lasting complaints. Beyond reduced sensitivity, it remains unclear how smell impairment relates to broader symptom patterns and whether changes extend to the affective evaluation of odors. At the same time, the effectiveness of current rehabilitation strategies is often limited by low adherence to training protocols. We analyzed a well-characterized clinical cohort of individuals with post-acute sequelae of COVID-19 several years after infection, integrating psychophysical olfactory testing, multidomain cognitive assessment, and patient-reported outcomes. Using complementary data-driven modeling approaches, we identified cognitive performance, particularly in verbal fluency tasks, as closely aligned with individual differences in olfactory ability. Additional contributions from respiratory and overall symptom burden were observed. Notably, alterations were not restricted to detection deficits: individuals with reduced olfactory performance showed a blunted evaluative response to odors typically perceived as highly pleasant, while responses to aversive stimuli were comparatively preserved. This selective shift was evident for specific odor categories, including food-related stimuli. Building on these findings, we developed a digitally supported olfactory training approach designed to enhance engagement and enable adaptive, user-specific training pathways. This strategy aims to improve the scalability and real-world effectiveness of rehabilitation for patients with persistent olfactory dysfunction.

S5-P22

Integration of main and accessory olfactory pathways in the rodent BAOT and avMeA

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At least two parallel sensory systems mediate olfaction in most mammals: the main and accessory olfactory systems. Specialized peripheral sensory organs detect chemosensory cues and relay them separately to the main and accessory olfactory bulbs, where olfactory information is processed in parallel. Therefore, we investigate downstream circuits and ask whether they integrate main and accessory olfactory information. First, we trace axonal projections of main and accessory olfactory bulb principal neurons. To this end, we combine stereotactic microinjections with targeted expression of recombinant adeno-associated viral genomes to identify downstream target regions of both pathways. We identify two amygdaloid nuclei, namely the bed nucleus of the accessory olfactory tract (BAOT) and the anteroventral medial amygdala (avMeA), as shared projection targets of the main and accessory olfactory pathways. Second, we employ single-cell patch-clamp recordings in acute brain slices to describe the physiology and morphology of neurons constituting these amygdaloid nuclei. By comprehensively quantifying electrophysiological and morphological features, we reveal distinct cell types in BAOT and avMeA. Third, we dissect functional connectivity between main and accessory olfactory bulb projections and these BAOT / avMeA cell types. To this end, we combine optogenetic activation of axonal terminals with single-cell patch-clamp recordings. Together, we describe both the anatomical and functional convergence of main and accessory olfactory pathways in BAOT and avMeA. Moreover, we extend these findings by systematically describing cell types and connectivity in both nuclei based on electrophysiological and morphological characteristics.

S5-P23

Survival of the senses: Trp73 regulation in olfactory neuron development

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The p53-family transcription factor p73 (Trp73), is known to regulate transcription, and neuronal development and survival. Its role in the specification of sensory neuron subtypes remains poorly understood. The Main Olfactory System (MOS) and Accessory Olfactory System (AOS) offer complementary models to study Trp73's function during olfactory maturation processes. In this study, we investigated the role of Trp73 in the olfactory epithelium (OE) and vomeronasal organ (VNO). The VNO neuroepithelium mainly consists of vomeronasal sensory neurons (VSNs) that express receptors of the V1R or V2R gene families. Using single-cell RNA sequencing, BaseScope™, in situ hybridization, and a Trp73 loss-of-function mouse model, we examined isoform-specific expression and developmental outcomes of neurons in the absence of Trp73. During development, Trp73 mRNA expression is highly concentrated in the VNO, with minimal expression in the OE. Consistent with literature, Trp73 knockout mice showed reduced numbers of neurons in the OE and a smaller olfactory bulb. The V1R⁺ and V2R-expressing neurons of the VNO are spatially segregated in the apical and basal regions, detect distinct ligands, and project to the anterior or posterior accessory olfactory bulb (AOB), respectively. VSN development is governed by a divergent transcriptional hierarchy: V2R-basal VSNs require Notch signaling followed by expression of Bcl11b and then AP-2ε (Tfap2e), whereas V1R-apical VSNs express Meis2. Trp73 mRNA expression coincides with this dichotomy's divergence. In Trp73-KO mice, the VSN developmental sequence was disrupted, preventing subtype specification and maturation. Consequently, VSNs failed to form connections with the AOB. Our findings highlight Trp73 as a key regulator of gene networks necessary for VSN subtype-specification and circuit. Statistical analyses were conducted using Welch's t-tests (N ≥ 3 per genotype). Work supported by NIH NICHD awards 2R01HD097331 and R01HD114827 (PEF).

S5-P24

How didn't you notice it: are human social chemosignals unconscious or unattended?Roxanne Mangampo¹, Benjamin Young^{1,2}¹ University of Nevada Reno, Neuroscience² University of Nevada Reno, Philosophy

Research on human social chemosignals (HSC) continues to uncover how our embodied odorant bouquet effects swaths of daily life often without our noticing. The range of HSC phenomena include kinship recognition, social acquaintance selection, the perception of health related odors, psychological states (emotions & personality traits), and social interactions. And in addition to the sheer range of interpersonal phenomena that we don't notice being modulate by smell, often these effects are claimed to occur unconsciously. However, we worry that given findings by Forrester and Spence (2018) on attentional load yielding failures to report perceiving a consciously accessible smell (inattentional anosmia) that the failure to notice HSC might be attributed to a lack of attention. Subjects might be consciously perceiving these bodily odors, yet fail to further attend to them or the resultant effect on behavior. Thus, the lack of subjective reportability might be attributable to a lack of attention. To examine our alternative thesis, we conducted a literature review examining experimental studies on HSC from which we identified a subset 62 target studies. We then further split the findings into claims regarding the perceived stimulus and/or the behavioral effect. Based on the survey of the literature, we argue that our thesis is a viable alternative explanation that has not been ruled out. Therefore, we conclude that skepticism is merited in some instances of the claim the human social chemosignaling occurs unconsciously. We think that our research provides motivation to further study the relation between olfactory conscious/unconscious processing and attention in general, and in particular, their interplay for HSC under laboratory conditions and within naturalistic settings where the multi-modal perceptual array can affect the cognitive load of navigating social interactions.

S5-P25

An in-house directory of vomeronasal type-1 receptor structures in domestic animals: a computational approach

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Vomeronasal type-1 receptors (V1Rs) are expressed in the vomeronasal organ (VNO), play a crucial role in detecting pheromone signals in domestic animals such as cats, dogs, goats, sheep, pigs, cattle and horses. Studying the structural modeling of vomeronasal receptors is essential for understanding the fundamental mechanisms of chemical communication, the combinatorial recognition of putative volatile molecules, and the influence of these chemical signals on animal behavior. Experimental structures of vomeronasal receptors are currently unavailable, making in silico approaches necessary to construct V1R models. AlphaFold predictions, however, exhibit low confidence in the intracellular and extracellular loops, highlighting the need to evaluate alternative computational approaches to obtain more accurate V1R structural models. Over the past several years, various computational tools have been employed to construct and refine structurally validated models of V1Rs in these species. Our team has developed refined structural models for intact sequences and established a repository of V1R structures in domestic animals. The analysis of V1R sequence conservation, receptor diversity, motif identification, and structural optimization offers valuable insights into the refined V1R models. Furthermore, the catalogue of intact V1R models enables the analysis of molecular interactions with volatile ligands and provides insights into the stability and specificity of combinatorial ligand recognition based on the receptor structures.

S5-P26

Olfactory Function in Tracheostomy PatientsSarah Steigleder¹¹ University of Tübingen/ Schön Klinik Bad Aibling

Tracheostomy patients frequently suffer from anosmia or other olfactory disorders, commonly attributed to the absence of nasal airflow caused by the tracheal cannula. Despite the clinical relevance of this condition, the precise relationship between tracheostomy and olfactory dysfunction remains largely unexamined. In this study, we investigated whether tracheostomy patients retain the ability to perceive odors when nasal airflow is artificially restored. In 40 tracheostomy patients, odorants were administered bilaterally at a controlled airflow of 7 LPM. Stimuli were selected based on normative data from a healthy sample, ensuring established differences in valence and intensity and enabling direct comparisons between patient and control groups as well as within-patient analyses across odors. Alongside pleasantness and intensity ratings, respiratory behavior was recorded to examine whether inhalatory preparation influences odor perception. Preliminary analyses suggest that tracheostomy patients are able to discriminate both aversive (butyric acid) and pleasant odors (orange oil, lemon oil, isoamyl acetate) from a neutral mineral oil baseline. The trigeminal component of olfactory perception – as indicated by responses to eucalyptus – appears to be better preserved than purely olfactory processing. Overall, however, valence perception is markedly reduced compared to non-tracheotomized healthy controls. These findings contribute to the understanding of olfactory mechanisms in tracheostomy patients and may have implications for the clinical management of smell disorders in this population.

S5-P27

Early olfactory dysfunction and alteration in neural lipids in a Niemann-Pick type C1 mouse model

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Niemann-Pick type C1 (NPC1) disease is a rare lysosomal disorder, where cholesterol export is impaired due to variants of NPC cholesterol transporter 1. Loss-of-function mutations cause intracellular accumulation of unesterified cholesterol, due to impaired mobilization and re-esterification of LDL cholesterol, especially in liver, spleen, lung, and brain. Lipid dysmetabolism causes visceral symptoms, neurodegeneration and reduced life spans in patients. As olfactory dysfunction frequently precedes symptoms in neurodegenerative conditions, we investigated whether the olfactory system might provide early insight into NPC1 disorder. Using NPC1 (I1061T) mice bearing the most common mutation found in patients, we analysed the olfactory epithelium (OE) and the olfactory bulb (OB) in wild-type, heterozygous, and knock-out mice at two time points, namely 36 and 60 days after birth. Our results show that neurodegeneration in the OE becomes dramatic at two months of age of mutant mice and involves reduced density of mature olfactory sensory neurons (OSNs), increased apoptosis, and neuroinflammation, while progenitor and immature OSN populations were preserved. Heterozygous mice exhibited subtler alterations. Matrix-assisted laser desorption/ionization - time-of-flight/mass spectrometry (MALDI-TOF/MS) analyses of the OE from mutant mice revealed phospholipid signatures consistent with neuroinflammation, while neural gangliosides GM2, which were considered NPC1 hallmarks, were detected in both OB and brain from mutant mice only. These structural changes affected the functionality of olfactory system such as reduced odorant responses in mutant mice compared to wild type. Additionally, we found deficits in odor-guided food-seeking tests in mutant, but not in the wild type and heterozygous mice. Our findings demonstrate that early olfactory dysfunction and neuroinflammation in NPC1 are remarkable pathological hallmarks that exacerbate the progression of neuronal dysfunction.

S5-P28

Bitter taste receptor expression and evolution in the eucalypt-eating koalas

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Koalas (*Phascolarctos cinereus*) are one of the unique marsupials (pouched mammals) inhabiting coastal areas of eastern and southern Australia. They are dietary specialists, who almost rely on leaves of eucalypts (*Eucalyptus* spp.). Since eucalypt leaves contain high amount of toxic secondary metabolites, koalas have been thought to evolve their chemosensory systems to detect less toxic leaves. The Koala Genome Consortium contributed to understanding genetic backgrounds of koalas' uniqueness and found their expanded bitter taste receptor gene (TAS2R) repertoire, which may allow them to optimize their leaf choice by minimizing toxins. In this study, we obtained deeper insight into koala TAS2R. First, we performed bulk RNA-seq in tongue (round and slit-like vallate papillae), gastrointestinal tract (esophagus, stomach, small intestine, cecum, and colon), and liver collected from multiple koalas after their natural death. Many TAS2R genes were expressed on the tongue papilla and liver, whereas few TAS2R genes on the gastrointestinal tract. Next, nucleotide variation in the 29 TAS2R genes from 396 koala genomes were determined. They are broadly under balancing selection, suggesting that they have maintained functionally diverged alleles that allow a heterozygotic koala to recognize and regulate a greater number of bitter ligands. Using eucalypt species distribution data, genotype-environment association analyses revealed that TAS2R differentiation is not merely a product of geographic isolation but is significantly driven by local environmental adaptation. These results indicate that regional and individual preference to specific eucalypt species may be driven by TAS2R diversity. Further research such as functional cellular assay and behavioral experiments will demonstrate mechanism of koalas' taste. This study was financially supported by the Office of the NSW Chief Scientist and Engineer and the NSW Department of Planning and Environment, JST SPRING, and JSPS KAKENHI.

S5-P29
=====**Development of an Olfactory Stimulation Device Focused on Contamination and Its Performance Evaluation Using PTR-TOF/MS**Tatsu Kobayakawa¹, Naomi Gotow¹, Hirotaka Kaneda²¹ *National Institute of Advanced Industrial Science and Technology AIST*² *Kyushu Sangyo University*

While various olfactory stimulation devices exist, most lack the precision and stimulus purity required for scientific research or medical examination. In humans or animals experiments, such devices must provide high stimulus accuracy. Specifically, when switching between multiple odors, the system must ensure that only the target odor is delivered and that “odorless” intervals are genuinely neutral. Despite this necessity, few studies have evaluated device performance from the perspective of stimulus quality. In this study, we developed a new olfactory stimulation device designed to minimize interference among odors and to ensure the presentation of odorless air between stimuli. The system features (1) structural isolation, separating flow paths for odor stimuli and odorless air; (2) active cleaning, in which the tubing is continuously flushed with air during non-stimulus intervals; and (3) precise odor preparation, using odorless bags and syringes to ensure consistent and accurate odor concentrations. To evaluate device performance, gas emitted from the device was continuously monitored using PTR-TOF/MS at one-second intervals. The analysis showed that residual odors from previous stimuli remained in the tubing and decayed gradually. By modeling the data under the assumption of two adsorption processes—adsorption within the olfactory stimulation device and that within the PTR-TOF/MS system itself—we could determine the appropriate duration of odorless air presentation required for residual concentrations to fall below the odor threshold. Although some contamination was observed when alternating between two different odors, the concentrations of these non-target molecules also decayed exponentially and, on average, remained below odor threshold. These findings demonstrate that quantitative performance evaluation focused on stimulus quality is an essential requirement for the development and use of reliable olfactory stimulation devices in research.

S5-P30

Unveiling the flavor network distribution and specialization for a population living with obesity using functional MRI Multivoxel Pattern Analysis.

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Flavor perception emerges from the integration of at least olfactory and gustatory signals, yet the organization of the complex network involved remains unclear. Furthermore, whether this network is altered or more specialized in individuals living with obesity remains to be investigated. Using fMRI multivoxel pattern analysis, this study aimed to characterize the flavor brain network and to assess differences between normal-weight (NW) individuals living with obesity (OB) (N = 30 and 31 per group). Functional brain activity was analyzed using an F1-based metric (sensitivity/specificity-based classification) to identify activated voxels in response to taste stimulation (salt diluted in water), aroma stimulation (smoked bacon aroma diluted in water), and flavor (combining both). The network was built within 104 regions of interest using searchlight analysis, considering voxels decoding the flavor condition with a minimum F1 value of 65%, excluding those also decoding odor or taste alone. Overall, the searchlight analysis shows that voxels across almost all of the ROI decoded the flavor condition to different extents, suggesting that the flavor network is distributed across a large number of functional areas. Within these ROIs, voxels that specifically decoded flavor varied between subjects; we found very few overlaps of flavor-decoding voxels across subjects. Furthermore, the OB group decoded flavor better (higher mean F1 values) in the right parahippocampal gyrus, corresponding to the temporal agranular insular cortex (without any differences in the number of decoding voxels compared to the NW group). Increased specificity in the parahippocampal gyrus, adjacent to the primary gustatory cortex, further indicates enhanced flavor encoding in the OB group. This possibly reflects the retrieval of flavor memory engrams due to higher exposure to flavor. Therefore, we will discuss whether these results reflect a larger “exposure effect” rather than an altered network.

S5-P31

Olfactory receptor diversification across ontogeny in the hydrothermal vent shrimp *Rimicaris exoculata*

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Chemoreception is a critical sensory modality for navigating extreme environments, yet its ontogenetic elaboration remains poorly understood in non-model organisms. The hydrothermal vent shrimp *Rimicaris exoculata* offers a unique model: across its life cycle, individuals transition from pelagic dispersive larvae navigating the open water column to detect distant vent-derived chemical cues (Mn, Fe gradients) to adults forming dense aggregations at the vent fluid interface, where multimodal chemosensory integration governs habitat selection, symbiont monitoring, and trophic regulation. Using comparative transcriptomics across four alvinocaridid species, we characterised the diversity and tissue-specific expression of ionotropic receptors (IRs), the main olfactory receptor family in crustaceans, alongside gustatory receptors (GRs) and TRP channels. *R. exoculata* exhibits an extreme expansion of IR genes compared to coastal decapods, particularly in olfactory-associated subfamilies, while GR diversity is drastically reduced. Strikingly, IR expression profiles follow a clear ontogenetic trajectory: a simplified olfactory repertoire in larvae shifts progressively toward a dense, diversified IR complement in adults, correlated with ecological niche transitions and symbiosis acquisition. These results provide the first molecular evidence for ontogeny-driven olfactory diversification in a deep-sea crustacean, opening new perspectives on chemoreceptor evolution under extreme selective pressure.

S5-P32

Odor vocabularies diverge on codability and abstraction but converge on valence and edibility

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Olfaction has been called “the muted sense” due to the difficulty in describing odors. Yet some studies indicate that culturally distinct languages differ in their capacity to express odor experiences. A few languages contain large inventories of abstract odor terms that express odor qualities directly, independent of their sources, and thus have odor vocabularies that are high in abstraction. This allows for a high between-speaker consistency-i.e., codability-in the linguistic encoding of odors. Do cross-linguistic differences in abstraction and codability coexist with cross-cultural similarities among odor vocabularies on a deeper semantic level, differentiating odors primarily in terms of valence (pleasant vs. unpleasant), on the one hand, and edibility (edible vs. inedible), on the other? No comprehensive cross-linguistic comparison of such odor-semantic organization exists to date. We compare odor vocabularies of eight languages (Swedish, English, Polish, Russian, Spanish, Turkish, Hindi and Cantonese) from five different language families. In an on-line survey, over 50 speakers of each language (N = 456) provided odor descriptors of 121 imagined odors. We analyzed the abstraction, codability, valence and edibility of the resulting vocabularies, comparing their between-odor and between-language variability, on the one hand, and their language pair convergence, on the other. We find that odor vocabularies primarily diverge in codability and abstraction, but converge to a higher degree in valence and edibility. These findings indicate that although languages differ with respect to their encoding and descriptive strategies of odors, they share the same odor-semantic organization. We also present an ongoing study, aimed to further validate these findings, in which we collect odor descriptors and perceptual ratings of 60 odors from a large sample of participants (N = 2040) in 25 different languages from 14 different language families.

S5-P33

Tasting order, not concentration, governs detection of the kokumi peptide γ -EVG in a simple umami matrix

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Objectives: Kokumi γ -glutamyl peptides enhance mouthfulness and the basic tastes via the calcium-sensing receptor (CaSR). γ -Glutamyl-valyl-glycine (γ -EVG) is the most potent known kokumi peptide, but prior demonstrations used trained panels and flavour-rich foods. We asked whether γ -EVG umami enhancement is concentration-dependent in a simple aqueous umami matrix, and whether untrained consumers detect it. **Methods:** Two independent cohorts of Danish adults (N=69, N=74) completed forced-choice triangle tests in a fixed umami matrix (as in Oshu et al, 2010) with added γ -EVG at 0.005–0.04 g/kg, presented in ascending or reversed (order-effect) sequences, followed by Check-All-That-Apply (CATA) attribute profiling (8 attributes). Discrimination was tested against chance (Gussing model, $\alpha=0.05$); CATA frequencies by Cochran's Q. **Results:** No single γ -EVG concentration was discriminated from control in either cohort, and detection did not rise with concentration (ascending cohort $p=0.082, 0.644, 0.876$ for 0.005, 0.01, 0.04 g/kg). Pooling tastings by position, γ -EVG was discriminable on the first tasting (59/143 correct, $p=0.029$) but not on subsequent tastings (59/212, $p=0.963$), independent of concentration. CATA showed γ -EVG samples were sensorially active, umami (~45%), salt (~34%), mouthfeel (~29%) most frequent, but attribute selection did not differ between first and later tastings (Cochran's Q, n.s.) or scale with dose. **Conclusions:** In a simple aqueous umami matrix with untrained consumers, γ -EVG detection was governed by tasting order rather than concentration (i.e., discriminable on first exposure only) consistent with persistent, slowly dissociating CaSR activation. **Funding:** Ajinomoto Co., Inc.

S5-P34

The Lateral Habenula Integrates Sexual Olfactory Cues to Regulate Male Mouse BehaviorVincent Hellier¹, Louise Sionneau¹, Chantal Porte¹, Pablo Chamero¹, Helene Vacher¹¹ *Laboratoire de Physiologie de la Reproduction et des Comportements, UMR 0085 INRAE-CNRS-University of Tours, Nouzilly, France*

The perception of social cues, particularly olfactory signals, is essential for regulating sexual behavior in rodents. The lateral habenula (LHb), an epithalamic structure, has been implicated in the integration of social olfactory information, as well as in the regulation of motivation and the expression of sexual behavior. However, its role within the olfactory neural circuitry, and specifically in processing sexual olfactory cues, remains unclear. Here, we show that the LHb of both sexually naïve and experienced male mice is activated by sexual odors present in urine. Moreover, kainic acid lesions of the LHb impair the ability to prefer female over male odors and disrupt sexual behavior. Interestingly, LHb neurons activated by urine also express estrogen receptor alpha, suggesting a potential link with aromatase-expressing neurons. Using retrograde viral tracing, we demonstrate that aromatase neurons from the medial amygdala (MeA) and the bed nucleus of the stria terminalis (BNST), two regions critical for processing social odors and regulating sexual behavior, project directly to the LHb. Immunodetection of the immediate early gene *Egr1* further reveals that these MeA and BNST aromatase neurons projecting to the LHb are activated following exposure to female urine. Together, our findings suggest that, in males, the LHb acts as a processing hub that integrates sexual olfactory signals through its connections with the aromatase-dependent olfactory neural circuit. This work was supported by the INRAE Phase Department (France).

S5-P35

Do the living environment and the genetic background influence the mouse VNO histological structure? Preliminary results.

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The role of the vomeronasal organ (VNO) in animals is well known, and the mouse model is the most used to study its function. Recent studies described some causes leading to spontaneous VNO alterations in animals, such as aging or living environment conditions. The impact of genetic diversity on the VNO condition remains unexplored, despite its strong influence on behavior is well documented. This communication aims to hypothesize that genetic background, coupled with environmental and hygienic conditions, may also affect VNO structure in mice. Across studies, we histologically assessed the VNO in genetically intact C57BL/6JRj and RjOrl:SWISS mice and in two transgenic models: Alzheimer's (5XFAD) and genetic vasectomy (Prm1EGFP). We did not observe any histological differences between C57BL/6JRj and RjOrl:SWISS mice' vomeronasal sensory epithelium (VNSE) ($p=0.286$). But, scoring alterations from 0 (absence) to 3 (strong) in the VNSE, C57BL/6JRj mice presented fewer alteration signs (mean score = 1.1) than 5XFAD transgenic mice (mean score = 1.7), and even than wild type 5XFAD mice (mean score = 1.6). Also, genetically intact mice VNOs presented less glycogen accumulation (4.2% positivity) than 5XFAD mice (7.5%) and WT-5XFAD mice (7.7%). Furthermore, the VNSE of Prm1EGFP mice, with or without expression of genetic mutation, seemed to show less cellular density, and thus more vacuolization (mean score 2.1 and 1.56, respectively), than usually observed in non-modified mice (mean score=1.1). Overall, our preliminary results may pave the way for further studies that should evaluate the off-target effects of genetic manipulation on VNO structure together with the impact of housing conditions and sanitary status, which vary widely between laboratories. This presentation aims to encourage reflection on animal model choice in behavioral studies and to open discussion on potential biases that may arise from strain-related VNO alterations and their possible impact on analyses.